



Medical Coverage Policy

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Bariatric Surgery and Procedures

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Related Coverage Resources

[Gastric Pacing/Gastric Electrical Stimulation \(GES\)](#)

[Panniculectomy and Abdominoplasty](#)

[Surgical Treatments for Obstructive Sleep](#)

[Apnea](#)

[Sleep Management](#)

[Vagus Nerve Stimulation \(VNS\)](#)

INSTRUCTIONS FOR USE

The following Coverage Policy applies to health benefit plans administered by Cigna Companies. Certain Cigna Companies and/or lines of business only provide utilization review services to clients and do not make coverage determinations. References to standard benefit plan language and coverage determinations do not apply to those clients. Coverage Policies are intended to provide guidance in interpreting certain standard benefit plans administered by Cigna Companies. Please

note, the terms of a customer's particular benefit plan document [Group Service Agreement, Evidence of Coverage, Certificate of Coverage, Summary Plan Description (SPD) or similar plan document] may differ significantly from the standard benefit plans upon which these Coverage Policies are based. For example, a customer's benefit plan document may contain a specific exclusion related to a topic addressed in a Coverage Policy. In the event of a conflict, a customer's benefit plan document always supersedes the information in the Coverage Policies. In the absence of a controlling federal or state coverage mandate, benefits are ultimately determined by the terms of the applicable benefit plan document. Coverage determinations in each specific instance require consideration of 1) the terms of the applicable benefit plan document in effect on the date of service; 2) any applicable laws/regulations; 3) any relevant collateral source materials including Coverage Policies and; 4) the specific facts of the particular situation. Each coverage request should be reviewed on its own merits. Medical directors are expected to exercise clinical judgment where appropriate and have discretion in making individual coverage determinations. Where coverage for care or services does not depend on specific circumstances, reimbursement will only be provided if a requested service(s) is submitted in accordance with the relevant criteria outlined in the applicable Coverage Policy, including covered diagnosis and/or procedure code(s). Reimbursement is not allowed for services when billed for conditions or diagnoses that are not covered under this Coverage Policy (see "Coding Information" below). When billing, providers must use the most appropriate codes as of the effective date of the submission. Claims submitted for services that are not accompanied by covered code(s) under the applicable Coverage Policy will be denied as not covered. Coverage Policies relate exclusively to the administration of health benefit plans. Coverage Policies are not recommendations for treatment and should never be used as treatment guidelines. In certain markets, delegated vendor guidelines may be used to support medical necessity and other coverage determinations.

Overview

This Coverage Policy addresses bariatric surgery and procedures for the treatment of obesity and morbid obesity.

Coverage Policy

Coverage for bariatric surgery or revision of a bariatric surgical procedure varies across plans and may be governed by state mandates. Refer to the customer's benefit plan document for coverage details.

This coverage policy statement is organized as follows:

- 1) Criteria that applies to Adults only**
- 2) Criteria that applies to Adolescents only**
- 3) Criteria that applies to Adults and Adolescents**

Adults

Bariatric surgery for the treatment of obesity in an adult (age $\geq$ 18 years) using a covered procedure outlined below is considered medically necessary when ALL of the following criteria are met:

- **EITHER** of the following:

- BMI (Body Mass Index) ≥ 35 kg/m² (class 2 obesity) (BMI ≥ 27.5 kg/m² in Asians- when ethnicity is confirmed by provider attestation)
- BMI (Body Mass Index) 30–34.9 kg/m² (class 1 obesity) (BMI 25–27.4 kg/m² in Asians- when ethnicity is confirmed by provider attestation) with at least one clinically significant obesity-related comorbidity, including but not limited to the following:
 - mechanical arthropathy in a weight-bearing joint (symptomatic degenerative joint disease in a weight bearing joint)
 - diabetes mellitus
 - poorly controlled hypertension (systolic blood pressure at least 140 mm Hg or diastolic blood pressure 90 mm Hg or greater, despite optimal medical management)
 - hyperlipidemia
 - coronary artery disease
 - lower extremity lymphatic or venous obstruction
 - obstructive sleep apnea
 - pulmonary hypertension
 - evidence of fatty liver disease (i.e., metabolic dysfunction-associated steatotic liver disease [MASLD; previously termed nonalcoholic fatty liver disease (NAFLD)] or metabolic dysfunction-associated steatohepatitis [MASH; previously termed nonalcoholic steatohepatitis (NASH)])
 - gastroesophageal reflux disease (GERD) refractory to medical therapy
- A thorough multidisciplinary evaluation within the previous 12 months which includes ALL of the following:
 - a description of the proposed procedure(s)
 - documentation of failure of weight loss by medical management
 - unequivocal clearance for bariatric surgery by a mental health provider
 - a nutritional evaluation by a physician, physician assistant, nurse practitioner or registered dietitian

Initial Bariatric Surgical Procedures (Adults)

When the specific medical necessity criteria noted above for bariatric surgery for an adult have been met, ANY of the following open or laparoscopic bariatric surgical procedures for the treatment of obesity is considered medically necessary:

Procedure	Open CPT® Codes	Laparoscopic CPT® Codes
Vertical band gastroplasty	43842	43659
Adjustable silicone gastric banding (e.g., LAP-BAND®, REALIZE™)	43843	43770
Sleeve gastrectomy as a stand-alone or staged procedure	43843	43775
Roux-en-Y gastric bypass (roux limb less than 150 cm)	43846	43644
Roux-en-Y gastric bypass (roux limb greater than 150 cm)	43847	43645
Biliopancreatic Diversion with Duodenal Switch (BPD/DS)	43845	43659 or 44799
Biliopancreatic Diversion (BPD) without DS	43633	43659 or 44799

Single-anastomosis duodenal-ileal bypass with Sleeve gastrectomy (SADI-S) (i.e. Loop duodenal switch)	43999	43659
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Procedure	CPT® Code
Endoscopic sleeve gastroplasty	43889

Adjustment of a silicone gastric banding is considered medically necessary to control the rate of weight loss and/or treat symptoms secondary to gastric restriction following a medically necessary adjustable silicone gastric banding procedure.

The following bariatric surgical procedures for the treatment of obesity, when performed alone or in conjunction with another bariatric surgical procedure are considered not medically necessary:

Procedure	CPT® Code(s)
Gastric electrical stimulation (GES) or gastric pacing	43881 and 64590
Vagus nerve stimulation	64553 OR 64568 OR 0908T

The following bariatric surgical procedures for the treatment of obesity, when performed alone or in conjunction with another bariatric surgical procedure are considered experimental, investigational or unproven:

Procedure	CPT® and HCPCS Code(s)
Band over bypass	43770, 43843, 43999
Band over sleeve	43770, 43843, 43999
Fobi-Pouch (limiting proximal gastric pouch)	43659, 43843, 43999
Intestinal bypass (jejunioileal bypass)	44238, 44799
Intragastric balloon (e.g., Orbera™, ReShape™, Obalon)	43290, 43999, 0813T
Laparoscopic greater curvature plication	43659
Mini-gastric bypass/ One Anastomosis Gastric Bypass (OAGB)/Loop gastric bypass	43659, 43843
Endoscopic bariatric surgery procedures , including but not limited to the following: ➤ Natural Orifice Transluminal Endoscopic Surgery (NOTES) ➤ restorative obesity surgery, endoluminal (ROSE) ➤ StomaphyX™ ➤ duodenojejunal bypass liner (e.g., Endobarrier™) ➤ transoral gastroplasty (e.g., TOGA®)	43289, 43499
Transoral Outlet Reduction (TORe)	43289, 43499, C9785
Roux-en-Y gastric bypass combined with simultaneous gastric banding	43644 or 43645 and 43770 OR 43846 or 43847 and 43843 or 43999
Stomach aspiration therapy (e.g., AspireAssist®)	43659, 43999

Procedure	CPT® and HCPCS Code(s)
Vagus nerve blocking (e.g., Maestro®)	64999

Reoperation and Revisional Bariatric Surgery (Adults)

Replacement of an adjustable silicone gastric band or separate or concurrent band removal and conversion to a second bariatric surgical procedure is considered medically necessary if there is evidence of band slippage or band component malfunction and the faulty component cannot be repaired

Gastric band removal is considered medically necessary for gastrointestinal symptoms (e.g., persistent nausea and/or vomiting, gastroesophageal reflux) with or without imaging evidence of obstruction.

The following procedures are considered medically necessary when the individual develops a major complication from a previous bariatric surgical procedure (e.g., stricture, obstruction, erosion, gastric prolapse, ulceration, fistula formation, esophageal dilatation, gastroesophageal reflux disease refractory to medical therapy):

- surgical repair or reversal (i.e., takedown)
- conversion to a medically necessary bariatric surgical procedure

In the absence of a major complication, revision of a previous bariatric surgical procedure or conversion to another procedure for an adult is considered medically necessary when ALL of the following are met:

- Weight loss failure $\geq$ two years following a previous bariatric surgical procedure
- Individual must meet the initial medical necessity criteria for surgery
- The requested procedure includes ANY of the following:

Procedure	Open CPT® Codes	Laparoscopic CPT® Codes
Vertical band gastroplasty	43842	43659
Adjustable silicone gastric banding (e.g., LAP-BAND®, REALIZE™)	43843	43770
Sleeve gastrectomy as a stand-alone or staged procedure	43843	43775
Roux-en-Y gastric bypass (roux limb less than 150 cm)	43846	43644
Roux-en-Y gastric bypass (roux limb greater than 150 cm)	43847	43645
Biliopancreatic Diversion with Duodenal Switch (BPD/DS)	43845	43659 or 44799
Biliopancreatic Diversion (BPD) without DS	43633	43659 or 44799
Revision of gastrojejunal anastomosis (gastrojejunostomy)	43860	43659
Single-anastomosis duodenal-ileal bypass with Sleeve gastrectomy (SADI-S) (i.e. Loop duodenal switch)	43999	43659

Surgical reversal (i.e., takedown), revision of a previous bariatric surgical procedure or conversion to another bariatric surgical procedure is considered not medically necessary for EITHER of the following:

- Inadequate weight loss due to individual noncompliance with postoperative nutrition and exercise recommendations
- ANY other indication

Adolescents

Bariatric surgery for the treatment of morbid obesity in an adolescent (age 11–17 years) is considered medically necessary using a covered procedure outlined below when ALL of the following criteria are met:

- The individual has evidence of EITHER of the following:
 - BMI (Body Mass Index) $\geq 40 \text{ kg/m}^2$ or $\geq 140\%$ of the 95th percentile (class 3 obesity) (whichever is lower)
 - BMI (Body Mass Index) $35\text{--}39.9 \text{ kg/m}^2$ or $\geq 120\%$ of the 95th percentile (class 2 obesity) (whichever is lower) with at least one clinically significant obesity-related comorbidity, including but not limited to the following:
 - coronary artery disease
 - diabetes mellitus
 - idiopathic intracranial hypertension
 - poorly controlled hypertension (systolic blood pressure at least 140 mm Hg or diastolic blood pressure 90 mm Hg or greater, despite optimal medical management)
 - obstructive sleep apnea
 - gastroesophageal reflux
 - metabolic dysfunction-associated steatohepatitis (MASH; previously termed nonalcoholic steatohepatitis [NASH])
- A thorough multidisciplinary evaluation within the previous 12 months which includes ALL of the following:
 - a description of the proposed procedure(s)
 - documentation of failure of weight loss by medical management
 - unequivocal clearance for bariatric surgery by a mental health provider
 - a nutritional evaluation by a physician, physician assistant, nurse practitioner or registered dietitian

Initial Bariatric Surgical Procedures (Adolescents)

When the specific medical necessity criteria noted above for bariatric surgery for an adolescent have been met, ANY of the following open or laparoscopic bariatric surgical procedures for the treatment of morbid obesity is considered medically necessary:

Procedure	Open CPT® Codes	Laparoscopic CPT® Codes
Sleeve gastrectomy	43843	43775
Roux-en-Y gastric bypass (roux limb less than 150 CM)	43846	43644

Roux-en-Y gastric bypass (roux limb greater than 150 CM)	43847	43645
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All other bariatric surgical procedures for the treatment of morbid obesity in an adolescent are considered not medically necessary.

Reoperation and Revisional Bariatric Surgery (Adolescents)

The following procedures are considered medically necessary when the adolescent develops a major complication from a previous bariatric surgical procedure (e.g., stricture, obstruction, erosion, gastric prolapse, ulceration, fistula formation, esophageal dilatation, gastroesophageal reflux disease refractory to medical therapy):

- surgical repair
- conversion to a medically necessary bariatric surgical procedure (i.e., Roux-en-Y or sleeve gastrectomy)

In the absence of a major complication, revision of a previous bariatric surgical procedure or conversion to another procedure for an adolescent is considered medically necessary when ALL of the following are met:

- Weight loss failure $\geq$ two years following a previous bariatric surgical procedure
- Individual must meet the initial medical necessity criteria for surgery
- The requested procedure includes ANY of the following:

Procedure	Open CPT® Codes	Laparoscopic CPT® Codes
Sleeve gastrectomy	43843	43775
Roux-en-Y gastric bypass (roux limb less than 150 CM)	43846	43644
Roux-en-Y gastric bypass (roux limb greater than 150 CM)	43847	43645
Revision of gastrojejunal anastomosis (gastrojejunostomy)	43860	43659

Surgical reversal (i.e., takedown), revision of a previous bariatric surgical procedure or conversion to another bariatric surgical procedure is considered not medically necessary for EITHER of the following:

- Inadequate weight loss due to individual noncompliance with postoperative nutrition and exercise recommendations
- ANY other indication

Adults and Adolescents

The following procedures are considered not medically necessary for reporting bariatric surgery treatment of obesity or morbid obesity:

Procedure	CPT® Code(s)
Gastrectomy, partial, distal; with gastroduodenostomy	43631
Gastrectomy, partial, distal; with gastrojejunostomy	43632

Gastrectomy, partial, distal; with formation of intestinal pouch	43634
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Bariatric Surgery for the Treatment of Other Conditions

Bariatric surgery is considered not medically necessary for the primary treatment of any condition other than obesity or morbid obesity.

Liver Biopsy or Upper Endoscopy

The following procedure performed in conjunction with a bariatric surgery is considered not medically necessary:

- liver biopsy in the absence of signs or symptoms of liver disease (e.g., elevated liver enzymes, enlarged liver, abnormal intraoperative findings)

When performed concurrently as part of a bariatric surgical procedure EACH of the following is considered integral to the procedure and not separately reimbursable:

- upper gastrointestinal endoscopy performed concurrent with a bariatric surgery procedure to confirm a surgical anastomosis or to establish anatomical landmarks

Coding Information

Notes:

1. This list of codes may not be all-inclusive since the American Medical Association (AMA) and Centers for Medicare & Medicaid Services (CMS) code updates may occur more frequently than policy updates.
2. Deleted codes and codes which are not effective at the time the service is rendered may not be eligible for reimbursement.

Initial Bariatric Surgical Procedures (Adults)

Considered Medically Necessary when criteria in the applicable policy statements listed above are met:

CPT®* Codes	Description
43633	Gastrectomy, partial, distal; with Roux-en-Y reconstruction
43644 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43659 [†]	Unlisted laparoscopy procedure, stomach
43770 [†]	Laparoscopy, surgical, gastric restrictive procedure; placement of adjustable gastric restrictive device (eg, gastric band and subcutaneous port components)
43775	Laparoscopy, surgical, gastric restrictive procedure; longitudinal gastrectomy (ie, sleeve gastrectomy)
43842	Gastric restrictive procedure, without gastric bypass, for morbid obesity; vertical-banded gastroplasty
43843 [†]	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastroplasty

CPT®* Codes	Description
43845	Gastric restrictive procedure with partial gastrectomy, pylorus-preserving duodenoileostomy and ileoileostomy (50 to 100 cm common channel) to limit absorption (biliopancreatic diversion with duodenal switch)
43846 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy
43847 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption
43889	Gastric restrictive procedure, transoral, endoscopic sleeve gastropasty (ESG), including argon plasma coagulation, when performed
43999 [†]	Unlisted procedure, stomach
44799 [†]	Unlisted procedure, small intestine

[†]Note: Considered Experimental/Investigational/Unproven when used to report any procedure listed in this policy as Experimental/Investigational/Unproven for the treatment of obesity

Considered Not Medically Necessary for the treatment of obesity, when performed alone or in conjunction with another bariatric surgical procedure:

CPT®* Codes	Description
43881	Implantation or replacement of gastric neurostimulator electrodes, antrum, open
64553	Percutaneous implantation of neurostimulator electrode array; cranial nerve
64568	Open implantation of cranial nerve (eg, vagus nerve) neurostimulator electrode array and pulse generator
64590	Insertion or replacement of peripheral or gastric neurostimulator pulse generator or receiver, direct or inductive coupling
0908T	Open implantation of integrated neurostimulation system, vagus nerve, including analysis and programming, when performed

Considered Experimental/Investigational/Unproven when used to report any procedure listed as Experimental/Investigational/Unproven for the treatment of obesity:

CPT®* Codes	Description
43289	Unlisted laparoscopy procedure, esophagus
43290	Esophagogastroduodenoscopy, flexible, transoral; with deployment of intragastric bariatric balloon
43499	Unlisted procedure, esophagus
43644 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43659 [†]	Unlisted laparoscopy procedure, stomach
43770 [†]	Laparoscopy, surgical, gastric restrictive procedure; placement of adjustable gastric restrictive device (eg, gastric band and subcutaneous port components)
43843 [†]	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastropasty
43846 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy

CPT®* Codes	Description
43847 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption
43999 [†]	Unlisted procedure, stomach
44238 [†]	Unlisted laparoscopy procedure, intestine (except rectum)
44799 [†]	Unlisted procedure, small intestine
64999	Unlisted procedure, nervous system
0813T	Esophagogastroduodenoscopy, flexible, transoral, with volume adjustment of intragastric bariatric balloon

[†]Note: Considered Medically Necessary when criteria in the applicable policy statements listed above are met for bariatric surgery for an adult for the treatment of obesity

HCPCS Codes	Description
C9785	Endoscopic outlet reduction, gastric pouch application, with endoscopy and intraluminal tube insertion, if performed, including all system and tissue anchoring components

Reoperation and Revisional Bariatric Surgery (Adults)

Considered Medically Necessary when criteria in the applicable policy statements listed above are met:

CPT®* Codes	Description
43633	Gastrectomy, partial, distal; with Roux-en-Y reconstruction
43644 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43659 [†]	Unlisted laparoscopy procedure, stomach
43770 [†]	Laparoscopy, surgical, gastric restrictive procedure; placement of adjustable gastric restrictive device (eg, gastric band and subcutaneous port components)
43771	Laparoscopy, surgical, gastric restrictive procedure; revision of adjustable gastric restrictive device component only
43772	Laparoscopy, surgical, gastric restrictive procedure; removal of adjustable gastric restrictive device component only
43773	Laparoscopy, surgical, gastric restrictive procedure; removal and replacement of adjustable gastric restrictive device component only
43774	Laparoscopy, surgical, gastric restrictive procedure; removal of adjustable gastric restrictive device and subcutaneous port components
43775	Laparoscopy, surgical, gastric restrictive procedure; longitudinal gastrectomy (ie, sleeve gastrectomy)
43842	Gastric restrictive procedure, without gastric bypass, for morbid obesity; vertical-banded gastroplasty
43843 [†]	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastroplasty
43845	Gastric restrictive procedure with partial gastrectomy, pylorus-preserving duodenoileostomy and ileoileostomy (50 to 100 cm common channel) to limit absorption (biliopancreatic diversion with duodenal switch)

CPT®* Codes	Description
43846 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy
43847 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption
43848	Revision, open, of gastric restrictive procedure for morbid obesity, other than adjustable gastric restrictive device (separate procedure)
43860	Revision of gastrojejunal anastomosis (gastrojejunostomy) with reconstruction, with or without partial gastrectomy or intestine resection; without vagotomy
43865	Revision of gastrojejunal anastomosis (gastrojejunostomy) with reconstruction, with or without partial gastrectomy or intestine resection; with vagotomy
43886	Gastric restrictive procedure, open; revision of subcutaneous port component only
43888	Gastric restrictive procedure, open; removal and replacement of subcutaneous port component only
43999 [†]	Unlisted procedure, stomach
44799 [†]	Unlisted procedure, small intestine

[†]Note: Considered Experimental/Investigational/Unproven when used to report any procedure listed in this policy as Experimental/Investigational/Unproven for the treatment of obesity

Initial Bariatric Surgical Procedures (Adolescents)

Considered Medically Necessary when criteria in the applicable policy statements listed above are met:

CPT®* Codes	Description
43644 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43775	Laparoscopy, surgical, gastric restrictive procedure; longitudinal gastrectomy (ie, sleeve gastrectomy)
43843 [†]	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastroplasty
43846 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy
43847 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption

[†]Note: Considered Experimental/Investigational/Unproven when used to report any procedure listed in this policy as Experimental/Investigational/Unproven for the treatment of morbid obesity

Reoperation and Revisional Bariatric Surgery (Adolescents)

Considered Medically Necessary when criteria in the applicable policy statements listed above are met:

CPT®* Codes	Description
43644 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645 [†]	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43659 [†]	Unlisted laparoscopy procedure, stomach
43775	Laparoscopy, surgical, gastric restrictive procedure; longitudinal gastrectomy (ie, sleeve gastrectomy)
43843 [†]	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastroplasty
43846 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy
43847 [†]	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption
43860	Revision of gastrojejunal anastomosis (gastrojejunostomy) with reconstruction, with or without partial gastrectomy or intestine resection; without vagotomy

[†]Note: Considered Experimental/Investigational/Unproven when used to report any procedure listed in this policy as Experimental/Investigational/Unproven for the treatment of obesity

Adults and Adolescents

Considered Not Medically Necessary when used to report bariatric surgery treatment of obesity or morbid obesity:

CPT®* Codes	Description
43631	Gastrectomy, partial, distal; with gastroduodenostomy
43632	Gastrectomy, partial, distal; with gastrojejunostomy
43634	Gastrectomy, partial, distal; with formation of intestinal pouch

Bariatric Surgery for the Treatment of Other Conditions

Considered Not Medically Necessary when performed solely for the treatment of any condition other than obesity or morbid obesity:

CPT®* Codes	Description
43644	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and Roux-en-Y gastroenterostomy (roux limb 150 cm or less)
43645	Laparoscopy, surgical, gastric restrictive procedure; with gastric bypass and small intestine reconstruction to limit absorption
43770	Laparoscopy, surgical, gastric restrictive procedure; placement of adjustable gastric restrictive device (eg, gastric band and subcutaneous port components)
43771	Laparoscopy, surgical, gastric restrictive procedure; revision of adjustable gastric restrictive device component only
43772	Laparoscopy, surgical, gastric restrictive procedure; removal of adjustable gastric restrictive device component only
43773	Laparoscopy, surgical, gastric restrictive procedure; removal and replacement of adjustable gastric restrictive device component only

CPT®* Codes	Description
43774	Laparoscopy, surgical, gastric restrictive procedure; removal of adjustable gastric restrictive device and subcutaneous port components
43775	Laparoscopy, surgical, gastric restrictive procedure; longitudinal gastrectomy (ie, sleeve gastrectomy)
43842	Gastric restrictive procedure, without gastric bypass, for morbid obesity; Vertical-banded gastroplasty
43843	Gastric restrictive procedure, without gastric bypass, for morbid obesity; other than vertical-banded gastroplasty
43845	Gastric restrictive procedure with partial gastrectomy, pylorus-preserving duodenoileostomy and ileoileostomy (50 to 100 cm common channel) to limit absorption (biliopancreatic diversion with duodenal switch)
43846	Gastric restrictive procedure, with gastric bypass for morbid obesity; with short limb (150 cm or less) Roux-en-Y gastroenterostomy
43847	Gastric restrictive procedure, with gastric bypass for morbid obesity; with small intestine reconstruction to limit absorption
43848	Revision, open, of gastric restrictive procedure for morbid obesity, other than adjustable gastric restrictive device (separate procedure)
43860	Revision of gastrojejunal anastomosis (gastrojejunostomy) with reconstruction, with or without partial gastrectomy or intestine resection; without vagotomy
43865	Revision of gastrojejunal anastomosis (gastrojejunostomy) with reconstruction, with or without partial gastrectomy or intestine resection; with vagotomy
43886	Gastric restrictive procedure, open; revision of subcutaneous port component only
43888	Gastric restrictive procedure, open; removal and replacement of subcutaneous port component only

Liver Biopsy or Upper Endoscopy

Considered Not Medically Necessary when used to report liver biopsy performed in conjunction with a bariatric surgery in the absence of signs or symptoms of liver disease:

CPT®* Codes	Description
47379	Unlisted laparoscopic procedure, liver

Considered integral to the primary bariatric procedure when gastrointestinal endoscopy is concurrently performed to assess a surgical anastomosis or to establish anatomical landmarks:

CPT®* Codes	Description
43235	Esophagogastroduodenoscopy, flexible, transoral; diagnostic, including collection of specimen(s) by brushing or washing, when performed (separate procedure)

***Current Procedural Terminology (CPT®) ©2025 American Medical Association: Chicago, IL.**

General Background

Obesity and overweight are defined clinically using the body mass index (BMI). BMI is an objective measurement and is currently considered the most reproducible measurement of total body fat. In adults, excess body weight (EBW) is defined as the amount of weight that is in excess of the ideal body weight (IBW), or a BMI ≥ 25 kg/m². The National Heart, Lung and Blood Institute (NHLBI) (1998) clinical guidelines recommended that the BMI should be used to classify overweight and obesity and to estimate relative risk for disease compared to normal weight. The American Heart Association, American College of Cardiology and The Obesity Society (AHA/ACC/TOS) updated the NHLBI guidelines in 2013 (Jensen, et al., 2014). The guidelines defined the following classifications based on BMI:

Classification	BMI
Underweight	< 18.5 kg/m ²
Normal weight	18.5–24.9 kg/m ²
Overweight	25.0–29.9 kg/m ²
Obesity (Class 1)	30.0–34.9 kg/m ²
Obesity (Class 2)	35.0–39.9 kg/m ²
Extreme Obesity (Class 3)	≥ 40 kg/m ²

BMI is a direct calculation based on height and weight, regardless of gender:

$$\text{BMI} = \frac{\text{weight (kg)}}{\text{height (m)}^2} \text{ OR } \left[\frac{\text{weight (lb)}}{\text{height (in)}^2} \right] \times 703$$

Clinically severe or morbid obesity is defined as a BMI ≥ 40 kg/m² or a BMI of 35–39.9 kg/m² with comorbid conditions. Another group of individuals who have been identified are the super-obese. Super-obesity has been defined in the literature as a BMI > 50 kg/m². Comorbidities of morbid obesity that may be considered include any of the following:

- mechanical arthropathy (weight-related degenerative joint disease)
- type 2 diabetes
- clinically unmanageable hypertension (systolic blood pressure at least 140 mm Hg or diastolic blood pressure 90 mm Hg or greater, or if individual is taking antihypertensive agents)
- hyperlipidemia
- coronary artery disease
- lower extremity lymphatic or venous obstruction
- severe obstructive sleep apnea
- obesity-related pulmonary hypertension

Other severe obesity-related co-morbidities including obesity-hypoventilation syndrome (OHS), Pickwickian syndrome (a combination of obstructive sleep apnea [OSA] and OHS), metabolic dysfunction-associated steatotic liver disease [MASLD; previously termed nonalcoholic fatty liver disease (NAFLD)] or metabolic dysfunction-associated steatohepatitis [MASH; previously termed nonalcoholic steatohepatitis (NASH)] pseudotumor cerebri, gastroesophageal reflux disease (GERD), asthma, venous stasis disease, severe urinary incontinence, or considerably impaired quality of life, may also be considered for bariatric surgical intervention (Mechanick, et al., 2013, updated 2020).

Strategies for Weight Loss

Treatment of obesity is generally described as a two-part process: 1) assessment, including BMI measurement and risk factor identification; and 2) treatment/management. Obesity management includes primary weight loss, prevention of weight regain and the management of associated risk. During the assessment phase, the individual needs to be prepared for the comprehensive nature of the program, including realistic timelines and goals. Obesity is a chronic disease, therefore treatment should be regarded as a long-term issue. General recommendations for an overall weight-loss strategy include the following (Jensen and Bessesen, 2024; Goroll and Mulley, 2009):

- For overweight or obese patients not ready to lose weight, the best approach is to educate them about health risks, address other cardiovascular risk factors, and encourage the maintenance of their current weight.
- For motivated persons who are overweight (BMI 25 to 29.9 kg/m²) and have two or more obesity-related medical conditions or are frankly obese (BMI >30 kg/m²), a six-month goal of a 10% weight loss can be set (1 to 2 lb/week) and a program of diet, exercise, and behavioral therapy prescribed. If, after six months, the target weight is not achieved, one can consider adding pharmacologic therapy for those at greatest risk (BMI > 27 kg/m² plus two or more cardiovascular risk factors, or BMI > 30 kg/m²).
- For markedly obese persons at greatest risk (BMI > 35 kg/m² with two or more obesity-related medical conditions or BMI > 40 kg/m²), consider a surgical approach if serious and repeated attempts using the foregoing measures have been unsuccessful.

The NHLBI guidelines (1998) include the following recommendations regarding nonsurgical strategies for achieving weight loss and weight maintenance:

- Dietary Therapy:
 - Low-calorie diets are recommended for weight loss in overweight and obese persons. Reducing fat as part of a low-calorie diet is a practical way to reduce calories.
 - Optimally, dietary therapy should last at least six months, as many studies suggest that the rate of weight loss decreases after about six months. Shorter periods of dietary therapy typically result in lesser weight reductions.
 - The literature suggests that weight-loss and weight-maintenance therapies that provide a greater frequency of contacts between the individual and the practitioner and are provided over the long term should be put in place. This can lead to more successful weight loss and weight maintenance.
- Increased Physical Activity/Exercise is recommended as part of a comprehensive, weight-loss therapy and weight-maintenance program because it:
 - modestly contributes to weight loss in overweight and obese adults
 - may decrease abdominal fat
 - increases cardiorespiratory fitness
 - may help with maintenance of weight loss
- Combined Therapy: The combination of a reduced-calorie diet and increased physical activity is recommended, since it produces weight loss, decreases abdominal fat and increases cardiorespiratory fitness.
- Behavior Therapy: Is a useful adjunct when incorporated into treatment for weight loss and weight maintenance.

In addition, the NHLBI recommended that weight-loss drugs approved by the U.S. Food and Drug Administration (FDA) only be used as part of a comprehensive weight-loss program, including diet and physical activity for individuals with a BMI ≥ 30 with no concomitant obesity-related risk factors or diseases, or for individuals with a BMI ≥ 27 with concomitant obesity-related risk factors or diseases.

Clinical supervision is an essential component of dietary management. According to the NHLBI (1998), "frequent clinical encounters during the initial six months of weight reduction appear to facilitate reaching the goals of therapy". Nutritional counseling by a registered dietitian (RD) in the course of treatment for patients with overweight and obesity is optimal, as the RD is uniquely qualified to provide medical nutrition therapy to improve and maintain a healthy diet that meets nutrition needs and advances weight loss efforts to improve cardiometabolic outcomes (Raynor, et al., 2024). Lifestyle modification should include a referral to a registered dietitian or credible weight loss program/service for counseling in energy intake reduction and nutritional strategies with a weight reduction goal of 5–10% of total body weight. During the period of active weight loss, regular visits of at least once per month and preferably more often with a health professional for the purposes of reinforcement, encouragement, and monitoring will facilitate weight reduction (NHLBI, 1998). Physicians can also provide clinical oversight and monitoring of what are often complex comorbid conditions and can select the optimal and most medically appropriate weight management, nutritional and exercise strategies. Some commercially available diet programs do not consistently provide counselors who are trained and certified as registered dietitians or with other equivalent clinical training. However, diet programs/plans, such as Weight Watchers®, Jenny Craig® or similar plans are acceptable methods of dietary management if there is concurrent documentation of at least monthly clinical encounters with a physician.

Surgical Intervention

The NHLBI recommended weight-loss surgery as an option for carefully selected adult patients with clinically severe obesity (BMI of 40 or greater; or BMI of 35 or greater with serious comorbid conditions) when less-invasive methods of weight loss have failed and the patient is at high risk for obesity-associated morbidity or mortality. Surgical therapy for morbid obesity is not only effective in producing weight loss but is also effective in improving several significant complications of obesity, including diabetes, hypertension, dyslipidemia, and sleep apnea. The degree of benefit and the rates of morbidity and mortality of the various surgical procedures vary according to the procedure (Colquitt, et al., 2014; Bouldin, et al., 2006).

Access to a multidisciplinary team approach, involving a physician with a special interest in obesity; a surgeon with extensive experience in bariatric procedures, a dietitian or nutritionist; and a psychologist, psychiatrist or licensed mental health care provider interested in behavior modification and eating disorders, is optimal. A mental health evaluation should specifically address any mental health or substance abuse diagnoses, maladaptive eating behaviors, psychosocial functioning, the emotional readiness and ability of the patient to make and sustain lifestyle changes, and the adequacy of their support system. Any identified psychopathology, substance use disorder, or eating disorder must receive appropriate referral and treatment. Additionally, all patients should be informed about potential new psychosocial issues that may arise following bariatric surgery (American Society for Metabolic and Bariatric Surgery [ASMBS], 2021). Realistic expectations about the degree of weight loss, the compromises required by the patient and the positive effect on associated weight-related comorbidities and quality of life should be discussed and contrasted with the potential morbidity and operative mortality of bariatric surgery (Jensen and Bessesen, 2024).

With bariatric surgery procedures, patients lose an average of 50–60% of excess body weight and have a decrease in BMI of about 10 kg/m² during the first 12–24 postoperative months. Long-term studies show a tendency for a modest weight gain (5–7 kg) after the initial postoperative years;

long-term maintenance of an overall mean weight loss of about 50% of excess body weight can be expected.

In 2022, American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO) (Eisenberg et al., 2022) updated indications for metabolic and bariatric surgery (MBS), to replace previous guidelines established by the National Institutes of Health (NIH) in 1991. In 2024, the ASMBS/IFSO (De Luca, et al., 2024) published a grade of recommendation and level of evidence for each recommendation as follows:

Grade of recommendation	Level of evidence	Type of study
A	1a	Systematic review of [homogeneous] randomized controlled trials
A	1b	Individual randomized controlled trials [with narrow confidence intervals]
B	2a	Systematic review of [homogeneous] cohort studies of "exposed" and "unexposed" subjects
B	2b	Individual cohort study/low-quality randomized control studies
B	3a	Systematic review of [homogeneous] case-control studies
B	3b	Individual case-control studies
C	4	Case series, low-quality cohort, or case-control studies
D	5	Expert opinions based on nonsystematic reviews of results or mechanistic studies

ASMBS/IFSO updated guidelines on indications for metabolic and bariatric surgery with grade of recommendation and level of evidence are as follows:

- Metabolic and bariatric surgery (MBS) is recommended for patients with a body mass index (BMI) of 30–34.9 kg/m² with type 2 diabetes mellitus and/or other obesity-associated medical problems. (Grade of recommendation B, Level of evidence 2a)
- MBS is recommended for patients with a BMI of 35–40 kg/m² regardless of the presence, absence, or severity of obesity-associated medical problems (Grade of recommendation D, Level of evidence 5)
- Regarding BMI thresholds in the Asian population, access to MBS should not be denied solely based on the BMI (Grade of recommendation B, Level of evidence 2a,)
- Regarding MBS in the older population, there is no evidence to support an age limit (Grade of recommendation B, Level of evidence 2a)
- Regarding MBS for pediatric and adolescents, MBS is safe in the population younger than 18 years, produces durable weight loss, and improvement in obesity-associated medical problems (Grade of recommendation A, Level of evidence 1b)
- Regarding MBS prior to joint arthroplasty, MBS can be considered a bridge to joint arthroplasty in patients with BMI ≥30 kg/m² (Grade of recommendation B, Level of evidence 2b)
- In patients with severe obesity and an abdominal wall hernia, MBS-induced weight loss is suggested before hernia repair (Grade of recommendation B, Level of evidence 2b)
- Regarding MBS prior to organ transplantation, published data supports considering patients in need of solid organ transplant to undergo MBS first to improve their eligibility for transplantation (Grade of recommendation B, Level of evidence 2b)
- MBS is safe and effective in patients BMI ≥60 kg/m² (Grade of recommendation B, Level of evidence 2a)
- MBS is associated with a reduction of progression of metabolic dysfunction-associated liver disease (MAFLD) to cirrhosis (Grade of recommendation B, Level of evidence 2b)

- MBS can be a useful treatment adjunct in patients with obesity and heart failure (Grade of recommendation B, Level of evidence 2b)
- Multidisciplinary care is at present the unmodifiable core of pre- and postoperative obesity management (Grade of recommendation B, Level of evidence 2c)
- Revisional MBS induces satisfactory metabolic outcomes with acceptable rates of complications and mortality (Grade of recommendation B, Level of evidence 2b)

BMI Requirement

Selection criteria for studies have previously included BMI ranges for clinically severe or morbid obesity, as outlined by the NHLBI. However, the use of bariatric procedures in patients with lower BMI measurements, with or without comorbidities, has been evaluated. Prospective and large retrospective studies indicate that metabolic and bariatric surgery (MBS) is a viable treatment option for patients with class I obesity who fail to achieve significant or lasting weight loss or improvement in co-morbidities through nonsurgical methods (Varban et al., 2020; Aminian et al., 2018; Schauer et al., 2017; Ikramuddin et al., 2016; Noun et al., 2016).

The American Association of Clinical Endocrinologists/American College of Endocrinology (AAACE/ACE), the Obesity Society (TOS), American Society for Metabolic & Bariatric Surgery (ASMBS), Obesity Medicine Association (OMA), and American Society of Anesthesiologists (ASA) 2019 updated guidelines have a new guideline regarding adjusting body mass index criterion for bariatric procedures based on ethnicity (e.g., 18.5 to 22.9 kg/m² is normal range, 23 to 24.9 kg/m² overweight, and ≥ 25 kg/m² obesity for Asians) (Mechanick et al., 2020). This is based on epidemiologic studies that indicate the prevalence of diabetes and cardiovascular disease occur at lower BMI levels in the Asian population than in the white population (Mechanick, et al., 2020; Ntuk et al., 2014; WHO, 2000).

Preoperative Weight Loss

Some propose that participation in a pre-operative weight loss program may provide better postoperative outcomes and reduce or prevent perioperative surgical complications. Weight loss programs may also help to identify those individuals who will be committed to and compliant with the short-term, long-term and lifelong medical management, behavioral changes, lifestyle changes, and diet and physical exercise regimens required to ensure the long-term success of bariatric surgery. However, there is a lack of consensus by professional societies and a lack of evidence in the published peer-reviewed literature to support the clinical effectiveness of preoperative weight loss prior to bariatric surgery. The outcomes of the available evidence are limited and conflicting (Cassie, et al., 2011; National Heart, Lung and Blood Institute [NHLBI], 1998). Despite limited evidence-based support, it is optimal for patients to demonstrate good eating and exercise habits prior to undergoing bariatric surgery in preparation for the post-surgical regimen.

According to the NHLBI Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults (1998), the initial goal of weight-loss therapy should be to reduce body weight by approximately 10% from baseline. With success, further weight loss can be attempted, if indicated, through additional assessment. The NHLBI guidelines further stated that:

- Bariatric surgery is not considered a first-line treatment.
- Even the most severely obese individuals (i.e., super-obese with BMI over 50) can be helped by a preoperative weight loss through a program of reduced-calorie diet and exercise therapy.
- Optimally, dietary therapy should last at least six months.
- Moderate weight loss (i.e., 10% of initial body weight) can significantly decrease the severity of obesity-associated risk factors. It can also set the stage for further weight loss, if indicated.

Bariatric surgeons and centers have advocated for preoperative weight loss, as it is believed that patients who are able to achieve this weight loss are most likely to have successful outcomes after surgery. The benefits of a preoperative weight-loss program include all of the following:

- reduction of the severity of obesity-associated risk factors, such as blood pressure, glucose intolerance, cardiorespiratory function and pulmonary function
- reduction of operative morbidity and surgical risk
- improvement in surgical access with weight loss
- identification of those individuals who will be committed to and compliant with the short-term, long-term and lifelong medical management follow-up, behavioral changes, lifestyle changes, and diet and physical exercise regimen required to ensure the long-term success of this surgery

Literature Review

Studies in the published peer-reviewed medical literature evaluating the impact of preoperative weight loss on the outcomes of bariatric surgery have yielded mixed results. There is a lack of evidence to support the clinical effectiveness of participation in a structured preoperative weight loss program.

Benotti et al. (2009) reported on 881 patients undergoing open or laparoscopic gastric bypass. All preoperative patients completed a six-month multidisciplinary program that encouraged a 10% preoperative weight loss. Study analysis demonstrated that increasing preoperative weight loss was associated with reduced complication frequencies in the entire group for total complications ($p=0.004$) and most likely for major complications ($p=0.06$).

Solomon and colleagues (2009) conducted a prospective randomized controlled trial of patients who underwent laparoscopic Roux-en-Y gastric bypass (LRYGB) after being randomized to either the non-weight-loss group ($n=35$) or the weight-loss group ($n=26$). Patients in the weight-loss group were requested to lose 10% or more of their excess body weight prior to surgery. One-year follow-up data were available for 26 patients in the weight-loss group and 18 in the non-weight-loss group. The patients in the weight-loss group had a better weight loss profile in all categories. However there were no statistically significant differences between the two groups in patient weight, BMI, amount of excess weight-loss, change in BMI, and resolution of comorbidities.

Alami et al. (2007) performed a prospective randomized trial ($n=61$) of patients undergoing laparoscopic gastric bypass surgery. Patients were assigned preoperatively to either a weight loss group ($n=26$) with a 10% weight loss requirement or a group that had no weight loss requirements ($n=35$). The two groups were identical in terms of initial weight, BMI, and incidence of comorbidities. Perioperative complications, operative time, postoperative weight loss, and resolution of co-morbidities were analyzed. Of the 61 patients, data was available for 12 at one-year follow-up. Preoperative weight loss before LRYGB was found to be associated with a decrease in the operating room time ($p=0.0084$) and an improved percentage of excess weight loss in the short term ($p=0.0267$). Complication rates were similar in both groups. Preoperative weight loss was also not shown to have a statistically significant impact on the resolution of comorbidities. Study limitations include the small sample size and the number of patients lost to follow-up.

A study by Jamal et al. (2006) compared outcomes of gastric bypass patients undergoing a mandatory 13 weeks of preoperative dietary counseling (PDC) ($n=72$) to a group of patients without this requirement ($n=252$). The PDC group had a higher incidence of obstructive sleep apnea compared to the no-preoperative dietary counseling group ($p<0.04$). The two groups had similar incidences of obesity-related comorbidities. The dropout rate prior to surgery was reported to be 50% higher in the PDC group than in the no-preoperative dietary counseling group.

($p < 0.05$). The no-preoperative dietary counseling patients had a statistically greater percentage of excess weight loss ($p < 0.0001$), lower BMI ($p < 0.015$), and lower body weight ($p < 0.01$) at one-year follow-up. Resolution of major comorbidities, complication rates, 30-day postoperative mortality, and postoperative compliance with follow-up were similar in the two groups (Jamal, et al., 2006). Limitations to the study include its lack of randomization and the relatively short-term follow-up of one year which may not have been long enough to demonstrate differences in outcomes.

A systematic review and meta-analysis by Lau et al. (2022) explored the efficacy of lifestyle interventions on BMI prior to bariatric surgery. Included in the review were four studies ($n = 345$) published between 2012 and 2019. Three outcomes were examined: the effect of pre-operative lifestyle intervention on pre-surgery BMI reduction, on post-operative BMI reduction, and on post-operative outcome. Three studies were analyzed for lifestyle intervention impact on pre-surgical BMI reduction. The lifestyle intervention group had 1.44 BMI units less in comparison to the controls (95% CI: $-2.01, -0.86$). All four studies were included in the analysis of post-operative BMI after pre-operative lifestyle interventions. All studies reported participant weight loss but there was no additional weight loss benefit in the groups who received lifestyle intervention prior to surgery. The intervention groups lost mean -0.05 BMI units (95% CI: $-1.39, 1.29$) compared to control groups. A third outcome measure, the impact of pre-surgical BMI reduction on post-operative outcome, was only examined in one included study. Thus, no quantitative analysis was performed on this measure. The evaluated randomized controlled trial for this measure compared individuals who lost more than 5% of initial body weight preoperatively to individuals who lost less than 5% of their initial body weight. At six months post-surgery, the group $\geq 5\%$ lost more weight (25.7% total body weight loss vs. 22.3%, $p < 0.0006$), but this difference disappeared at the 12 and 24 month follow ups (12 months: 28.5% vs. 28.3% weight loss, $p = 0.33$; 24 months: 28.1% vs. 27.8% weight loss, $p = 0.37$). This systematic review and meta-analysis is limited by the small number of included studies and limitations of the studies themselves, including concern for bias in three of the trials due to drop out rate. The authors conclude that it is likely that lifestyle interventions prior to surgery confer some decreased risk during the procedure, but that post-surgical weight loss not clearly influenced by pre-surgical BMI reduction.

Professional Societies/Organizations

American Society for Metabolic and Bariatric Surgery (ASMBS): A 2016 position statement issued by ASMBS cited the lack of data from RCTs supporting mandated preoperative weight loss. The ASMBS stated that patients seeking surgical treatment for clinically severe obesity should be evaluated based on their initial BMI and co-morbid conditions (Kim, et al., 2016).

American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS): According to the guidelines for bariatric surgery from AACE/TOS/ASMBS all patients seeking bariatric surgery should have a comprehensive preoperative evaluation. A brief summary of personal weight loss attempts, commercial plans, and physician-supervised programs should be reviewed and documented, along with the greatest duration of weight loss and maintenance. This information is useful in substantiating that the patient has made reasonable attempts to control weight before considering obesity surgery. The guidelines stated that preoperative weight loss should be considered for patients in whom reduced liver volume can improve the technical aspects of surgery (Mechanick, et al., 2008, updated 2013, 2020).

American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): In a 2022 joint statement the indications for metabolic and bariatric surgery, the ASMBS and IFSO state the lack of data to support the practice of mandated preoperative weight loss. A multidisciplinary team

should be utilized to assess and manage the patient's modifiable risk factors with a goal of reducing risk of perioperative complications and improving outcomes (Eisenberg, et al., 2022).

Medical Clearance Recommendations

The American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS) guidelines on support for the bariatric surgery patient stated that all patients should undergo preoperative evaluation for obesity, related co-morbidities and causes of obesity, with special attention directed to those factors that could affect a recommendation for bariatric surgery. The preoperative evaluation must include a comprehensive medical history, psychosocial history, physical examination and appropriate laboratory testing to assess surgical risk. A gastrointestinal (GI) evaluation may be indicated preoperatively in symptomatic patients (e.g., H. pylori screening in areas of high prevalence; gallbladder evaluation and upper endoscopy). Patients should be followed by their primary care physician and have age and risk appropriate cancer screening before surgery. Recommended elements of medical clearance for bariatric surgery include the following (Mechanick, et al., 2013, updated 2020):

- In patients considered for bariatric surgery, chest radiograph and standardized screening for obstructive sleep apnea (with confirmatory polysomnography if screening tests are positive) should be considered.
- Tobacco use should be avoided at all times by all patients. In particular, patients who smoke cigarettes should stop, preferably at least six weeks before bariatric surgery.
- Noninvasive cardiac testing beyond an electrocardiogram is determined on the basis of the individual risk factors and findings on history and physical examination
- All patients should undergo evaluation of their ability to incorporate nutritional and behavioral changes before and after bariatric surgery.
- All patients should undergo an appropriate nutritional evaluation, including micronutrient measurements, before any bariatric surgical procedure. In comparison with purely restrictive procedures, more extensive perioperative nutritional evaluations are required for malabsorptive procedures.
- A psychosocial-behavioral evaluation, which assesses environmental, familial, and behavioral factors, should be required for all patients before bariatric surgery. Any patient considered for bariatric surgery with a known or suspected psychiatric illness, or substance abuse, or dependence, should undergo a formal mental health evaluation before performance of the surgical procedure.

Bariatric Surgical Procedures

Bariatric surgery for morbid obesity involves reducing the size of the gastric reservoir, contributing to the establishment of an energy deficit by restricting caloric intake. The goal of bariatric surgery is to induce and maintain permanent loss of at least half of the preoperative, excess body weight. This amount of weight loss should bring the patient to a weight at which many or most weight-related comorbidities are reverted or markedly ameliorated. The NHLBI report (1998) has recognized two types of operations that have proven to be effective: restrictive procedures that limit gastric volume and malabsorptive procedures which in addition to limiting food intake also alter digestion.

Gastric Bypass

Gastric bypass procedures combine the creation of a small stomach pouch to restrict food intake and construction of a bypass of the duodenum and other segments of the small intestine to produce malabsorption. The Roux-en-Y gastric bypass (RYGB) is the most commonly performed gastric bypass procedure. RYGB has also been less frequently performed for other indications such as gastroparesis. During RYGB, a small stomach pouch is created by stapling or by vertical

banding to restrict food intake. Next, a Y-shaped section of the small intestine consisting of two limbs and a common channel is attached to the pouch to allow food to bypass the duodenum and jejunum. This procedure results in reduced calorie and nutrient absorption. The degree of intended malabsorption is determined by the length of the Roux limb or common channel and varies as follows: standard (short-limb), 40 cm; long-limb, 75 cm; and very long-limb, 150 cm.

Complications of the RYGB include anastomotic leaking and strictures, nutritional deficiencies, and the dumping syndrome. The dumping syndrome occurs when a large amount of undigested food passes rapidly from the stomach into the small intestine and is characterized by abdominal pain, nausea, vomiting and weakness.

RYGB can be performed via open and laparoscopic approaches. A systematic review of the scientific literature on open and laparoscopic surgery for morbid obesity (Gentileschi, et al., 2002) concluded that laparoscopic Roux-en-Y is as safe as open RYGB. The overall body of evidence indicates that, in general, laparoscopic RYGB has been shown to achieve significant sustained weight loss with resolution of obesity-related comorbidities (Jan, et al., 2005; DeMaria, et al., 2002; Schauer, et al., 2000; Wittgrove and Clark, 2000). Evidence suggests that weight-loss outcomes are comparable to open gastric bypass at one year. In comparative trials, RYGB has been reported to be associated with substantially greater weight loss and reduction of comorbidities following surgery (Agrawal et al., 2025). It continues to be the surgical treatment of choice for morbid obesity (Weber, et al., 2004; Lee, et al., 2004).

Gastric Banding

In this restrictive procedure, a band made of special material (e.g., silicone, polypropylene mesh, Dacron vascular graft) is placed around the stomach near its upper end, creating a small pouch and a narrow passage into the larger remainder of the stomach. Adjustable gastric banding refers to bands in which the pressure can be changed without an invasive procedure. The open approach to gastric banding is considered obsolete in practice and has largely been replaced by laparoscopic techniques.

Laparoscopic Adjustable Silicone Gastric Banding (LASGB)

LASGB is a minimally invasive gastric restrictive procedure that involves the wrapping of a saline-filled band around an area of the stomach with the goal of limiting food consumption. The adjustable band can be inflated or deflated percutaneously via an access port (reservoir) attached to the band, based on weight changes. The access port is placed in or on the rectus muscle, allowing for noninvasive band adjustment. This adjustment process helps determine the rate of weight loss and is an essential part of LASGB therapy. Appropriate adjustments, made up to six times annually, are critical for successful outcomes (Buchwald, 2005). Adjustable gastric banding devices approved for marketing in the U.S. include the Bioenterics® LAP-BAND® Adjustable Gastric Banding (LAGB®) System (ReShape Lifesciences, Inc, San Clemente, CA), and the REALIZE™ Adjustable Gastric Band (Ethicon Endo-Surgery, Inc., Cincinnati, OH).

According to patient information provided by the manufacturer of the LAP-BAND, when the band is initially placed, it is usually left empty or only slightly inflated to allow time for adjustment to the device and healing. The first band adjustment is typically done approximately four to six weeks after the initial placement. There is no set schedule for adjustments. The surgeon decides when it is appropriate to adjust the band based on weight loss, amount of food the individual can eat, exercise and amount of fluid currently in the band. Adjustments can be made in the hospital or in a doctor's office. Fluoroscopy may be used to assist in locating the access port, or to guide the needle into the port. It is also used after the band has been adjusted to evaluate the pouch size and stoma size. During each adjustment, a very small amount of saline will be added to or removed from the band. The exact amount of fluid required to make the stoma the right size is unique for each person. More than one adjustment may be needed to achieve an ideal fill that will result in gradual weight loss. If a band is too loose, this may cause a patient to feel hungry or

dissatisfied with small meals. A band that is too tight may result in dysphagia, regurgitation, or maladaptive eating.

US Food and Drug Administration (FDA)

Both the LAP-BAND (now ReShape Lap-Band Adjustable Gastric Banding System) and the REALIZE Adjustable Gastric Band received Premarket Approval (PMA) from the U.S. Food and Drug Administration (FDA), which is the regulatory pathway required for Class III medical devices. Class III devices are considered high risk and require evidence of safety and effectiveness, typically through clinical studies.

Common Indication Attributes:

- **Intended Use:** Both devices are indicated for weight reduction in adults with obesity.
- **BMI Criteria:** Initially, indicated for patients with a BMI ≥ 40 kg/m², or BMI ≥ 35 kg/m² with one or more severe comorbid conditions, or those 100 lbs. or more over their ideal weight. In 2011, the LAP-BAND indication was expanded to include patients with BMI 30–35 kg/m² who also have an existing obesity-related condition (FDA, 2011).
- **Conservative Therapy Requirement:** Both devices are indicated only for patients who have failed more conservative weight reduction alternatives, such as supervised diet, exercise, and behavior modification programs.
- **Adjustment & Customization:** The bands are adjustable post-implantation to optimize weight loss and minimize complications. Adjustments are individualized and may require multiple visits.
- **Contraindications:** Both devices share similar contraindications, including inflammatory diseases of the GI tract, severe cardiopulmonary disease, portal hypertension, and cirrhosis of the liver.

Device or Product	Identifier	Manufacturer
LAP-BAND (ReShape Lap-Band)	P000008 P140012	ReShape Lifesciences, Inc
REALIZE Adjustable Gastric Band	P070009	Ethicon Endo- Surgery, Inc.

*FDA product codes: LTI

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Literature Review

Evidence in the published, peer-reviewed scientific literature suggests that laparoscopic adjustable gastric banding (LAGB) is a safe and effective surgical treatment option for patients with morbid obesity. Although a large number of studies have reported on the effectiveness of this technique, available evidence supporting the use of adjustable gastric banding is primarily in the form of retrospective reviews and prospective case series. Numerous case series have been published, with several studies including over 500 patients each. A limited number of randomized trials have been published, with few studies comparing adjustable gastric banding with established surgical approaches, such as gastric bypass. Well-designed comparative clinical trials comparing adjustable banding with established bariatric surgical procedures are limited. BMI inclusion criteria for studies have generally been within the guidelines set forth by the NHLBI (i.e., BMI ≥ 40 or 35–39.9 with an obesity related co-morbid condition). While a number of these studies and case series report a

substantial weight loss following laparoscopic banding, the percentage of excessive weight loss (EWL) after one year appears to be less than the percentage of EWL associated with gastric bypass procedures (O'Brien, et al., 2003). Reported success rates and results have been variable across studies.

Angrisani et al. (2007) performed a prospective, randomized comparison (n= 51) of LAGB with the LAP-BAND system and LRYGB. At five-year follow-up, the LRYGB patients had significantly lower weight and BMI and a greater percentage of excess weight loss than those in the LAGB group ($p<0.001$). Weight loss failure was observed in nine of 26 LAGB patients and in one of 24 LRYGB patients ($p<0.001$). These study results suggested that LRYGB resulted in a higher percentage of weight loss compared to LAGB.

Jan et al. (2005) studied a consecutive series of patients who underwent either LRYGB (n=219) or LAGB (n=154) over a three-year period by a single surgeon. The authors reported that the LAGB group had shorter operative times, less blood loss and shorter hospital stays as compared to the LRYGB group. The incidence of minor and major complications was reported to be similar in the two groups, with the morbidity potentially greater after LRYGB and the reoperation rate greater after LAGB group. Early weight loss was greater in the bypass group; however, it was noted that the difference appeared to diminish over time.

Several early studies reported high failure and complication rates associated with the banding procedure. Reported complications include both operative complications (splenic or esophageal injury) and late complications (band slippage, gastric erosion of the band, dilatation, reservoir deflation/leak, persistent vomiting, long-term failure to lose weight and gastric reflux) (Gustavsson, et al., 2002; Victorzon and Tolonen, 2001; Holeczy, et al., 2001).

More recent studies have reported varying rates of complications, with a focus on the more commonly occurring complications of band slippage and erosion. Rates of slippage have reportedly decreased with band improvements over time and changes in surgical technique. Himpens et al. (2011) presented long-term data from a case series of 82 patients who underwent LAGB. At 12-year follow-up, 54.3% of patients were available. Band erosion occurred in 28% of patients, with 17% of patients converting to laparoscopic Roux-en-Y gastric bypass. Overall, the mean excessive weight loss (EWL) was 42.8% (range, 24%-143%) at 12 years of follow-up. A mean EWL of 48% was found for patients who still had a band in place (51.4%).

Singhal et al. (2010) performed a meta-analysis (n=19 studies) of LAGB patients to examine the correlation between the occurrence rates for band erosion and slippage. The mean rates of erosion and slippage at two years or more of follow-up were found to be 1.03% and 4.93% respectively. The results demonstrated a statistically significant correlation between erosion and slippage rates ($p=0.48$; $p=0.032$).

Agrawal et al. (2025) reported results from a pragmatic, open-label, randomized controlled trial evaluating Roux-en-Y gastric bypass versus adjustable gastric banding and sleeve gastrectomy. The By-Band-Sleeve trial found that adjustable gastric banding led to significantly less weight loss and smaller quality-of-life gains than either Roux-en-Y gastric bypass or sleeve gastrectomy. 1,346 individuals were randomly assigned to one of these bariatric procedures (n = 463 in Roux-en-Y, 468 in adjustable gastric banding, and 420 in sleeve gastrectomy). Outcomes measured included proportion of individuals achieving greater than or equal to 50% excess weight loss and quality of life. At 3-year follow-up, 68% of RYGB and 41% of SG patients achieved $\geq 50\%$ excess weight loss, compared to only 25% with band ($p < 0.05$), and mean EQ-5D-5L quality of life scores improved to 0.72 (RYGB) and 0.68 (SG) vs 0.62 with band (both comparisons $p < 0.05$). Mean percentage total weight loss at 3 years was -26.8 (SD 13.5) for the Roux-en-Y gastric bypass, -14.0 (13.5) for adjustable gastric banding, and -19.4 (13.1) for the sleeve gastrectomy

($p < 0.05$ for RYGB or SG vs band). Physical health status and metabolic improvements also favored RYGB and SG over band. SG had significantly fewer total complications than band (incidence rate ratio vs band 0.78, 98% CI 0.62–0.98; $p < 0.05$), and about 14% of band patients required reoperations (band correction/removal or conversion to another procedure) within 3 years, a higher reoperation rate than in the RYGB or SG arms. Limitations of the study include the unblinded design, the addition of the SG group partway through the trial (extending recruitment to 6 years and causing unequal recruitment periods per group), some patient crossovers between procedures, and modest loss to follow-up and incomplete secondary outcome data.

Biliopancreatic Diversion with and without Duodenal Switch

As described originally by Scopinaro, the biliopancreatic diversion (BPD) is principally a malabsorptive procedure in which the distal two-thirds of the stomach are removed. The small pouch that remains is connected directly to the final segment of the small intestine, diverting bile and pancreatic juice into the distal ileum. Increased malabsorption and greater excess weight loss (EWL) occur, but at the expense of a higher incidence of both surgical and metabolic complications. These complications include: anastomotic ulceration, diarrhea, protein caloric malnutrition, metabolic bone disease and deficiencies in the fat-soluble vitamins, vitamin B₁₂, iron and calcium.

Hess adapted the procedure to include the duodenal switch (DS). The biliopancreatic diversion with duodenal switch (BPD/DS) incorporates both malabsorptive and restrictive mechanisms to minimize complications while still producing significant therapeutic weight loss. The procedure involves vertical subtotal gastrectomy with preservation of the pylorus. The first part of the duodenum is divided and attached to the terminal ileum. Sparing the pylorus significantly reduces the incidence of dumping syndrome, obstruction and stricture. Preservation of the early part of the duodenum results in a reduction in the incidence of vitamin and iron deficiencies. The majority of surgeons who perform BPD now incorporate DS (Neligan and Williams, 2005). In some centers, BPD/DS has been proposed as the procedure of choice for a subset of patients with a BMI > 50 or the super morbidly obese. The procedure is considered technically demanding with an operative mortality of 2% and major perioperative morbidity of 10%. Postoperative EWL is reported to range between 70% and 80%.

Literature Review

Biliopancreatic Diversion (BPD) without Duodenal Switch (DS): There is limited available evidence in the published literature evaluating the safety and effectiveness of BPD without duodenal switch (DS). Evidence supporting BPD without DS is primarily in the form of case series with up to five years follow-up (Guedea, et al. 2004), nonrandomized studies comparing outcomes based on the length of the common and alimentary limbs in the procedures (Gracia, et al., 2007) and retrospective reviews (Sethi, et al., 2016). Some studies reported favorable weight loss and remission of comorbidities following surgery. This established procedure has been largely replaced by BPD with duodenal switch (BPD/DS).

Biliopancreatic Diversion (BPD) with Duodenal Switch (DS): Randomized controlled trials, comparative studies and case series support BPD with DS for the treatment of obesity when medical management has failed. The results of the studies included statistically significant improvements in BMI and co-morbid conditions (e.g., total cholesterol, low-density lipoprotein cholesterol concentration, anthropometric measures) (Søvik, et al. 2011; Søvik, et al. 2010; Topart, et al., 2011; O'Rourke, et al., 2006, Prachand, et al., 2006; Hess, et al., 2005; Parikh, et al., 2005; Rabkin, et al., 2003; Anthone, et al., 2003, Verras, et al., 2023).

Professional Societies/Organizations

Society of American Gastrointestinal and Endoscopic Surgeons (SAGES): The SAGES (2008) guideline for laparoscopic bariatric surgery stated that in BPD, the common channel should be 60–100 cm, and the alimentary limb 200–360 cm. DS diminishes the most severe complications of BPD, including dumping syndrome and peptic ulceration of the anastomosis. BPD is effective in all BMI > 35 kg/m² subgroups, with durable weight loss and control of comorbidities beyond five years. Laparoscopic BPD provides equivalent weight loss, shorter hospital stay, and fewer complications than the open approach. BPD may result in greater weight loss and resolution of comorbidities than other bariatric surgeries, but with the highest mortality rate.

Single Anastomosis Duodenal-ileal Bypass with Sleeve Gastrectomy (i.e., Loop duodenal switch)

The biliopancreatic diversion with duodenal switch (BPD/DS), while proven to be efficacious for excessive weight loss (EWL) is technically difficult to perform and comes with possibility of long-term nutritional problems. Various modifications of the DS procedure have been introduced in an attempt to simplify the procedure and decrease the associated adverse effects. The single-anastomosis duodenal switch, also called stomach intestinal pylorus sparing surgery (SIPS), or the single loop DS, is similar to the standard DS procedure, with the exception of the small intestine being transected at one point instead of two. With this operation, the majority of the fundus is removed as in a sleeve gastrectomy, but basic stomach function remains. In addition, approximately one half of the upper small intestine is bypassed, resulting in a moderate decrease in calorie absorption. Weight loss is achieved both through restriction of food consumption and malabsorption. Another modification is the single anastomosis duodeno-ileal bypass with sleeve gastrectomy (SADI-S) which is based on the BPD in which a sleeve gastrectomy is followed by an end-to-side duodeno-ileal diversion. The preservation of the pylorus allows for reconstruction in one loop, which reduces operating time and needs no mesentery opening. In theory, single-anastomosis duodenoileal bypass with sleeve gastrectomy is a simplification of the DS that may mimic the standard BPD, but is faster and easier to perform. The single anastomosis duodenal-ileal bypass with sleeve gastrectomy and one anastomosis duodenal switch (SADI-S/OADS) is another proposed modification of duodenal switch anastomosing the duodenum directly to an omega loop of ileum 200 cm proximal to the ileo-cecal valve, eliminating the need for the Roux-en-Y jejunal-ileal anastomosis. Theoretical benefits over duodenal switch (DS) include reduction of the operative risk by eliminating one anastomosis with potentially similar outcomes in weight loss and health benefits. Other similar one anastomosis duodenal switch procedures reported in the literature include: SIPS (stomach intestinal pyloric sparing surgery), single anastomosis duodenal-jejunal bypass with sleeve gastrectomy (SADJB-SG), loop duodeno-jejunal bypass with sleeve gastrectomy (LDJBSG), single anastomosis duodenal switch, distal loop duodeno-ileostomy (DIOS) and proximal duodenojejunosomy (DJOS) (Brown, et al., 2018).

Literature Review

The SADI-S technique is a simplification of the biliopancreatic diversion with duodenal switch (BPD-DS) and provides similar outcomes to those reported after the classic DS (Kallies, et al., 2020). The evidence evaluating single anastomosis duodeno-ileal bypass, SADI-S or other modifications of this procedure for morbid obesity consists primarily of case series and retrospective reviews (Spinos, et al., 2021; Yashkov, et al., 2021; Zaveri, et al., 2019; Neichoy, et al., 2018; Surve, et al., 2018; Surve, et al., 2017; Cottam, et al., 2016; Mitzman, et al., 2016; Sánchez-Pernaute, et al., 2015; Lee, et al., 2014; Sánchez-Pernaute, et al., 2012).

Spinos, et al. (2021) conducted a systematic review to evaluate the effectiveness, complications and long-term overall nutritional status of patients who have undergone single-anastomosis duodenoileal bypass with sleeve gastrectomy/one anastomosis duodenal switch (SADI-S/OADS). A total of 14 studies reporting on the weight loss and metabolic/nutrient impact of SADI-S (five retrospective cohort and nine case series) met inclusion criteria. Narrative reviews, systematic reviews and meta-analyses, case reports, basic science papers, letters to the editor,

commentaries, and published abstracts were excluded. All papers reporting on any modifications of the standard SADI-S procedure were also excluded. A total of 1086 patients were included in the analysis: 68.4 % female, average age 45.0 ± 11.4 years with preoperative BMI 51.3 ± 9.5 kg/m². Race was not reported. The studies represented patients from a wide geographical distribution including the Netherlands, Spain, Portugal, Italy, Egypt, Czech Republic, Brazil, China, and the United States. The average body mass index (BMI) following SADI-S was 32.1 ± 6.7 kg/m². Mean total body weight (TBW) loss ranged from 11.3% to 17.3% at three months, 21.5% to 41.2% at 12 months, and 25.8% to 46.3% at 24 months. Mean excess body weight (EBW) loss ranged from 21.8% to 40.2% at three months, 60.9% to 91.0% at 12 months, and 44.3% to 86.0% at 24 months. Mean excess BMI (EBMI) ranged from 9.4% to 31.1% at three months, 17.9% to 86.6% at 12 months, and 19.5% to 80.8% at 24 months. The comorbidity resolution rates were 72.6% for diabetes mellitus (DM), 77.2% for dyslipidemia, 59% for hypertension (HTN), 54.8% for obstructive sleep apnea (OSA), and 25% for gastroesophageal reflux disease (GERD). Additionally, rates of improvement of comorbid conditions were 19.1% for DM, 27.1% for dyslipidemia, 31.7% for HTN, 28% for OSA, and 25% for GERD. The most common early (< 30 days) postoperative complications after SADI-S included the need for reoperation (3.1%), bleeding (1.1%), wound infection (1.0%), anastomotic leak (0.9%), and intrabdominal collection/abscess (0.6%). Late (> 30 days) postoperative complications were the need for reoperation (5.3%) and malnutrition/dumping syndrome (1.3%). Four deaths were noted: one in early postoperative period due to an anastomotic leak, three in the late postoperative period from cardiac arrest, untreated obstructed sleep apnea, and ventricular fibrillation. Only one study reported on the preoperative nutritional status of patients. All studies reported nutritional deficiencies postoperatively, specifically calcium, vitamin D, and PTH values. Additional abnormalities were observed in the total serum protein, albumin, zinc, selenium, and copper. Author noted limitations included heterogeneity in the technical aspects of the operations and reported outcomes; included studies were either retrospective cohort studies or case series, and short-term follow ups. SADI-S offers the benefits of a combined malabsorptive and restrictive bariatric operation, with fewer postoperative complications than the traditional standard biliopancreatic diversion with duodenal switch (BPD/DS).

Yashkov et al. (2021) conducted a retrospective review to compare the results of single-anastomosis duodenoileal bypass with sleeve gastrectomy (SADI-S) and Hess-Marceau's BPD/duodenal switch (RY-DS) operations at five years in 754 patients. The group that received SADI-S (n=226) had an average age of 42 ± 11 years, 30.1% were males and presented with a preoperative BMI of 48.9 ± 9 kg/m². The RY-DS group (n=528) presented with an average age of 39 ± 9.9 years, 20.8% were males and had a preoperative BMI of 47.9 ± 7.3 kg/m². Race or ethnicity was not detailed, however surgeries were performed by a single surgeon in the Russian Federation. Data extracted included weight loss parameters including total weight loss (TWL%), excess BMI loss (EBMIL%), excess weight loss (EWL%); remission of diabetes mellitus type 2; complications; and revision rate in the SADI-S group and was compared with the results of RY-DS group. At 12 months, EWL% (77% vs 73.3%) and TWL% (39.4% vs 38.9%) were better in the SADI-S ($p < 0.01$, and $p < 0.05$ respectively) than in the RY-DS group. EWL, TBWL, and EBMIL were comparable between the two groups at 24-36 months. RY-DS group had better weight loss (TBWL, EBMIL, EWL) in the fourth and fifth year. Diabetes remission occurred in 93.4% of SADI-S at three years and 98.6% of RY-DS patients at five years. Early complication rate was 2.65% in the SADI-S and 5.1% in the RY-DS groups. Protein deficiency (0.26% vs 0.55%) and small bowel obstruction rates (0.26% vs 0.4%) were lower after SADI-S, however bile reflux occurred in 7.5% of patients (n=17) and was a main reason (6 of 9) for revisions. No bile reflux occurred in the RY-DS group. Revisions occurred in 13.3% of RY-DS patients to improve results (n=31) and to treat side effects/complications (n=39). Author noted limitations of the study included the retrospective study design and that both surgeries were performed using an open technique. SADI-S was a simpler surgery to perform, had lower early complication rates, less protein deficiency, less small

bowel obstruction rates and less revisions while providing similar weight loss and diabetes remission up to three years when compared to the RY-DS procedure.

Moon et al. (2019) conducted a retrospective review to compare the safety and effectiveness of single-anastomosis duodenal-ileal bypass (SADI-S) to the traditional double-anastomosis duodenal switch (DS) on 185 patients at a single institution in the United States. In the SADI-S group (n=111), the mean age was 41.5 ± 9.5 years, 81 were female and 30 were male with a mean preoperative body mass index (BMI) of 56.3 kg/m^2 . The double-anastomosis DS patients (n=74) had a mean age of 40.8 ± 9.4 years with 57 female and 17 male patients with a mean BMI of 54.4 kg/m^2 . The majority of patients in both groups were white at 67.6% (n=75) and 75.7% (n=56) for SADI-S and double anastomosis DS, respectively. Follow ups were conducted at one, three, six, 12 months and yearly thereafter. There was significant loss to follow up by 24 months (n=23). Weight loss was comparable for the two groups. In the SADI-S group, percentage of total weight loss (TWL) was 22.0%, 38.5%, and 44.2% at 6, 12, and 24 months, respectively. TWL in the double anastomosis group was 20.2%, 38.0%, and 48.4% for the same time frame. The majority of patients had normal levels of vitamin A and E, however vitamin D levels were low in 40-60%. At last follow up, remission of diabetes occurred in all in SADI-S (54/54) and all in double anastomosis DS (32/32). Readmission rate within 30 days was lower for double-anastomosis DS (n=4) than SADI-S (n=13). Author noted study limitations included retrospective study design with significant loss to follow up as time progressed. An additional study limitation was that the majority of subjects were white women.

Neichoy et al. (2018) conducted a retrospective analysis on 225 patients who underwent stomach intestinal pylorus-sparing (SIPS) procedure to evaluate safety and effectiveness of the procedure. SIPS is a modification of the Roux-en-Y duodenal switch (RYDS) procedure. The mean preoperative body mass index (BMI) was $52.4 \pm 9.1 \text{ kg/m}^2$ and weight was 324 ± 71.7 pounds (lbs.). The majority of the patients were female (77%) with an average age of 49.3 ± 11.3 years. Race or ethnicity was not detailed. Follow ups were conducted at one, three, six, 12 months and yearly thereafter. Mean excess weight loss was 36.7% at three months, 50.6% at six months, 60.4% at nine months, 71.3% at 12 months, 81.1% at 18 months and 88.7% at 24 months. The average BMI points lost at 24 months was 26.6 ± 7.1 . Comorbidity resolution occurred in 94.4% of those with sleep apnea, 88.8% with type 2 diabetes, 68.4% with hypertension, 78.7% with hyperlipidemia, and 86% with gastroesophageal reflux. Short term complications included leak from the duodenoileostomy (DI) (n=5, 2%), stricture at the DI (n=3, 1%), small bowel injury (n=1) and death related to surgery (n=2). Long term complications included stricture at the DI (n=1), peripheral edema (n=3), diarrhea (n=5), malnutrition (n=3), dysphagia (n=2), superior mesenteric venous thrombosis (n=1), liver abscess (n=1), and death related to surgery (n=2). Author noted study limitations included the retrospective study design, small patient population, and lack of long term follow up.

Surve et al. (2017) conducted a retrospective review to compare the outcomes of 182 patients who underwent either biliopancreatic diversion with duodenal switch (BPD-DS) (n=62) or stomach intestinal pylorus sparing surgery (SIPS) (n=120) at a single private institution. The majority of patients were female (BPD-DS 61%; SIPS 65%) with an average age of 51.7 ± 12.3 years for BPD-DS and 49.1 ± 14 years for SIPS. Race or ethnicity was not detailed. Operative time for BPD-DS group was 136.9 ± 35.5 minutes compared to 69.9 ± 15.8 minutes for SIPS. Length of hospital stay for BPD-DS was 4.1 ± 6.2 days while SIPS was $2 \pm$ days. The percent excess weight loss (%EWL) and body mass index (BMI) lost for the BPD-DS group was 46.7% and 11.2 kg/m^2 at three months, 67.3% and 16.2 kg/m^2 at six months, 79.3% and 19.2 kg/m^2 at nine months, 86.6% and 21 kg/m^2 at 12 months, 92.7% and 22.7 kg/m^2 at 18 months, and 94.9% and 23.3 kg/m^2 at 24 months. The reported %EWL and BMI for the SIPS group was 44% and 10.2 kg/m^2 at three months, 62.1% and 14.3 kg/m^2 at six months, 72.7% and 16.8 kg/m^2 at nine months, 79.3% and 18.4 kg/m^2 at 12 months, 85% and 19.8 kg/m^2 at 18 months, and 87.1% and 20.3

kg/m² at 24 months. When comparing nutritional outcomes from baseline to 24 months, there was statistical difference (improvement) for glucose, HbA1C, insulin, cholesterol, triglyceride, vitamin D, and vitamin B1 for both procedures. There was no statistical difference between the two groups for postoperative nutritional data such as vitamins D, B1, B12, serum calcium, fasting blood glucose, glycosylated hemoglobin (HbA1C), insulin, serum albumin, serum total protein, and lipid panel. Short-term complication rate for the BPD-DS group was 20.9% and included intraabdominal abscess (n=2), anastomotic leak (n=2), sepsis (n=2), postoperative bleed (n=2), mild renal failure (n=2), duodenal stump leak (n=1), peritonitis (n=1), and small bowel obstruction (n=1). Short-term complication rate for the SIPS group was 1.6% and included acute blood loss anemia (n=1) and intraabdominal hematoma (n=1). Long-term complication rate for BPD-DS group was 32.2% and included diarrhea (n=7), malnutrition (n=5), hiatal hernia (n=3), sleeve stricture (n=2), liver failure (n=1) and common channel lengthening (n=1). Long term complication rate for the SIPS was 10.8% and included diarrhea (n=1), malnutrition (n=1), hiatal hernia (n=2), sleeve stricture (n=4), constipation (n=2), retrograde filling of the afferent limb (n=2), and common channel lengthening (n=1). No complications resulted in death. Author noted study limitations included the retrospective study design, small patient population, and lack of long term follow up. The SIPS is a simplified DS procedure. The SIPS eliminates one anastomosis and compared with BPD-DS has fewer perioperative and postoperative complications, shorter operative time and length of stay, and similar nutritional results at two years.

Cottam et al. (2015) conducted a retrospective review of a matched cohort of 108 patients who underwent either a Roux-en-Y gastric bypass (GBP) (n=54) or a single anastomosis loop duodenal switch (LDS) (n=54) to compare weight loss outcomes and complication rate. The cohort was matched based on gender and body mass index (BMI) within one point. The baseline BMI was 47.6 ± 8.8 kg/m² with 38 females and 16 males in each group. Average age for the GBP group was 46.7 ± 13.6 years and 51.9 ± 13 years for the LDS group. The percent of weight lost (%WL) for the GBP group was 19.5% at three months, 27.5% at six months, 32.7% at nine months, 36.1% at 12 months, 38.3% at 15 months and 39.6% at 18 months. The %WL for the LDS group was 19.2%, 26.8%, 32.3%, 36.3%, 39.1% and 41% at three, six, nine, 12, 15 and 24 months, respectively. The mean BMI at 18 months for the LDS group was 26.8 kg/m² compared to the GBP group with 29.5 kg/m². Short term complications for the GBP group included nausea (n=9), abdominal pain (n=4), diarrhea (n=1), incisional hernia (n=1), and diaphoresis (n=1). For the LDS group, short term complications reported were nausea (n=1) and abdominal wall spasms (n=1). Long term complications for the GBP group included nausea (n=17), ulcers (n=6), stricture (n=1), adhesions causing small bowel obstruction (n=3), perforation (n=1), intra-abdominal hemorrhaging (n=1), low pre-albumin (n=2), gastroesophageal reflux (GERD) (n=2), and renal failure (n=1). Reported long term complications for the LDS group included nausea (n=4), sleeve stricture (n=1), dilated fundus requiring reoperation (n=1), GERD (n=1), diarrhea (n=1), and miscounted small bowel requiring reoperation (n=1). Author noted study limitations included the retrospective study design, small patient population, lack of vitamin and mineral level analysis, lack of comorbidity analysis and short-term follow up. The LDS group had less short and term complications with statistically similar weight loss to the GBP group.

Professional Societies/Organizations

American Society for Metabolic and Bariatric Surgery (ASMBS): The ASMBS updated their guidelines in 2020 to endorse the SADI-S, a modification of classic Roux-en-Y DS, as an appropriate metabolic bariatric surgical procedure that meets appropriate standards for safety and benefit. The recommendation was based on current clinical knowledge, expert opinion and published peer-reviewed scientific evidence. The ASMBS reached the conclusion from the current review of the available peer-reviewed literature that SADI-S provides for similar outcomes to those reported after classic DS. The society states that concerns remain about intestinal adaptation, nutritional issues, optimal limb lengths, and long-term weight loss/regain after this procedure and recommends a cautious approach to the adoption of this procedure with attention

to ASMBS-published guidelines on nutritional and metabolic support of bariatric patients (Kallies, et al, 2020).

International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): In 2020, the IFSO confirmed their position on and updated their recommendations regarding single anastomosis duodenal-ileal bypass with sleeve gastrectomy and one anastomosis duodenal switch (SADI-S/OADS) (Brown, 2020). The position statement was based on current clinical knowledge, expert opinion and published peer-reviewed scientific evidence. A systematic review of the literature, updating the previous review, was conducted to summarize the current evidence to provide the most up-to-date information to guide practice. All studies that included any data or reported experiences with single anastomoses pylorus-preserving procedures were included. All study designs (i.e., case reports, retrospective reviews), study sizes and follow-up time frames were accepted. The update included an additional 25 new case series and three case reports with follow ups up to five years in four primary case series. Thirty-four case series were included. The geographical locations included the United States, Germany, Taiwan, Chile, Spain, China, Netherlands, Portugal, Italy, and Australia. The majority of subjects for each study were women (50.6%–85%). In total, there were 4,540 subjects but this included overlapping patients. For follow ups that occurred between 12–24 months, mean total body weight loss (TBWL) ranged from 23.6%–39.0% at 12 months, 39.6%–42.9% at 18 months, and 22.8%–47.8% at two years. Mean excess weight loss (EWL) ranged from 62.4%–102% at 12 months. Three studies reported follow-ups at five years for TBWL ranging from 22% to 38% with the follow-up rate from 78%–100%. Changes in type 2 diabetes diagnosis and treatment were reported in 28 case series. There was a significant improvement in both HBA1c and requirement for hypoglycemic agents. Early complications were uncommon and included anastomotic leaks, bleeding and nausea. Longer term complications were gastroesophageal reflux disease (GERD), bile reflux, flatulence, dumping syndrome, and nutritional issues. Post-operative nutritional issues included malnutrition, hypoalbuminemia, vitamin D deficiency, hypocalcemia, hyperparathyroidism, and iron deficiency. The IFSO states there is current evidence now out to five years in four primary case series that confirm that SADI-S/OADS can help a person with morbid obesity achieve and maintain significant weight loss with an improvement in metabolic health. Based on the existing data, the IFSO recommended the following:

- “SADI-S/OADS offers substantial weight loss that is maintained into the medium term.
- SADI-S/OADS provides an improvement in metabolic health that is maintained into the medium term.
- Nutritional deficiencies are emerging as long-term safety concerns for the SADI-S/OADS procedure and patients undergoing this procedure need to be aware of this, and counseled to stay in long-term multidisciplinary care.
- Surgeons performing the SADI-S/OADS, as well as other bariatric/metabolic procedures, are encouraged to participate in a national or international registry so that data may be more effectively identified.
- IFSO supports the SADI-S/OADS as a recognized bariatric/metabolic procedure, but highly encourages randomized controlled trials in the near future.”

Sleeve Gastrectomy (SG)

SG, also known as partial or vertical gastrectomy, is a restrictive procedure that is now being proposed as a definitive procedure for morbid obesity or as the first procedure in a staged surgical approach for those with very high BMI (BMI (>60 kg/m²). Weight loss following SG is thought to reduce the risk of a subsequent, more extensive procedure, such as biliopancreatic diversion, in very high-risk patients. It has been suggested that the hormone ghrelin may play a role in the weight loss associated with SG. Although resection of the fundus may lower ghrelin levels by reducing the volume of ghrelin-producing cells, low levels of this hormone after surgery may be due to the paracrine effect of gastrointestinal hormones such as glucagon-like peptide-1 (GLP-1), GLP, ghrelin, and other hormones.

SG can be an open or laparoscopic procedure and involves the resection of the greater curvature of the stomach with the remainder resembling a tube or sleeve. The resulting decrease in stomach size inhibits distention of the stomach so that early satiety is achieved. Preservation of the pyloric sphincter prevents the dumping syndrome. Other advantages of this procedure include the lack of intestinal anastomosis and no implantation of a foreign body. Major complications associated with SG include staple-line leak and postoperative hemorrhage.

Literature Review

The percentage of excessive weight loss (%EWL) for laparoscopic sleeve gastrectomy (LSG) has been reported to vary from 33%–90% and to be sustained for up to three years. The rate of complications has ranged from 0%–29% (average 11.2%), and the mortality rate from 0–3.3%. Rates of resolution or improvement of comorbidities after SG have been found to range from 45%–95.3%. Safety and effectiveness are comparable to other established bariatric procedures, with %EWL at three years, comorbidity resolution, complication and mortality rates for RYGB being 66%, 65–84%, 9.5%, 0.56%, respectively, and for LAGB, 55%, 41–59%, 6.5%, 0.47%, respectively (Shi, et al., 2010). A number of studies including systematic reviews (Brethauer, et al., 2009), randomized controlled trials and multiple case series (Hedberg, et al., 2026; Himpens, et al., 2010; Peterli, et al., 2009; Strain, et al., 2009; Arias, et al., 2009; Fuks, et al., 2009; Karamanakos, et al., 2008; Felberbauer, et al., 2008; Nocca, et al., 2008; Vidal, et al., 2007; Hamoui, et al., 2006; Silecchia, et al., 2006; Himpens, et al., 2006; Cottam, et al., 2006) support the safety and efficacy of SG.

Kraljevic et al. (2025) published long-term findings from the SM-BOSS multicenter randomized clinical trial comparing laparoscopic sleeve gastrectomy with Roux-en-Y gastric bypass. 217 adults with severe obesity were randomized 1:1 to laparoscopic sleeve gastrectomy (SG) (n = 107) or Roux-en-Y gastric bypass (RYGB) (n = 110). Inclusion criteria were BMI ≥ 40 or ≥ 35 with comorbidity, age 18–65; exclusions were prior bariatric surgery, severe GERD, large hiatal hernia, and presence of inflammatory bowel disease. Data was collected over a 10-year follow up period. The primary outcome was percentage excess BMI loss at 10 years. Secondary outcomes included total weight loss at 10 years and comorbid disease changes. Intention-to-treat analysis showed similar long-term weight outcomes: 60.6% (25.9) after SG and 65.2% (26.0) after RYGB ($p = .29$), whereas per-protocol results (excluding conversions) revealed RYGB achieving moderately greater excess weight loss (65.9% in RYGB vs 56.1% in SG, $p = .048$) with no significant difference in total weight loss (25.5% in SG vs 27.7% in RYGB). Both SG and RYGB significantly reduced BMI and attained high rates of metabolic disease remission (e.g., type 2 diabetes, dyslipidemia, hypertension) without significant between-group differences, and both procedures improved quality of life with similar complication profiles. However, individuals who underwent SG had a significantly higher late conversion rate to another procedure for inadequate weight loss or reflux (~30% vs ~5% after RYGB, $p < 0.001$) and more new-onset GERD ($p < 0.05$), whereas RYGB provided better long-term reflux control and slightly more durable weight outcomes. Limitations included attrition and crossover at 10-year follow-up (reducing statistical power), and an unblinded study design. In summary, SG produced substantial, sustained weight loss and comparable improvements in obesity-related comorbidities to RYGB at a decade post-surgery.

Shi et al. (2011) performed a systematic review of the literature (n=15 studies; 940 patients) analyzing outcomes of LSG compared to benchmark clinical data from LAGB and LRYGB. The %EWL for LSG varied from 33% to 90% and appeared to be sustained up to three years. The mortality rate was 0%–3.3% and major complications ranged from 0%–29% (average 12.1%). It was summarized that early, non-randomized data suggest that LSG is efficacious in the surgical management of morbid obesity.

Brethauer et al. (2009) performed a systematic review (n=36 studies) of the evidence on SG. Studies included a single nonrandomized matched cohort analysis, RCTs (n=2 studies) and uncontrolled case series (n=33 studies). The mean BMI in all 36 studies was 51.2 kg/m². The mean baseline BMI was 46.9 kg/m² for the high-risk patients (range 49.1–69.0) and 60.4 kg/m² for the primary SG patients (range 37.2–54.5). The follow-up period ranged from 3–60 months. The mean %EWL after SG reported in 24 studies was 33–85%, with an overall mean %EWL of 55.4%. The mean postoperative BMI was reported in 26 studies and decreased from a baseline mean of 51.2 kg/m² to 37.1 kg/m² postoperatively. Improvement or remission of type 2 diabetes was found in more than 70% of patients. Significant improvements were also seen in hypertension and hyperlipidemia, as well as in sleep apnea and joint pain. The major postoperative complication rate ranged from 0%–23.8%. The most frequent complications seen were leaks (2.2%) and bleeding requiring re-operation or transfusion (1.2%). Study data for high-risk staged and primary subgroups are listed in the following table:

Variable	High-risk patients/ Staged approach	Primary Procedure
Mean preoperative BMI	60.0	46.6
Mean postoperative BMI	44.9	32.2
Follow-up range	4-60 months	3-36 months
Mean %EWL	46.9%	60.4%
Mean Complication rate	9.4%	6.2%
Mortality rate	0.24%	0.17%

The authors summarized that although the long-term data are limited, based on the volume of available evidence, LSG is an effective weight loss procedure that can be performed safely as a first stage or primary procedure (Brethauer, et al., 2009).

The growing volume of studies in the published peer-reviewed medical literature suggests that the safety and effectiveness rates for SG are comparable to those for other accepted bariatric procedures such as RYGB and LAGB. There is sufficient evidence to support the use of SG as a stand-alone procedure or as the first of a two-stage procedure. Long-term data are needed to further define the role of SG for the treatment of morbid obesity.

Professional Societies/Organizations

American Society for Metabolic and Bariatric Surgery (ASMBS): The ASMBS updated position on sleeve gastrectomy (2017) stated that there is substantial, published long-term outcome data, including comparative studies, that confirm that sleeve gastrectomy (SG) provides significant and durable weight loss, improvements in medical co-morbidities, improved quality of life, and low complication and mortality rates for obesity treatment. SG and Roux-en-Y (RYGB) appear similar in terms of initial early weight loss and improvement of most weight-related co-morbid conditions. The effect of SG on gastroesophageal reflux disease (GERD), however, is less clear, because GERD improvement is less predictable and GERD may worsen or develop de novo. ASMBS recognized SG as an acceptable option for a primary bariatric procedure or as a first-stage procedure in high-risk patients as part of a planned, staged approach. Long-term weight regain can occur after SG and may require one or more of a variety of reinterventions. The guideline also discussed SG as a weight loss option for adolescents and stated that weight loss surgery (WLS) is becoming more accepted for this age group. Based on safety and efficacy data, there is a trend toward SG as the procedure of choice for adolescents, although both RYGB and SG are routinely performed in teen WLS programs. However, ASMBS noted that since there is almost no literature on the outcomes of adults who underwent WLS as teens, this area merits further study.

American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS): According to the AACE/TOS/ASMBS guidelines, a first-stage SG may be performed in high-risk patients to induce an initial weight loss (25 to 45 kg), with the possibility of then performing a second-stage RYGB or BPD/DS after the patient's operative risk has improved (Mechanick, et al., 2008). The 2013 update to these guidelines states that the LSG has become widely accepted as a primary bariatric operation and is no longer considered investigational (Mechanick, et al., 2013, updated 2020).

Society of American Gastrointestinal and Endoscopic Surgeons (SAGES): The 2008 SAGES guideline for laparoscopic bariatric surgery stated that SG is validated as providing effective weight loss and resolution of comorbidities to 3–5 years.

Endoscopic sleeve gastroplasty (ESG): Endoscopic sleeve gastroplasty is a procedure in which the volume of the stomach is reduced by approximately 70% through plication of the greater curvature of the stomach using an endoscopic suturing device (OverStitch, Apollo Endosurgery, Austin, TX). Suturing commences at the junction of the gastric body and antrum, progressing proximally towards the fundus, which is generally partially reduced while maintaining a small pouch to facilitate fundal accommodation. The process creates a tube of the gastric body, thereby altering feelings of fullness and satisfaction.

Literature Review

Abu Dayyeh et al. (2022) published the Multicenter ESG Randomized Interventional Trial (MERIT), a prospective, randomized trial, that compared ESG with lifestyle modifications (ESG group) with lifestyle modifications alone (control group) and was limited to patients with class I or class II obesity. At 52 weeks, primary and secondary outcome results were as follows:

- mean percentage of excess weight loss (EWL) was 49.2% for the ESG group and 3.2% for the control group ($p < 0.0001$)
- Mean percentage of total bodyweight loss was 13.6% for the ESG group and 0.8% for the control group ($p < 0.0001$)
- 25% or more of EWL was obtained by 77% of ESG group (59/77) and 12% of the control group (13/110) ($p < 0.0001$)
- improvement in one or more metabolic comorbidities occurred in 41 out of 51 individuals in ESG group and 28 out of 62 individuals in control group

Bhandari, et al. (2023) conducted a prospective cohort study of 612 individuals. At four years, the mean percentage total body weight loss was 18.19%; excess weight loss (% EWL) was 49.30%; with 90% of participants-maintaining a percentage of total weight loss of $\geq 5\%$ and 70% of patients maintaining an EWL.

Sharaiha et al. (2021) conducted a prospective cohort study of 216 patients. At five years, mean % total body weight loss (TBWL) was 15.9% ($p < 0.001$), 90% with 5% TBWL and 61% with 10% TBWL. At five years, mean excess body weight loss (% EWL) was 45.3% ($p < 0.001$); 74% of patients maintained 25% EWL.

Neto et al. (2020) evaluated 233 patients who underwent ESG and used class I and class II obesity as the primary indication. The authors noted that the percentage of excess BMI loss was significantly greater among patients with class I obesity than among those with class II obesity at 6 (51.1% versus 43.7%) and 12 months (60.2% versus 49.2%), supporting an indication for ESG in patients with a BMI of 30–35 kg/m².

Vargas et al. (2025) performed a systematic review evaluating outcomes after endoscopic sleeve gastroplasty. The review included a total of 38 studies (including prospective cohorts,

retrospective studies, case series, and one multicenter randomized trial) with populations of adults undergoing endoscopic sleeve gastropasty. Studies of pediatric populations were excluded. Across included studies, total sample sizes varied widely (ranging from small case series of approximately 21–35 participants to large cohorts exceeding 1000 patients), and most studies were single-arm without consistent comparator groups. Where comparators were included, ESG was evaluated against non-surgical or surgical alternatives. Outcomes assessed included total body weight loss (TBWL), excess weight loss (EWL), BMI reduction, and procedure-related complications, with follow-up durations ranging from short-term (3–6 months) to long-term (up to 5 years). Reported outcomes demonstrated consistent and clinically meaningful weight loss, with TBWL typically ranging from approximately 13% to 20% at 12 months and 15.9%–20.9% at 2–5 years, and EWL ranging from 13% to 79%. Several studies reported improvements in comorbidities, including remission or significant improvement in type 2 diabetes. The overall safety profile was favorable, with low rates of serious adverse events related to surgical bariatric procedures. Key limitations of the evidence base include substantial heterogeneity in study design, populations, and ESG techniques; predominance of retrospective and non-randomized studies; lack of standardized comparators; and limited generalizability due to variability in methodology and outcome reporting.

Marincola et al. (2021) conducted a systematic review and meta-analysis comparing endoscopic sleeve gastropasty (ESG) and laparoscopic sleeve gastrectomy (LSG). The authors concluded the analysis showed a moderate superiority of LSG versus ESG. No difference in terms of safety was shown between the two groups. ESG is a less-invasive, repeatable, and reversible.

Hedjoudje et al. (2020) conducted a systematic review and meta-analysis to evaluate the safety and efficacy of endoscopic sleeve gastropasty (ESG) for the treatment of obesity. Eight studies (n=1772) met inclusion criteria including three retrospective studies and five prospective case series. The patient's mean age was 38.3 ± 10.9 years, 62.6% were male with baseline body mass index (BMI) of 33.3 ± 4.5 kg/m² and 43.0 ± 8.9 kg/m². The pooled mean total body weight loss (TBWL) was 8.8% ($p \leq 0.01$) at one month, 11.2% ($p \leq 0.01$) at three months, 15.1% ($p < 0.01$) at six months, 16.1% ($p = 0.08$) at nine months, 16.5% ($p < 0.01$) at 12 months, and 17.1% ($p = 0.03$) at 18–24 months. The pooled decrease in BMI was 3.0 kg/m² ($p < 0.01$) at one month, 3.9 kg/m² ($p < 0.01$) at three months, 5.6 kg/m² ($p < 0.01$) at six months, 5.7 kg/m² ($p = 0.12$) at nine months, 6.1 kg/m² ($p < 0.01$) at 12 months, and 6.5 kg/m² ($p < 0.01$) at 18–24 months. The pooled relative excess weight loss (EWL) was 32.4% ($p < 0.01$) at one month, 47.1% ($p < 0.01$) at three months, 57.7% ($p = 0.01$) at six months, 66.2% ($p = 0.07$) at nine months, 61.8% ($p < 0.01$) at 12 months, and 66.9% ($p = 0.04$) at 18–24 months. The pooled estimate of post-ESG severe adverse event rate was 2.2%. No deaths were reported. The pooled rates of reported severe adverse events included 1.08% (n=18) pain or nausea requiring hospitalization, 0.56% (n=9) upper gastrointestinal bleeding, 0.48% (n=8) perigastric leak or collection, 0.06% (n=1) pulmonary embolism, and 0.06% (n=1) pneumoperitoneum. Author-noted limitations of the meta-analyses included the high degree of heterogeneity among included studies, risk of bias in non-randomized studies, short term follow-up, lack of evaluation of comorbidities, lack of reporting of mild adverse events, and different methods used among studies to report the % EWL.

Additional studies investigating the OverStitch Endoscopic Suturing System are primarily in the form of retrospective reviews and prospective case series with small patient populations (n=4–77) and short-term follow-ups (6–20 months) (Kumar, et al., 2018; Abu Dayyeh, et al., 2017; Pauli, et al., 2013).

US Food and Drug Administration

The OverStitch Endoscopic Suture System is classified as a Class II medical device. Class II devices are considered to present moderate to high risk to the patient and/or user and typically require special controls in addition to general controls to provide reasonable assurance of safety and effectiveness. The OverStitch Endoscopic Suture System is intended for the endoscopic

placement of suture(s) and approximation of soft tissue (FDA, 2023; FDA, 2008). The device enables physicians to perform various types of tissue apposition within the gastrointestinal tract and peritoneal cavity.

Device or Product	Identifier	Manufacturer
OverStitch Endoscopic Suture System	K081853	Apollo Endosurgery, Inc.
OverStitch NXT Endoscopic Suturing System	K231553	Apollo Endosurgery, Inc.

*FDA product codes: OCW, HCE

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Professional Societies/Organizations

The International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO) Bariatric Endoscopy Committee, conducted a systematic review and meta-analysis in 2024 and endorses endoscopic sleeve gastropasty (ESG) as an effective and safe treatment for obesity, including class 1 and 2 (Abu Dayyeh, et al., 2024).

Vertical Banded Gastroplasty (VBG)

This restrictive procedure uses both a band and staples to create a small stomach pouch. The pouch limits the amount of food that can be eaten at one time and slows passage of the food into the remainder of the stomach and gastrointestinal tract. VBG may be performed using an open or laparoscopic approach. Complications of VBG include esophageal reflux, leaking or rupture along the staple line, stretching of the stomach pouch from overeating.

Although reoperation rates have been reported to be higher for VBG, the available evidence in the form of RCTs, nonrandomized comparative trials, and case series (Miller, et al., 2007; Nocca, et al., 2007; Olbers, et al., 2005; Lee, et al., 2004; Morino, et al., 2003) report that substantial weight loss can be achieved with this restrictive procedure. VBG has been largely replaced by adjustable silicone gastric banding however and is now rarely performed (Centers for Medicare and Medicaid Services [CMS], 2006).

Other Bariatric Surgical Procedures

Fobi-Pouch

The Fobi-Pouch, limiting proximal gastric pouch, has been proposed by one investigator as an alternative to traditional bariatric surgery. The procedure involves a small (less than 25 ml) vertical banded pouch, a Silastic® ring around the stomach creating a stoma, and a gastroenterostomy to a Roux-en-Y limb. Published evidence supporting the use of this procedure is in the form of one descriptive article (Fobi and Lee, 1998) and one case series (Fobi, et al., 2002; n=50). Current evidence available in the published, peer-reviewed scientific literature indicates that the safety and efficacy of this procedure have not been established.

Gastric Pacing/Gastric Electrical Stimulation (GES)

GES is being investigated as a treatment for morbidly obese patients. It is thought that GES may cause increased satiety resulting in decreased food intake and weight loss. The exact mechanism by which gastric pacing impacts eating and behavior is unclear. There is currently insufficient

evidence in the literature to support the use of GES for the treatment of obesity. Please refer to the Gastric Pacing/Gastric Electrical Stimulation (GES) Coverage Policy for additional information.

Gastroplasty

Gastroplasty, also referred to as stomach stapling, is the prototypical restrictive procedure. A simple gastroplasty involves the stapling of the upper portion of the stomach horizontally. A small opening is left for food to pass through to the lower portion. The outlet of the pouch is restricted by a band, which slows emptying, allowing the person to feel full after only a few bites of food. It has been reported in the literature that those who have undergone this procedure seldom experience any satisfaction from eating, and tend to eat more, causing vomiting and tearing of the staple line. The available literature also reports that horizontal stapling alone has led to poor long-term weight loss. Because many simple gastroplasty patients have eventually required some type of revision operation in order to achieve successful weight loss, this procedure has largely been abandoned.

Intestinal/Jejunioleal Bypass

In a jejunioleal or intestinal bypass the proximal jejunum is joined to the distal ileum, bypassing a large segment of the small bowel. Various technical modifications of the jejunioleal anastomosis have been developed, all bypassing extensive length of small intestine and leading to inevitable malabsorption of protein, carbohydrate, lipids, and vitamins. However, unabsorbed fatty acids entering the colon has caused significant diarrhea in patients who have undergone this procedure. Other long-term complications have been observed in jejunioleal bypass patients, the most serious of which is irreversible hepatic cirrhosis (Morris, et al., 2017; Collins, et al., 2007). Because of these complications, jejunioleal bypass has fallen out of favor and is no longer one of the more commonly performed bariatric procedures.

Intragastric Balloon (IGB)

Treatment with the IGB has been proposed as a temporary aid for obese patients who have had unsatisfactory results in their medical management of obesity and for super-obese patients with higher surgical risk. The IGB technique allows the reduction of the gastric reservoir capacity, causing a premature sensation of satiety, facilitating the consumption of smaller amounts of food (Fernandes, et al., 2007). The balloon is typically removed within six months of insertion. Adverse effects associated with the intragastric balloon include gastric erosion, reflux, and obstruction. FDA approved balloons include: Orbera™ Intragastric Balloon System (Apollo Endosurgery Inc, Austin, TX, United States), the ReShape® Integrated Dual Balloon System (ReShape Medical, Inc., San Clemente, CA, United States, acquired by Apollo Endosurgery in 2018), Obalon (Obalon® Therapeutics, Inc., acquired by ReShape Lifesciences™, San Clemente, CA in July 2021), Spatz3 Adjustable Balloon System (Spatz Fgia, Inc. Ft. Lauderdale, FL), Allurion™ intragastric balloon system (Allurion Technologies, Inc.), and the Transpyloric Shuttle/TransPyloric Shuttle Delivery Device (TPS) (BAROnove, Inc. San Carols, CA).

US Food and Drug Administration

Most intragastric balloon systems and similar endoscopic weight loss devices are regulated as Class III medical devices and require Premarket Approval (PMA) from the U.S. Food and Drug Administration (FDA) due to their intended use for weight loss and the associated risks. PMA is the most stringent regulatory pathway, requiring evidence of safety and effectiveness, typically through clinical studies. Some accessory devices or components may be cleared via the 510(k) pathway, but the primary balloon systems themselves are PMA devices.

Common Indication Attributes:

FDA-approved intragastric balloon systems are indicated as adjuncts to weight reduction in adults with obesity (typically BMI 30–40 kg/m²) who have failed more conservative weight loss alternatives such as supervised diet, exercise, and behavior modification (FDA, 2019; FDA, 2015).

These devices are intended for temporary use (usually up to six months) and must be used in conjunction with a structured diet and behavior modification program. Device-specific instructions may include requirements for pre-placement screening, concurrent medication (e.g., proton pump inhibitors), and removal protocols. Indications may vary slightly by device, but all emphasize adjunctive use, temporary placement, and the need for ongoing medical supervision.

Safety Considerations:

The FDA has issued multiple safety communications regarding serious adverse events—including spontaneous hyperinflation, acute pancreatitis, and deaths—associated with liquid-filled intragastric balloons (e.g., Orbera, ReShape). Updated labeling now includes warnings about these risks. As of January 2019, the ReShape Balloon is no longer marketed in the U.S. (FDA, 2020).

Device or Product	Identifier	Manufacturer
Orbera™ Intragastric Balloon System	P140008	Boston Scientific Corporation
Spatz3 Adjustable Balloon System	P190012	Spatz Fgia, Inc.
Allurion™ Gastric Balloon System	P250023	Allurion Technologies, Inc.
Transpyloric Shuttle (TPS)	P180024	Baronova, Inc.

*FDA product codes: LTI

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Literature Review

Orbera: Ahmed and Ezzat (2018) conducted a randomized controlled trial to evaluate the effect of obesity on quality of life, and the influence of weight loss after insertion of BioEnteric balloon (Obera) (n=40 females) compared to Atkins diet (n=40 females). The weight lost in the first six months was highest in the treatment group in which 19 subjects (47.5%) lost 33 kg (p=0.00001), compared to the control group in whom 10 patients (25%) lost 17 kg of their body weight (p=0.00010). Patients in the both groups reported better quality of life following weight loss. Limitations of the study include the small, all female patient population, short-term follow-up and a shortage of the Atkins formula and foods in the area.

Fuller et al. (2013) conducted a randomized controlled trial (n=66) to evaluate the safety and efficacy of the Orbera IGB (n=31) compared to control (n=35) in obese individuals with metabolic syndrome. Eligible subjects were adults age 18–60 years with a BMI of 30–40 kg/m² for at least two years and had failed supervised weight reduction programs. Exclusion criteria included conditions of the gastrointestinal tract, prior gastric surgery or insertion of an IGB, hepatic or renal insufficiency, or psychiatric disorder. The primary outcome was percentage of weight loss at six months. Secondary outcomes included weight loss at 12 months and remission of metabolic syndrome. At 12 months, there was a significantly greater weight loss in the IGB group versus the control group (p=0.007). No significant difference in percentage of metabolic syndrome remission was found. Adverse events related to the gastrointestinal tract were common in the IGB group but predominantly resolved within two weeks.

Professional Societies/Organizations: The American Society for Gastrointestinal Endoscopy's (ASGE) Bariatric Endoscopy Task Force, a subcommittee of ASGE, conducted a systematic review and meta-analysis of the literature to evaluate endoscopic technologies for the treatment of obesity. The review included a meta-analysis of the available data on the Orbera. A total of 82

studies met inclusion criteria. Studies were primarily in the form of case series and retrospective reviews. Seven randomized controlled trials were also included. Based on a meta-analysis of 17 studies (n=1683 patients), the percentage of excess weight loss (%EWL) with the Orbera at 12 months was 25.44%. Three randomized, controlled trials compared %EWL in patients who received the Orbera (n=131) vs. control group (n=95). The mean difference in %EWL in patients who received the Orbera was significantly greater than controls at 26.9% ($p \leq 0.001$). The pooled percentage of total body weight loss (%TBWL) after Orbera implantation was 12.3%, 13.16%, and 11.27% at 3, 6, and 12 months, respectively. Subgroup analysis showed Orbera performed as well in higher BMI groups. The rates of adverse events pooled from 68 studies included 33.7% pain, 29% nausea, 18.3% gastrointestinal reflux disease (GERD), 12% erosion and 7.5% early removal. Serious side effects included 1.4% incidence of migration and 0.1% gastric perforation. Fifty percent (4/8) of gastric perforations occurred in patients who had undergone previous gastric surgeries. Four deaths were reported and were related to gastric perforation or an aspiration event. ASGE concluded that Orbera met the thresholds as a primary or bridge procedure with a mean %EWL of 25% at one year. ASGE noted that these recommendations should not be taken to imply that these devices could be used on their own without appropriate screening, dietary, and lifestyle intervention support, nor should they be used without consideration of surgical therapy. Author-noted limitations of the meta-analyses included the high degree of heterogeneity among included studies, risk of bias in non-randomized studies, and different methods used among studies to report the %EWL (Metropolitan Life Tables vs. BMI 25 method) (ASGE Bariatric Endoscopy Task Force, et al., 2015).

Obalon: Studies investigating the safety and efficacy of the Obalon gastric balloon in children, adolescents and adults are primarily in the form of case series with small patient populations (n=10–17) (De Peppo, et al., 2017; Nobili, et al., 2015; Mion, et al., 2013).

Sullivan et al. (2018) conducted a randomized controlled trial to assess the safety and efficacy of the Obalon balloon. Patients were randomized to treatment with the balloon plus lifestyle therapy (n=198) or to the control group using sham plus lifestyle therapy (n=198). Adults aged 22–64 years who were weight stable for 12 months, with a BMI 30–40 kg/m² who made at least one attempt to lose weight through a medically or nonmedically supervised weight loss program without success were eligible for inclusion. Primary outcome measures were the difference in mean percentage of total body weight loss (%TBWL). Secondary outcomes included weight loss maintenance from weeks 24–48 and changes in cardiometabolic risk factors from baseline to week 24. All balloons were removed at week 24. A total of 160 treatment group patients continued study participation through week 48 after unblinding, and 128 patients in the control group received the balloon and completed study testing through week 48. The %TWL and total weight loss was significantly more in the treatment group ($p=0.0085$; $p<0.001$, respectively). Body mass index changes in the treatment and control groups were 2.5 ± 1.8 and 1.3 ± 1.8 kg/m² ($p<0.0001$), respectively. The responder rate in the treatment group was 66.7% ($p<0.0001$). Weight loss maintenance in all patients at 48 weeks was 88.5%. Weight loss for the control patients who crossed over was $3.6 \pm 4.4\%$ at week 24 and $7.0 \pm 6.2\%$ at week 48 (n=128). Systolic blood pressure ($p=0.020$), plasma total cholesterol concentration ($p=0.0214$), plasma triglyceride concentration ($p=0.0049$), and plasma glucose concentration ($p=0.008$) were significantly improved in the treatment groups compared to controls. Adverse events in the balloon group included numerous gastrointestinal events (e.g., nausea, vomiting, dyspepsia, abdominal distention, diarrhea). There were 43 bleeding events. One balloon deflation occurred. The balloons used in this study used a different gas formulation than earlier balloons. The authors noted several limitations of the study including: the study was not powered to determine effects on metabolic outcomes; over 85% of participants in this study were women; short-term follow-up of six months; and the study did not investigate the mechanism of action of weight loss (weight loss increased after the third balloon placement).

Spatz3 Adjustable Balloon System: Xia et al. (2025) published a systematic review and meta-analysis of available literature addressing adjustable intragastric balloons (aIGB, Spatz3). The review included 12 studies (2 randomized trials with lifestyle modification control arms and 10 observational studies) encompassing a total of 4,981 individuals. Inclusion criteria required studies to involve adult individuals with the Spatz3 aIGB and sample sizes > 80, while animal studies, non-English publications, and overlapping data (e.g., duplicate patient cohorts) were excluded. Outcomes assessed included total body weight loss (TWL) as the primary efficacy endpoint, data on post-implantation balloon volume adjustments (upward or downward), and safety endpoints such as intolerance leading to early removal, gastric ulcers, balloon migration, balloon leaks, esophagitis, and other serious complications. The aIGB therapy achieved a pooled mean TWL of 16.4% (95% confidence interval 0.153–0.175) after an average balloon implantation duration of 9.9 months ($p < 0.001$). 34.2% of individuals chose upward balloon volume adjustments during treatment, resulting in an additional mean weight loss of 6.3 kg. 9.2% required downward volume adjustments to manage intolerance, with 90.8% of those individuals experiencing relief of intolerance symptoms ($p = 0.045$). For safety outcomes, intolerance requiring early balloon removal within the first 3 months occurred in 5.7% of individuals, and gastric ulcers were reported in 1.1%. Other infrequent adverse events included balloon migration causing intestinal obstruction (reported in 0.04–1.8% of cases), balloon deflation or leakage (0.5–7.3% incidence across different studies), and esophagitis (14.0% in one large series). No treatment-related mortality was observed among the individuals in the included studies; however, nine isolated serious complications were described in case reports (e.g., gastric perforation, small bowel obstruction, pancreatitis, or balloon hyperinflation) including one fatal gastric wall tear in a 45-year-old individual. Limitations of this evidence base include the reliance on a limited number of RCTs (most data were observational), substantial heterogeneity across study results, short follow-up duration (typically ≤ 12 months of balloon therapy), and incomplete information on certain patient factors (e.g., adherence and medical history), which together limit the certainty and generalizability of the findings.

Allurion Gastric Ballon System: Silva et al. (2024) conducted a systematic review and meta-analysis to assess the safety and efficacy of swallowable liquid-filled intragastric balloons (Allurion™). The review included 11 cohort studies with a total of 2,107 participants. No randomized controlled trials or parallel comparator arms were included, and all analyses were based on pre- versus post-intervention comparisons within a single-arm design. Eligible studies enrolled adults aged ≥ 18 years with obesity, while exclusion criteria included non-obesity populations and small case series (< 10 individuals), with additional study-level exclusions for non-interventional designs or abstracts without full data. The intervention consisted of the Allurion™ swallowable intragastric balloon, typically maintained for approximately 16 weeks, without an external comparator. Outcomes assessed included body mass index (BMI), total weight loss (TWL), excess weight loss (EWL), and metabolic parameters (e.g., glucose, lipids), with follow-up durations generally limited to approximately 4 months corresponding to the device treatment period. Pooled analyses demonstrated statistically significant reductions in BMI (mean difference -4.75 kg/m^2 ; 95% CI: -5.02 ; -4.47), %TWL (mean -12.47% ; 95% CI: -13.77 ; -11.17), and %EWL (average -48.04% ; 95% CI: -50.61 ; -45.48), along with improvements in selected metabolic measures (e.g., glucose reduction reported as statistically significant), although HDL changes were not statistically significant. Adverse events were reported as generally mild and transient. 4.76% of individuals required therapeutic intervention for adverse events. There were 4 reported cases of hyperinflation. The overall quality of evidence was graded as low to very low due to study design limitations. Key limitations of this systematic review include the absence of randomized or controlled comparative data, reliance on observational cohort designs with inherent risk of bias and confounding, heterogeneity across studies, small sample sizes, short-term follow-up limited to the treatment duration, and potential publication bias, all of which limit the strength and generalizability of conclusions regarding long-term efficacy and comparative effectiveness. The authors cautioned, “the limitation of long-term data is a significant factor in the literature on

this endoscopic therapy, hindering the validation of its efficacy for both weight loss and long-term metabolic benefits”.

TransPyloric Shuttle (TPS): The evidence includes one single-center feasibility study that investigated the safety and efficacy of TPS (n=20) (Marinos, et al., 2014).

Multiple Intra gastric Balloons: The evidence evaluating the safety and efficacy of the IGB includes technology assessments, meta-analyses, RCTs and case series, primarily with relatively small sample sizes.

Zheng et al. (2015) performed a systemic review and meta-analysis of the evidence (n=11 RCTs) for the safety and efficacy of IGBs for the treatment of obesity. All studies incorporated conservative therapy with the IGB treatment. Sample sizes ranged from 22–128 patients, and mean baseline BMIs ranged from 35.0–50.4 kg/m². IGBs were compared to behavioral modification, pharmacotherapy, and observation without treatment. Results were calculated with weighted mean differences which favored IGB for weight loss (p < 0.01). Statistically significant differences in favor of IGB were also found for excessive weight loss and BMI reduction. Adverse events were primarily vomiting and abdominal pain. No fatalities were reported. The results of this review are limited by the lack of blinding and the short-term follow-ups. It was concluded that IGBs with conservative therapy are a safe and effective obesity treatment in the short term. However, well designed follow-up RCTs are needed to evaluate long-term safety and efficacy.

Imaz et al. (2008) performed a meta-analysis of 15 studies (n=3608) on IGB for the treatment of obesity. The efficacy at balloon removal was estimated with a meta-analysis of two RCTs (n=75 patients) that compared balloon versus placebo. The estimates for weight lost at balloon removal were 14.7 kg, 12.2% of initial weight, and 5.7 kg/m², 32.1% of excess weight. These differences in weight lost between the IGB and placebo groups were 6.7 kg, 1.5% of initial weight, 3.2 kg/m², and 17.6% of excess weight. The majority of complications were reported to be mild and the early removal rate was 4.2%. In the opinion of the authors, the available evidence demonstrates that IGB is an effective treatment to lose weight in the short-term but does not verify the maintenance of this weight loss over the long term (Imaz, et al., 2008).

In a Cochrane review of the evidence for IGB, Fernandes et al. (2007) included nine randomized, controlled clinical trials (n=395) that compared IGB to no treatment, diet and a combination of balloon placement and diet. According to the authors, results indicated that compared with conventional management, the IGB did not show convincing evidence of a greater weight loss. Although few serious complications of intragastric balloon placement occurred, the relative risks for minor complications like gastric ulcers and erosions were significantly raised (Fernandes, et al., 2007).

A larger case series conducted by Genco et al. (2005) evaluated 2515 patients with a mean BMI of 44.4 who underwent intragastric balloon placement. The balloon was removed after six months. Mortality, complications, BMI, percentage excess weight loss (EWL), BMI loss and comorbidities were evaluated. The overall complication rate was reported to be 2.8%, including the death of two patients. Gastric perforation occurred in five patients (0.19%), four of whom had undergone previous gastric surgery. A total of 19 gastric obstructions (0.76%) presented in the first week after balloon positioning and were successfully treated by balloon removal. Preoperative comorbidities resolved in 617 (44.3%) of 1394 patients. After six months, mean BMI was 35.4 and the EWL was 33.9%. BMI loss was reported to be 4.9 (range 0–25). Despite the complications noted, it was concluded that intragastric balloon is an effective procedure with reduced comorbidities and satisfactory weight loss within a follow-up period of six months. Previous gastric surgery was noted to be a contraindication to intragastric balloon placement.

Currently, the available evidence in the published, peer-reviewed scientific literature is insufficient to establish the safety and efficacy of intragastric balloons.

Laparoscopic Greater Curvature Plication

Laparoscopic greater curvature plication, also referred to as gastric plication or gastric imbrication, is being investigated as a less invasive surgical procedure for obesity. The procedure is similar to laparoscopic sleeve gastrectomy (LSG) but does not involve removal of stomach tissue. The stomach is folded and sewn and therefore the procedure is theoretically reversible. A combination of gastric banding with greater curvature gastric plication has also been described in the literature. This procedure is similar to laparoscopic gastric plication but includes placement of the adjustable gastric band. This combined technique has been suggested to augment the early weight loss after gastric banding with possible decrease in the need for band adjustments (ASMBS, 2011; reviewed 2015).

Literature Review

Evidence evaluating the safety and effectiveness of laparoscopic greater curvature plication, with or without adjustable gastric banding, consists primarily of case series with patient populations ranging from 26-244 and follow-ups of 12 months to five years (Doležalova-Kormanova, et al. 2017; Kim, et al., 2015; Niazi, et al., 2013; Taha, 2012; Talebpour, et al., 2012; Skrekas, et al., 2011; Ramos, et al., 2010). Outcomes of percentage of excessive weight loss (EWL), operative timeframes, and resolution of comorbidities have been reported. Limitations in these studies include lack of a randomized controlled design and short-term follow-up.

Talebpour et al. (2018) conducted a randomized controlled trial (n=70) to compare the safety and effectiveness of laparoscopic gastric plication (LGP) (n=35) vs. laparoscopic sleeve gastrectomy (LSG) (n=35). Primary outcome measures were BMI reduction and the percentage of excess weight loss (%EWL) and total body weight loss (%TWL). Postoperative complications were the secondary outcome. At the 24 months' follow-up there was no significant difference in the lower BMI and mean %EWL between the two groups. The mean BMI was significantly lower in both groups compared to baseline ($p < 0.05$). Seven patients in LSG and one in LGP were readmitted to the hospital ($p = 0.024$). Readmissions in the LSG group were due to two cases of leakage, one case of a draining abscess, two cases of cholecystectomy, and two patients developed acute coronary syndrome. In the LGP group, one patient was readmitted to perform an abdominoplasty. Diarrhea was significantly more frequent in the LGP group (6 vs. 1, $p = 0.046$). Serious events were more common in the LSG group including two cases of leakage and one case of draining abscess formation, whereas in the LGP group, one patient died from pulmonary thromboembolism. Hospital stay was 6.06 ± 1.53 days in the LGP group and 7.46 ± 1.93 days in the LSG group, statistically significant ($p = 0.001$). Limitations of the study include the small patient population and short-term follow-up. Randomized controlled trials with larger patient populations and long-term follow-up are needed to establish the safety and efficacy of LGP.

Ye et al. (2017) conducted a systematic review to compare the safety and efficacy of laparoscopic sleeve gastrectomy (LSG) and laparoscopic greater curvature plication (LGCP). Three randomized controlled trials, four retrospective studies, and one prospective study (n=536) met inclusion criteria. This meta-analysis showed a significantly greater percentage of weight loss (%EWL) after LSG compared to LGCP at 3 months ($p = 0.02$), six months ($p < 0.01$), 12 months ($p < 0.01$) and three years ($p < 0.01$). Based on five studies LSG was associated with a significantly shorter postoperative hospital stay ($p < 0.01$). No significant differences were found in operation time ($p = 0.06$), adverse events ($p = 0.06$), and the resolution of obesity-related hypertension ($p = 0.57$) and diabetes mellitus ($p = 0.31$). Adverse events were defined as postoperative major morbidities, including leaks, stenosis, bleeding, and any other reason of reoperation. Author-noted limitations of this analysis included: possible publication bias due to the inclusion of only the eight studies; the heterogeneity of included patients, the surgeons' experience, and the duration of observation

which might have reduced the reliability of the effect size; and five studies were non-randomized, controlled trials, which may have caused selection and detection bias. A multicenter randomized controlled trial with long-term follow-up is needed to validate these findings.

A systematic review (n=521 patients) by Kourkoulos et al. (2012) included prospective case series (n=8 studies) and case reports (n=2 studies). Inclusion criteria in five studies were age over 18 years old and BMI > 40 or BMI > 35 with at least one comorbidity. Inclusion criteria were not defined in the one study with a minimum BMI of 36, as well as a second study in which minimum BMI was 30. The inclusion criteria for the remaining study included age 18–62 years, BMI of 32–35 kg/m², and a history of GERD and obesity for more than five years with unsuccessful attempts at conservative weight-loss therapy. This study was aimed at demonstrating the efficacy of LGCP with Nissen fundoplication in obese patients with GERD. Universal exclusion criteria included pregnancy, previous bariatric or gastric surgery, hiatal hernia, uncontrolled diabetes, cardiovascular risks, a history of eating disorders (e.g., bulimia), medical therapy for weight loss within the previous two months, or any other condition that constituted a significant risk of undergoing the procedure. A BMI > 50 was defined as an exclusion criterion in two studies. Outcomes of weight loss and complications were assessed. Reported percentage of excessive weight loss in all studies was found to be approximately 50% at six months, 60–65% at 12 months, and 60–65% at 24 months. The total complication rate was 15.1%. The reoperation rate was 3% and the rate of conversion to another procedure was 0.2%. Mortality was zero at 24 months. The authors concluded that the literature on gastric plication and its modifications is limited. More data are required and randomized control trials must be completed in order to reach safe conclusions.

Another systematic review (n=307 patients) by Abdelbaki et al. (2012) also included prospective case series (n=5 studies) reviewed by Kourkoulos et al. (2012) as described above, and case reports (n=2 studies). The age range of patients was 23 to 59 years. At 12 months of follow up, excess weight loss (EWL) ranged from 23.3% to 67%. Patients were followed for more than two years in two studies with EWL rates of 57% and 65%. One study showed inadequate weight loss (< EWL 50 %) in 29/135 subjects (21.48%) and failure (< EWL 30%) of weight loss in 8/135 (5.9%). Complications including gastric leaks and perforations, developed in 25/307 patients (8%), with a complication rate range of 7%–15.3%. Prospective randomized controlled trials with long-term follow-up comparing gastric plication to other well-established bariatric procedures are needed to prove the reliability and metabolic effectiveness of this new procedure.

There is insufficient evidence in the published, peer-reviewed medical literature demonstrating safety and effectiveness of gastric plication. Well-designed studies with larger patient populations comparing this technique to established bariatric procedures are needed to draw firm conclusions regarding the overall safety, efficacy and impact on health outcomes.

Professional Societies/Organizations

American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS): According to the 2013 updated guidelines from the American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS), while there are several short-term studies demonstrating relative safety and effectiveness of greater curvature plication with outcomes intermediate between LAGB and SG, more robust comparative data and conclusive data evaluating the durability of this procedure will be needed before specific recommendations can be made (Mechanick, et al., 2013, updated 2020).

American Society for Metabolic and Bariatric Surgery (ASMBS): The ASMBS (2011; reviewed 2015) policy statement on laparoscopic gastric plication explained that the quantity (n=4

studies, < 300 patients) and quality (prospective or retrospective case series) of the available data is insufficient to draw any definitive conclusions regarding the safety and efficacy of this procedure. The Society supports the following recommendations regarding gastric plication alone or in combination with adjustable gastric band placement for the treatment of obesity:

1. Gastric plication procedures should be considered investigational at this time. This procedure should be performed under a study protocol with third party oversight (local or regional ethics committee, Institutional Review Board, Data Monitoring and Safety Board, or equivalent authority) to ensure continuous evaluation of patient safety and to review adverse events and outcomes.
2. Reporting of short- and long-term safety and efficacy outcomes in the medical literature and scientific meetings is strongly encouraged. Data for these procedures should also be reported to a program's center of excellence database.
3. Any marketing or advertisement for this procedure should include a statement to the effect that this is an investigational procedure.
4. The ASMBS supports research conducted under an IRB protocol as it pertains to investigational procedures and devices. Investigator meetings held to facilitate research are necessary and supported, as is the reporting of all data through BOLD, Bariatric NSQIP or a specific research database. The ASMBS does not support CME courses on investigational procedures and devices held for bariatric surgeons for the purpose of use of investigational procedures outside an IRB research protocol.

Mini-Gastric Bypass/One Anastomosis Gastric Bypass (OAGB)/Loop Gastric Bypass

The mini-gastric bypass, also called the Omega loop gastric bypass and single- or one-anastomosis gastric bypass (OAGB), has been proposed as a bariatric surgery method. The controversial procedure is performed laparoscopically and is similar to the Roux-en-Y. An endoscopic stapler is used to divide the stomach into two parts, and a new, narrow stomach pouch is formed. The larger part of your stomach remains in the body and continues to produce digestive juices to help with digestion but will no longer come in contact with food. Once the new stomach is formed, it is connected (anastomosed) to a two-meter loop of bowel consisting of the duodenum and part of the jejunum. By bypassing two meters of the small intestine, the food passes from the small stomach pouch directly into the small bowel where it meets the digestive juices from the detached portion of the stomach. Complications include biliary reflux and esophagitis. Ongoing concerns following mini-gastric bypass include gastric and esophageal bile reflux, marginal ulcer, poor follow-up and remnant gastric cancer (Wang, et al., 2017). Some patients who undergo loop gastric bypass develop symptomatic bile reflux gastritis and esophagitis, necessitating conversion to RYGB (Salameh, 2006). The loop gastric bypass as developed years ago has generally been abandoned by many bariatric surgeons.

Literature Review

Evidence supporting the use of the mini-gastric bypass is primarily in the form of retrospective reviews (n=1200–1520) and small case series (Carbajo, et al., Jan 2017; Carbajo, et al., May 2017; Taha, et al., 2017).

Guetta et al. (2026) published findings from a retrospective, single-center cohort study in which all included individuals underwent sleeve gastrectomy (SG, intervention) or one-anastomosis gastric bypass (OAGB, comparator) from 2009 through 2024. Propensity score matching (1:1) was performed to minimize confounding, yielding a matched cohort of 2,382 individuals (1,191 in each surgical arm) for analysis. Follow-up was extended to 60 days postoperatively (beyond the standard 30-day period) to capture complications occurring between days 30 and 60; the primary outcome was any complication within 60 days (with complications classified by Clavien–Dindo severity as a secondary outcome). In the matched analysis, the overall 60-day complication incidence was significantly higher after SG than OAGB (15.4% vs 11.8%, p=0.012). SG patients

experienced significantly more intermediate-severity complications (Clavien–Dindo grade II: 5.1% vs 2.4%, $p < 0.001$; IIIa: 1.3% vs 0.3%, $p=0.003$; IVb: 0.7% vs 0.1%, $p=0.039$) and higher rates of specific complications such as leak, abscess, pleural/abdominal fluid collection, bleeding, and dysphagia/vomiting (all $p < 0.05$) compared to OAGB, whereas OAGB patients had more marginal ulcers (0.7% vs 0%, $p=0.008$) and respiratory complaints (2.2% vs 0.6%, $p < 0.001$). Severe complications (Clavien–Dindo grade IIIb–V) and 60-day mortality were rare in both groups and did not differ significantly between SG and OAGB. Limitations include the non-randomized retrospective design, single-center setting, potential temporal biases (different eras of procedure adoption), reliance on utilization-based triggers for complication identification, and incomplete capture of complications managed outside the study institution.

The Omega Loop Versus Roux-en-Y Gastric Bypass (YOMEGA) was a multicenter, randomized controlled trial conducted by Robert et al. (2019) to compare the safety and efficacy of one anastomosis gastric bypass (OAGB) ($n=129$) to Roux-en-Y gastric bypass (RYGB) ($n=124$). Patients were eligible for inclusion if they were aged 18–65 years, had a BMI ≥ 40 kg/m² or a BMI ≥ 35 kg/m² with at least one comorbidity (e.g., type 2 diabetes, high blood pressure, obstructive sleep apnea, dyslipidemia, arthritis) and had undergone a multidisciplinary evaluation by the bariatric team. Exclusion criteria were a history of esophagitis, Barrett’s esophagus, severe gastro-esophageal reflux disease resistant to proton-pump inhibitors, and previous bariatric surgery. The primary outcome measure was the percentage excess BMI loss two years after surgery. Secondary outcomes included: weight and BMI, early and late complications, mean length of hospital stay; duration of surgery; quality of life within two years of surgery; the incidence of gastro-esophageal reflux disease and diarrhea, steatorrhea at six months, dumping syndrome at each follow-up visit, metabolic profile, and histological modifications of gastric and esophageal mucosa. Two years postoperatively, the mean percentage excess BMI loss was -87.9% (SD 23.6) in the OAGB group and -85.8% (SD 23.1) in the RYGB group. The mean difference of percentage excess BMI loss was -3.3% in favor of OAGB. The mean operative time was significantly shorter in the OAGB group ($p<0.001$). The median duration of hospital stay was five days for both groups. There was no significant difference between the groups in HbA1C at two years. The mean decrease in HbA1C at two years was significantly greater in the OAGB group ($p=0.0037$). The proportions of type 2 diabetes remission were not significantly different between the treatment groups ($p=0.28$). There was no significant difference between the groups in the values and decrease of fasting glycemia, triglycerides, LDL cholesterol, HDL cholesterol, and total cholesterol. The incidence of diarrhea was significantly higher in the OAGB group ($p=0.04$), as was median steatorrhea at six months ($p=0.0002$). There was no significant difference in dumping syndrome at two years ($p=0.82$). The improvement in quality of life was not significantly different between the groups. Intraoperative complications occurred in four (3%) RYGB patients (three hemorrhages; one bowel injury) compared with eight (7%) OAGB patients (four hemorrhages; two bowel injuries; two stapling of the nasogastric tube). A total of 66 serious adverse events associated with surgery were reported. There were significantly more serious adverse events in the OAGB group vs. the RYGB group (42 vs. 24, respectively) ($p=0.042$) and nutritional complications (nine vs. no patients, respectively) ($p=0.0034$). A limitation of the study includes the number of patients lost to follow-up. Five patients did not undergo their assigned surgery, and after undergoing their surgery, 14 patients were excluded from the per-protocol analysis. There were four revisions from OAGB to RYGB. Additional limitations include the initial small patient population and the short-term follow-up of two years. The results of this study showed that OAGB was not inferior to RYGB in terms of percentage excess BMI loss at two years, using a 200 cm biliopancreatic limb in the OAGB group and a 150 cm alimentary limb and 50 cm biliopancreatic limb in the RYGB group. However, there were more serious adverse events following OAGB including nutritional complications. Prospective studies with long-term follow-up are needed to conclusively identify the safety and efficacy of OAGB.

Li et al. (2023) conducted a systematic review and meta-analysis of randomized controlled trials (RCTs) comparing one-anastomosis gastric bypass (OAGB) with Roux-en-Y gastric bypass (RYGB) for the treatment of obesity, identifying eight prospective RCTs encompassing a total of 931 participants. Inclusion criteria required English-language RCTs comparing OAGB and RYGB with ≥ 1 month follow-up and reporting weight-related outcomes, while exclusion criteria eliminated withdrawn articles, animal studies, and studies lacking full text. The intervention was OAGB and the comparator was RYGB. Outcomes assessed included weight loss metrics (BMI, percent of excess weight loss [%EWL], and percent of excess BMI loss [%EBMIL]) at multiple timepoints (6 months, 12 months, 2 years, and up to 5 years), perioperative complications (intraoperative, early, and late), remission of obesity-related comorbidities (type 2 diabetes, hypertension, hyperlipidemia, gastroesophageal reflux disease), and operative time. Follow-up durations varied across included trials, ranging from 3 months to 5 years. Comparative analysis demonstrated statistically significant advantages for OAGB in several domains, including greater %EWL at 12 months (MD 3.55; $p = 0.009$), lower BMI at 12 months (MD -2.03 ; $p = 0.01$) and 2 years (MD -2.97 ; $p < 0.00001$), fewer early postoperative complications (RR 0.45; $p = 0.04$), higher hyperlipidemia remission (RR 1.39; $p < 0.0001$), and shorter operative time (MD -36.95 minutes; $p < 0.00001$). No statistically significant differences ($p \geq 0.05$) were observed for BMI or %EWL at 6 months, %EBMIL at 2 years, BMI at 5 years, intraoperative or late complications, or remission of diabetes, hypertension, or GERD. Limitations include heterogeneity in outcome reporting across studies, small numbers of trials contributing to certain endpoints, variable follow-up durations limiting long-term conclusions, potential bias due to inadequate reporting of blinding, and overall limited statistical power for some outcomes. The authors conclude there is no obvious advantage or disadvantage between the two techniques and that more randomized controlled trials are needed.

Wang et al. (2018) conducted a systematic review and meta-analysis of the literature to compare the safety and efficacy of mini gastric bypass (MGB) ($n=4558$) vs. Roux-en-Y (RYGB) ($n=3934$). Ten cohort studies and one randomized controlled trial were included. A significant difference was seen in favor of MGB vs. RYGB in the one-year excess weight loss percentage (%EWL) ($p=0.007$) (five studies); two-year %EWL ($p=0.003$) (two studies); type 2 diabetes remission rate ($p=0.002$), and shorter operative time ($p<0.00001$) (four studies). There were no significant differences in remission rate of hypertension ($p=0.57$) (four studies), mortality ($p=0.43$) (four studies), leakage rate ($p=0.48$) (four studies), gastroesophageal reflux disease ($p=0.47$) (three studies) and hospital stay ($p=0.89$) (three studies). Limitations of the study included: the lack of randomized controlled trial, possible selection bias of studies, small patient populations, high heterogeneity between studies, and the short-term follow-ups. The authors cautioned that due to the small sample sizes and biased data, the results of this analysis should be treated with caution. Larger sample size and multi-center RCTs are needed to compare the safety and efficacy of these two procedures.

One randomized open comparison study ($n=80$) compared mini-gastric bypass to laparoscopic Roux-en-Y (Lee, et al., 2005). The authors reported similar efficacy in terms of excess weight loss (EWL) at two years. Based on a retrospective review ($n=2678$) intraoperative and early complication rates were 0.5% and 3.1%, respectively. Follow-ups that occurred at five years was 62.6%. Adverse events included perioperative bleeding, postoperative duodenal-gastro-esophageal reflux, gastric or anastomotic leak, and abscess or infection (Musella, et al., 2017). Comparative studies with large patient populations and long-term follow-up are needed to support the safety and effectiveness of mini-gastric bypass.

Wang et al. (2017) conducted a systematic review and meta-analysis to investigate the safety and effectiveness of laparoscopic mini-gastric bypass (MGB) versus laparoscopic sleeve gastrectomy (SG). Comparative studies, patients aged 20–70 years, with at least one of the desired outcome measures were included. Two randomized controlled trials and 12 cohort studies met inclusion

criteria. The primary endpoints were one-year percentage excess weight loss (%EWL), five-year %EWL, and remission rate of comorbidities (type 2 diabetes, hypertension, obstructive sleep apnea [OSA], osteoarthritis). The secondary endpoints included overall early complications rate, leakage rate, postoperative bleeding rate, overall late complications rate, ulcer rate, vomiting rate, anemia rate, GERD rate, hospital stay, and revision rate. The last endpoint was operation time. Meta-analysis included 1998 MGB patients and 1864 SG patients. Two studies included patients with a preoperative BMI > 50 kg/m² and three studies included type 2 diabetics. Results of meta-analyses included the following:

- MGB one-year %EWL ranged from 58%–79.3%; SG ranged from 45%–71.4%
- MGB five year %EWL ranged from 68%–78.2%; SG ranged from 51.2%–68.7%
- MGB had a higher one-year %EWL than SG group (p=0.005) (seven studies)
- MGB had a higher five-year %EWL (p > 0.001) (two studies)
- MGB had a higher remission rate of type 2 diabetes (p=0.002) (eight studies)
- MGB had a higher remission rate of hypertension (p=0.02) (six studies)
- MGB had a higher remission rate of OSA (p=0.03) (three studies)
- MGB had a lower remission rate of osteoarthritis (p=0.008) (two studies)
- MGB's overall comorbidity remission rates were 86% for type 2 diabetes, 75% for hypertension, 93% for OSA and 68% for osteoarthritis; SG's overall comorbidity remission rates were 65% for type 2 diabetes, 60% for hypertension, 76% for OSA and 88% for osteoarthritis.
- There were no significant differences in overall early complication rates (six studies), bleed rates (p=0.095) (six studies), vomiting rates (p=0.36) (three studies), anemia rates (p=0.17) (two studies) and operation times (p=0.58) (four studies).
- MGB had a lower leakage rate (p=0.02) (five studies), lower overall late complication rate (p=0.02) (three studies), lower GERD rate (p=0.006) (four studies), shorter hospital stay (p=0.05) and a lower revision rate (p < 0.001) (five studies).
- MGB had a higher ulcer rate (p=0.001) (six studies)

The authors noted that due to the biased data, small patient populations, short-term follow-ups and heterogeneity of the studies, the results may be unreliable. Multicenter randomized controlled trials with large patient populations and long-term follow-ups are needed to compare the effectiveness and safety between mini-gastric bypass and sleeve gastrectomy. Future studies should also investigate the rate of bile reflux and remnant gastric cancer following MGB.

The American Society for Metabolic and Bariatric Surgery (ASMBS) (Parikh, et al., 2018) conducted a literature review on the one-anastomosis gastric bypass (OAGB) or mini gastric bypass. Eleven review articles and 58 clinical studies of which 93% were retrospective reviews were included. A mean follow-up of 60 months was reported in 86% of the studies. Four randomized controlled trials (RCT) involving 341 patients were also included. From the review, ASMBS concluded that OAGB had a relatively short operative time, low complication rate, and excellent weight loss outcomes. However, the retrospective nature of the studies and lack of long-term follow-up data limited the current evidence regarding OAGB, particularly in regard to concerns about long-term nutritional deficiencies due to the hypoabsorptive nature of the procedure, as well as issues specific to the loop gastroenterostomy configuration, such as bile reflux and its potential long-term carcinogenic effects. Well-designed prospective studies with long-term follow-ups are needed to establish the safety and efficacy of OAGB.

Natural Orifice Transluminal Endoscopic Surgery (NOTES)

NOTES, also referred to as endoscopic (oral)-assisted, endoluminal, or transoral incisionless surgery, involves the use of natural orifice access (e.g., mouth, anus) to perform a surgical procedure which potentially reduces or eliminates the trauma of access incisions. The NOTES technique is currently being investigated for use in a range of procedures including bariatric procedures such as gastric bypass (Schauer, et al., 2007).

Gys et al. (2019) conducted a systematic review and meta-analysis of the published literature to evaluate the efficacy and safety of endoscopic gastric plication or gastroplasty for morbid obesity. Endoscopic gastroplasty is a natural orifice transluminal endoscopic surgery (NOTES) which is considered a minimally invasive and mostly restrictive procedure. Twenty two studies (n=2475 patients, mean age 41.2 years, mean baseline BMI 37.8 ± 4.1 kg/m²) met the inclusion criteria of a clinical trial with the primary intervention of endoscopic gastroplasty. Seven different techniques were used among the 22 studies. Eight studies (n=1721) used the OverStitch Endoscopic Sleeve Gastroplasty (ESG). Five studies (n=465) used primary obesity surgery endoluminal (POSE™). Four studies (n=128) used the transoral (vertical) gastroplasty (TOGa®) technique. One study (n=64) used endoluminal vertical gastroplasty (EVG Bard EndoCinch™ suturing system). One study (n=18) used transoral gastric volume reduction (TGVR RESTORe Suturing System). One study (n=17) used the 360°, fully flexible articulating circular endoscopic stapler (ACE). Two studies (n=62) used the endoluminal suturing device (Endomina). Studies were excluded if weight loss was not a primary outcome, had indications other than morbid obesity or concomitant surgical procedures were performed. The primary outcome measured was weight loss at six months or more after the procedure. The secondary outcome was the occurrence of adverse events or complications including insufficient weight loss or regain. The mean follow-up was 13 months. The procedures included in the meta-analysis were endoscopic sleeve gastroplasty (ESG) and the primary obesity surgery endolumenal (POSE™). The average pooled percentage excess weight loss (%EWL) at six months (p = 0.02) and 12 months (p = 0.04) favored POSE™. The %EWL for ESG was $57.9 \pm 3.8\%$ at six months and $44.4 \pm 2.1\%$ at 12 months. %EWL for POSE™ was $68.3 \pm 3.8\%$ at six months and $44.9 \pm 2.1\%$ at 12 months. The TOGa® studies reported 35 %EWL after six months and 43 %EWL after 12 months. The EVG Bard EndoCinch™ suturing system reported 58.1 ± 19.9 %EWL after 12 months. TGVR RESTORe Suturing System reported 27.7 ± 21.9 %EWL after 6–12 months follow-up. ACE reported a median %EWL of 34.9 after 12 months. Endomina reported 32%EWL at six months and 29% EWL at 12 months. ESG reported adverse events were pneumothorax (n=2), perigastric collection (n=8), pulmonary embolism (n=2), intraluminal bleeding (n=5), and leakage (n=1). POSE™ adverse events included intraluminal bleeding (n=2), extra gastric bleeding (n=1), and hepatic abscess (n=1). TOGa® studies reported post-procedural COPD exacerbation as the only major adverse event. The EVG Bard EndoCinch™, TGVR RESTORe Suturing System, ACE, and Endomina reported no major adverse events. Other frequent adverse events reported for any device used include abdominal pain, sore throat, and/or nausea with or without vomiting. Minor adverse events not requiring intervention included superficial phlebitis, temporomandibular dysfunction, pharyngitis, esophagitis, (mild) mouth trauma, and self-limiting intraluminal bleeding. An author-noted limitation of the review was that the devices used varied in techniques. Limitations of the study include the small patient populations and relatively short-term follow-up. High quality studies with larger patient populations and long-term follow-up are needed to determine efficacy and safety of these devices.

Khan et al. (2019) conducted a systematic review and meta-analysis to evaluate and compare the efficiency of three endoscopic procedures for the treatment of obesity (endoscopic sleeve gastroplasty [ESG], AspireAssist [AA], primary obesity surgery endolumenal [POSE]). Twelve studies (n=1149 patients) met the inclusion criteria. Studies were included if they used one of the three endoscopic procedures (ESG, POSE, or AA), reported on either percent excess weight loss (%EWL) or percent total body weight loss (%TBWL), had follow-ups of six months or more, and a minimum of ten patients in the study. Studies were excluded if they were suspected of patient overlap based on inclusion dates reported. Primary outcomes measured were percent excess weight loss (%EWL) and percent total body weight loss (%TBWL) and were reported at six, 12, and 24 months. Four observational studies (n=369) evaluated the performance of ESG. Pooled mean %EWL was 49.67 (p=0.22) at six months, 52.75 at 12 months, and 60.40 at 24 months. Pooled mean %TBWL was 16.01 (p < 0.001) at six months, 17.41 at 12 months, and 19.61 at 24 months. Two randomized control trials (RCT) and two observational studies evaluated the AA

(n=307). Pooled mean %EWL at six and 12 months was 43.25 and 50.85, respectively. Pooled mean %TBWL was 15.37 at 12 months and 20.10 at 24 months. Two RCT and two observational studies evaluated POSE (n=447). Pooled mean %EWL was 43.79 and 44.91 at six and 12 months. Pooled mean %TBWL at six months 13.82 (p=0.12) and 12 months 10.98 (p < 0.001). ESG showed more weight loss than POSE at six and 12 months. No differences in weight loss between ESG and AA at 12 and 24 months was reported. There was no difference in weight loss between AA and POSE at 12 months. The most common adverse events reported were abdominal pain, nausea and vomiting. Adverse events for AA included peristomal infections and granulation tissue formation. Adverse events for ESG were reported in one patient each with pulmonary embolism, pneumoperitoneum, and perigastric fluid collection. Author-noted limitations included: heterogeneity of the studies, inclusion of small studies and inability to evaluate effects on comorbidities such as diabetes, hypertension and lipid profiles. Well-controlled studies with larger patient populations are needed to support the current limited evidence on weight loss in both the short- and long-term, as well as safety for these devices.

Restorative Obesity Surgery, Endoluminal (ROSE): ROSE is an endoscopic-assisted procedure that is being investigated for the treatment of weight regain following gastric bypass surgery that is caused by a gradual expansion of the gastric pouch. The stomach is accessed orally via an endoscope and the stomach pouch is reduced in size using a device such as the StomaphyX™ endoluminal fastener and delivery system (EndoGastric Solutions, Inc., Redmond, WA). StomaphyX is described as a non-invasive weight loss procedure to reduce the size of a patient's stomach without any incisions.

StomaphyX: The StomaphyX is inserted endoscopically and used to create permanent folds (plications) by repeatedly suctioning and fastening parts of the stomach wall using H-shaped durable fasteners. The folds make the area within the stomach smaller, reducing the amount of food the patient can eat. The folds can also serve to slow the draining of food into the lower part of the stomach, prolonging feelings of fullness to further facilitate weight loss.

US Food and Drug Administration

Endoluminal suturing systems such as StomaphyX are regulated as Class II devices and are cleared through the FDA 510(k) process based on substantial equivalence to predicate devices (FDA, 2007). Their indications typically involve endoscopic, trans-oral tissue approximation and ligation within the gastrointestinal tract, including applications for tissue closure and treatment of conditions like GERD. The 510(k) pathway does not require premarket demonstration of clinical effectiveness, but devices must meet safety and performance standards.

Device or Product	Identifier	Manufacturer
StomaphyX	K062875	EndoGastric Solutions, Inc.

*FDA product codes: OCW

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Literature Review

Evidence investigating the safety and efficacy of StomaphyX is lacking.

Professional Societies and Organizations

American Society for Metabolic and Bariatric Surgery (ASMBS): According to ASMBS there are currently a number of endoluminal innovations and novel devices and technologies in different

stages of development or application for the treatment of obesity, including provisional interventions. The Society noted that the theoretical goals of these therapies include decreasing the invasiveness, risk, and barriers to acceptance of effective treatment of obesity. However, these outcomes cannot be assumed and must be proven. The Society stated that the use of novel technologies should be limited to clinical trials done in accordance with the ethical guidelines and designed to evaluate the risk and efficacy of the intervention (ASMBS, 2009).

Duodenal-jejunal Bypass Liner: The duodenojejunal bypass liner (DJBL) is an endoscopically placed and removable intestinal liner. The EndoBarrier™ Gastrointestinal Liner (GI Dynamics, Lexington, MA) is described as a non-surgical, physical barrier that enables food to bypass portions of the intestine. This device is proposed for bariatric preoperative weight loss but has not been approved by the FDA.

Literature Review

Evidence in the published peer reviewed medical literature evaluating the safety and effectiveness of the endoscopic duodenal-jejunal bypass liner is limited to few studies with small sample sizes and short-term follow-up. Koehestanie et al. (2014) conducted a multicenter RCT of obese patients with T2DM assigned to treatment with DJBL implantation (n=38) versus control (n=39). Patient eligibility criteria included adults between 18 and 65 years of age, BMI between 30 and 50 kg/m², and T2DM for less than 10 years. Exclusion criteria were weight loss of more than 4.5 kg within 12 weeks before screening, anticoagulation therapy, and weight loss medication. After six months' follow-up, a statistically significant decrease in body weight was observed in favor of the DJBL group (p < 0.05). EWL was also greater in the DJBL versus control group (p < 0.05). HbA1c levels decreased to 7.0% in the DJBL group compared with 7.9% in the control group (p < 0.05). In the DJBL group, 76.3% of the patients had at least one adverse event (e.g., gastrointestinal complaints) compared to 59% of the patients in the control group. Although study results suggest DJBL implantation may be effective in improving HbA1c levels and may result in EWL, the study is limited by its small sample size and short-term follow-up.

An RCT (n=41) by Schouten et al. (2010) compared patients who received the endoscopically placed duodenal-jejunal bypass sleeve or EndoBarrier Gastrointestinal Liner (n=30), to a diet control group (n=11). Successful implantation occurred in 26 patients. Mean EWL after three months was 19.0% for device patients versus 6.9% for control patients (p < 0.002). All patients had at least one adverse event, primarily abdominal pain and nausea during the first week after implantation.

Professional Societies/Organizations

American Society for Gastrointestinal Endoscopy (ASGE): The Bariatric Endoscopy Task Force, a subcommittee of ASGE, conducted a systematic review and meta-analysis of the literature to evaluate endoscopic technologies for the treatment of obesity. The review included a meta-analysis of the available data on the EndoBarrier duodenal jejunal bypass sleeve (DJBS). Regarding EndoBarrier DJBS, six randomized controlled trials and six prospective studies met inclusion criteria. Three studies (n=105) reported 35.3% excess weight loss (EWL) at 12 months following implantation. Four randomized controlled trials compared 12–24 weeks of treatment with EndoBarrier (n=90) vs. sham or control (n=84). The mean %EWL difference compared with a control group was significant at 9.4%. The studies were associated with a high degree of heterogeneity. Compared to baseline, the EndoBarrier demonstrated a significant improvement in HbA1C from -0.7 at 12 weeks (p=0.16), -1.7 at 24 weeks (p<0.001) weeks and -1.5 at 52 weeks (p < 0.001) following implantation. There was a statistically significant improvement in HbA1C when EndoBarrier was compared to controls (p=0.001). Adverse events included 58.7% pain, 39.4% nausea/vomiting, 18.37% early removal, 4.93% migration, 3.9% GI bleeding and 3.47% sleeve obstruction. The Task Force noted that enrollment in the multicenter U.S. pivotal trial was placed on hold in March 2015 by the U.S. Food and Drug Administration after four cases of hepatic

abscess occurred among the 325 patients already enrolled. Author-noted limitations of the meta-analyses included the high degree of heterogeneity among included studies, risk of bias in non-randomized studies, and different methods used among studies to report the %EWL (Metropolitan Life Tables vs. BMI 25 method) (ASGE Bariatric Endoscopy Task Force, et al., 2015). Randomized controlled trials with larger patient populations and long-term data are necessary to support the safety and efficacy of EndoBarrier DJBS.

Transoral Gastroplasty (TG): Transoral gastroplasty, also known as vertical sutured gastroplasty or endoluminal vertical, involves the use of endoscopically guided staplers that create a stapled restrictive pouch along the lesser curvature of the stomach. The TOGA® system (Satiety Inc., Palo Alto, CA) developed for this procedure has not been FDA-approved. Currently there is insufficient evidence in the published peer-reviewed medical literature evaluating the safety and effectiveness of this procedure.

Endoscopic Closure Devices: Endoscopic closure devices are used in a variety of surgical procedures including bariatric surgery. The devices are proposed for endoscopic closure of acute and chronic gastrointestinal (GI) wall defects, including spontaneous and iatrogenic perforations, anastomotic leaks, and chronic fistulae. They may also allow closure of enterotomies created for NOTES procedures.

The American Society for Gastrointestinal Endoscopy's (ASGE) (2012) report on emerging technology stated that further prospective studies are needed to define the role of these devices in the closure of GI wall defects.

Currently there is insufficient evidence in the peer-reviewed medical literature to support the use of transluminal endoscopic surgical procedures and devices including the ROSE procedure, StomaphyX, transoral gastroplasty, DJBL, and endoscopic closure devices for the management of severe obesity.

Transoral outlet reduction (TORe): Transoral outlet reduction (TORe) is an endoscopic procedure used in patients with weight gain post Roux-en-Y gastric bypass (RYGB). Weight regain after Roux-en-Y gastric bypass (RYGB) correlates with dilated gastrojejunal anastomosis (GJA). Endoscopic sutured transoral outlet reduction (TORe) has predominantly been performed by either placing interrupted sutures at the GJA or the creation of a purse-string suture. This procedure has also been performed using various techniques including plication devices (Stomaphyx; Endogastric Solutions, California, United States), suction based superficial suturing devices (Endocinch; C.R. Bard, Murray Hill, New Jersey, United States) or with the full thickness suturing devices (Overstitch; Apollo Endosurgery, Texas, United States) (Dhindsa, et al. 2020).

Literature Review

Evidence in the published peer reviewed medical literature evaluating the safety and effectiveness of transoral outlet reduction is insufficient to support its use. The evidence primarily consists of retrospective reviews and case series with small patient populations (n=20-44) and short-term follow-ups (9-12 months) (Fayad, et al., 2019; Laterza, et al., 2017; Catalano, et al., 2016; Goyal, et al., 2015).

Beran et al. (2025) performed a systematic review and meta-analysis of the literature addressing TORe after RYGB to evaluate its safety and efficacy. Six studies were included (n = 333). The primary outcomes were clinical success (defined as sustained symptom improvement without need for repeat TORe or surgical revision at final follow-up) and change in Sigstad's score; secondary outcomes included need for surgery, repeat TORe, and serious adverse events. Follow-up durations ranged from 3 months to 3.4 years across studies. Pooled analysis demonstrated a clinical success rate of 83% (95% CI 71-90) and a statistically significant improvement in

Sigstad's score (mean difference -11.12, 95% CI -15.33 to -6.91; $p < 0.001$). Rates of subsequent surgery (5.8%), repeat TORe (6.9%), and serious adverse events (3%) were low but derived from uncontrolled data. Limitations include the absence of randomized controlled trials, reliance on observational designs with risk of selection bias and unmeasured confounding, significant heterogeneity across studies, inconsistent definitions of clinical success, variable and often short follow-up durations limiting long-term durability assessment, and incomplete or inconsistent reporting of standardized outcome measures. Collectively, the evidence is limited by non-comparative design and methodological heterogeneity, reducing the strength of conclusions regarding comparative effectiveness.

Roux-en-Y Gastric Bypass (RYGB) Combined with Gastric Banding

The combination of RYGB with a banding procedure is being investigated as a treatment to enhance weight loss and avoid weight regain. The evidence evaluating this combined procedure is currently limited. A prospective randomized double-blind trial ($n=90$) by Bessler et al (2007) compared banded and nonbanded open gastric bypass for the treatment of super obesity. No significant differences were found in the overall number of complications, resolution of co-morbidities, or % excess weight loss (EWL) at six, 12, and 24 months (43.1% versus 24.7%, 64.0% versus 57.4%, and 64.2% versus 57.2%, respectively) postoperatively. The banded patients had achieved significantly greater %EWL at 36 months (73.4% versus 57.7%; $p < 0.05$). The incidence of intolerance to meat and bread was greater in the banded group.

The available evidence for gastric bypass combined with simultaneous gastric banding is insufficient to support safety and efficacy for the treatment of obesity, and to demonstrate a clinical benefit with improved long-term outcomes.

Stomach Aspiration Therapy

Aspiration therapy is being investigated as a weight loss method for patients with Class II and III obesity. This therapy involves the endoscopic percutaneous placement of a gastrostomy tube which drains gastric contents after meal consumption. Aspiration therapy requires the use of the AspireAssist system that allows instillation of fluid into the stomach and partial aspiration of ingested meals (Sullivan, 2013).

U.S. Food and Drug Administration

The AspireAssist® System received FDA approval through the Premarket Approval (PMA) process on June 14, 2016. The PMA pathway is the most stringent regulatory process for medical devices and is reserved for Class III devices, which are considered high risk and require evidence of safety and effectiveness, typically through clinical studies. The AspireAssist System is indicated for long-term weight reduction in adults aged 22 years and older with a body mass index (BMI) of 35–55 kg/m² who have failed to achieve and maintain weight loss with non-surgical therapy (FDA, 2016). The device is intended for use in conjunction with lifestyle therapy and continuous medical monitoring. It is not intended for short-term use or as a stand-alone therapy. Contraindications include the following:

- previous abdominal surgery that significantly increases the medical risks of gastrostomy tube placement
- esophageal stricture, pseudo-obstruction, severe gastroparesis or gastric outlet obstruction
- inflammatory bowel disease
- history of refractory gastric ulcers
- ulcers, bleeding lesions, or tumors discovered during endoscopic examination
- uncontrolled hypertension (blood pressure > 160/100)
- history or evidence of serious pulmonary or cardiovascular disease, including acute coronary syndrome, heart failure requiring medications, or NYHA (New York Heart Association) class III1 or IV2 heart failure

Device or Product	Identifier	Manufacturer
AspireAssist® System	P150024	Aspire Bariatrics, Inc.

*FDA product codes: OYE

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Literature Review

Nyström et al. (2018) conducted a post-market registry/observational study (n=201) to evaluate the safety and efficacy of the AspireAssist System at five European centers. Subjects were age ≥ 18 years with a BMI of 35.0–70.0 kg/m² and had failed conservative weight loss methods. Following AspireAssist implantation lifestyle/cognitive behavior therapy was provided and varied from center to center. Follow-ups occurred monthly or as medically warranted during the first year. After the first year, follow-ups occurred every 3–6 months with some visits conducted electronically or telephonically. Mean weight loss outcomes included: 18.2% $\pm$ 9.4% (n=155) at one year; 19.8% $\pm$ 11.3% (n=82) at two years; 21.3% $\pm$ 9.6% (n=24) at three years and 19.2% $\pm$ 13.1% (n=12) at four years. Clinically significant reductions at year one were observed in glycated hemoglobin (HbA1C) (p < 0.0001) (n=57), triglycerides (p < 0.001) (p=53), total cholesterol (p < 0.01) (n=53) and blood pressure. Of the 199 successful gastrostomies, 47 participants discontinued aspiration therapy and had their gastrostomy tubes removed along with 17, 18, 9, 2, and 1 subjects in the first, second, third, fourth, and fifth year, respectively. Reasons for discontinuing the therapy included: achievement of weight loss, lack of weight loss, inability or unwillingness to adhere to therapy, discomfort and/or fatigue with the therapy. Five subjects pursued other bariatric surgeries. Periprocedural complications included pain, possible/actual wound infections, and benign pneumoperitoneum. Postoperative complications included: gastric leakage, stomal irritation/granulation tissue; infection/possible infection; buried bumper; and A-Tube rotation. Author-noted limitations of the study included: lack of a control group; only two centers reported cardiometabolic data; short-term follow-up; and number of subjects lost to follow-up. Another limitation was the variation in the lifestyle/cognitive behavior therapy at each center. Randomized controlled trials with large patient populations and long-term follow-up are needed to support the safety and efficacy of the AspireAssist.

The pivotal study for FDA-approval was an RCT (n=207) published by Thompson et al (2017). In this study, patients were randomized to receive treatment with aspiration therapy plus lifestyle counseling (n=137) or lifestyle counseling alone (n=70). Inclusion criteria were age 21–65 years old and a BMI of 35.0–55.0 kg/m². Exclusion criteria included previous bariatric surgery, serious cardiovascular disease, use of medications that cause clinically significant weight gain or loss, and a history of an eating disorder. The first co-primary end point was mean percentage of excessive weight loss (% EWL) at 52 weeks, with success defined as at least a 10% difference in %EWL between the AspireAssist and Lifestyle Counseling groups. The second co-primary end point was the proportion of participants who achieved at least 25% EWL at 52 weeks. Success was defined as at least 50% of the AspireAssist group achieving at least 25% EWL. Secondary end points included change in percentage of total body weight from baseline and the proportion of participants who achieved a reduction in total body weight of 10% or more. After enrollment, 29 AspireAssist and 29 Lifestyle Counseling participants withdrew from the study leaving 82 AspireAssist (74% of those enrolled) and 31 Lifestyle Counseling participants (52% of those enrolled) who completed the entire 52-week study. Both co-primary end points were met: 1) %EWL in the AspireAssist group was 22% greater than the %EWL achieved in the Lifestyle Counseling only group, and 2) 59% of the AspireAssist group lost at least 25% of EBW. Adverse

events were primarily associated with the gastrostomy tubes and included the development of peristomal granulation tissue (40.5%) and abdominal pain (37.8%). Serious adverse events were severe abdominal pain, peritonitis, pre-pyloric ulcer, and A-tube replacement due to Skin-Port malfunction, each occurring in one patient (0.9%). Acknowledged study limitations include the lack of blinding which was not possible, the short-term follow-up period, and the number of patients lost to follow-up (28%). Study results suggested that aspiration therapy may be effective in achieving weight loss. However, safety issues surround the required gastrostomy tube placement and additional well-designed studies with longer follow-up are needed to define the role of this weight-loss therapy.

Norén and Forssell (2016) conducted a prospective observational study (n=25) the AspireAssist system for treatment of obesity, and its effect on patient's quality of life. Inclusion criteria were BMI ≥ 35.0 kg/m² and age from 25 to 65 years. Exclusion criteria were myocardial infarction during the last three months, known malignancy, chronic liver or kidney disease, prior major surgery in the upper gastrointestinal tract, psychiatric disease including substance abuse, or eating disorder. Participants had the option to continue therapy for an additional year. Follow-up of 12 months was completed by 20/25 patients. The mean extreme weight loss (EWL) was 54.4% at 12 months and 61.5% at 24 months. In diabetic patients (n=7), there was a significant reduction in HbA1c level from a median of 47 to a median of 42 (p=0.03). The primary adverse effect was moderate to severe pain. Quality of life measured by EQ-5D and VAS was reported to significantly increase during treatment. Study limitations include the non-randomized controlled design, small patient population, and short-term follow-up.

A randomized controlled pilot study (n=18 subjects) by Sullivan et al. (2013) assigned individuals diagnosed with obesity in a 2:1 ratio to undergo aspiration therapy for one year plus lifestyle therapy (n=11) or lifestyle therapy alone (n=7). Lifestyle intervention comprised a 15-session diet and behavioral education program. Adults with a BMI between 40.0 and 50.0 kg/m² or between 35.0 and 39.9 kg/m² with comorbidities were selected. Exclusion criteria were evidence of an eating disorder or major depression, history of gastrointestinal disease or previous gastric surgery that would increase the risk of A-Tube placement, uncontrolled hypertension, sleep apnea, fasting serum glucose level ≥ 105 mg/dL, diabetes, serum triglyceride level > 400 mg/dL or pregnancy/lactation. One-year follow-up was completed by 10/11 aspiration therapy subjects and 4/7 subjects who received lifestyle therapy only. The percentage of weight loss and excess weight loss (EWL) in the aspiration therapy group was significantly greater than in the lifestyle therapy group (p=0.02, p=0.036 respectively) at 52 weeks. No significant change in the percentage of weight loss or EWL occurred from week 52 to week 104 in the subjects (n=7) who continued aspiration therapy. The use of aspiration therapy was not reported to induce any adverse eating behaviors. The adverse events included peristomal pain and irritation. No serious adverse events occurred in either group. These study results indicate that aspiration therapy may be associated with weight loss in obese patients. However it is difficult to draw conclusions regarding safety and efficacy due to the small number of patients included and lack of long-term follow-up.

There is a paucity of evidence in the published peer-reviewed medical literature evaluating the safety and effectiveness of stomach aspiration therapy. Studies primarily include small patient populations and short-term follow-ups. Additional well-designed, long-term studies are needed to support this treatment for Class II and III obesity.

Vagus Nerve Blocking

Vagus nerve blocking (VNB) or vagal blocking therapy has been investigated as a treatment for obesity. In vagal blocking for obesity control (VBLOC) (e.g., Maestro) an implanted neurogenerator discharges high-frequency, low-energy electrical pulses to block vagus nerve signals in the abdominal region, inhibiting gastric motility and increasing satiety (feeling full). The procedure involves the placement of two leads around the vagal nerve trunks via laparoscopy. An

external device programs the generator. Early clinical trial results suggest that VNB may achieve excess weight loss (EWL) that is comparable to approximately half of that achievable by LAGB.

U.S. Food and Drug Administration

The Maestro® Rechargeable System received approval through the FDA's Premarket Approval (PMA) process (FDA, 2015), which is the most stringent regulatory pathway for medical devices. PMA is required for Class III devices, which are considered high risk and must demonstrate safety and effectiveness, typically through clinical studies. According to the FDA (2015), the Maestro system is "indicated for use in weight reduction in patients aged 18 years through adulthood who have a Body Mass Index (BMI) of 40 to 45 kg/m², or a BMI of 35 to 39.9 kg/m² with one or more obesity related co-morbid conditions, and have failed at least one supervised weight management program within the past five years." Contraindications are as follows:

- cirrhosis of the liver, portal hypertension, esophageal varices or a clinically significant hiatal hernia
- planned magnetic resonance imaging (MRI)
- planned ultrasound diathermy
- high risk for surgical complications
- permanently implanted, electrical powered medical device, or gastrointestinal device or prosthesis (e.g., pacemakers, implanted defibrillators, or neurostimulators)

Potential adverse of the device include allergic reaction to the implanted material and damage to the vagal nerve trunks. The FDA-approval was based on one pilot and two pivotal studies (i.e., EMPOWER, ReCharge) (FDA, 2015).

Device or Product	Identifier	Manufacturer
Maestro® Rechargeable System	P130019	Reshape Lifesciences, Inc.

*FDA product codes: PIM

Note: Coverage decisions are not based solely on FDA approval. Device or product names are provided for example purposes only. Their inclusion does not indicate endorsement or preference for any specific brand or model. This list is not intended to reflect all available products or technologies.

Literature Review

Evidence in the published peer-reviewed medical literature evaluating vagus nerve blocking (VNB) for severe obesity consists of RCTs and case series. Morton et al. (2016) conducted an RCT (n=84) to evaluate the safety and efficacy of vagal blocking device (vBloc) in patients with moderate obesity and comorbidities. This sub-group from the FDA ReCharge trial was randomized to vBloc (n=53) or sham (n=31). Obesity-related comorbidities included dyslipidemia (73%), hypertension (58%), sleep apnea (33%), and type 2 diabetes (8%). The vBloc group achieved 33% excess weight loss (EWL) compared to 19% EWL in the sham group at 12 months (p < 0.0001). Common adverse events of vBloc through 12 months of follow-up were heartburn/dyspepsia and implant site pain; the majority of events were reported as mild or moderate.

Ikramuddin et al. (2014) conducted the ReCharge trial, a multicenter randomized, double-blind, sham-controlled study (n=239) of patients implanted with a nerve block device (Maestro Rechargeable System) using active (n=162) versus sham treatment (n=77). Inclusion criteria were a BMI of 40-45 or 35-40 with at least one obesity-related condition. The co-primary endpoints were percentage of excess weight loss (% EWL) at 12 months and the percentage of patients achieving ≥ 20% EWL and ≥ 25% EWL. At 12 months, 52% of patients in the vagal nerve block group achieved 20% or more excess weight loss and 38% achieved 25% or more excess

weight loss. In the sham group 32% of subjects achieved 20% or more loss and 23% achieved 25% or more loss. Efficacy endpoints were not met. A total of eight patients in the active therapy group required a revision procedure. Therapy-related serious adverse event rate in the vagal nerve block group was 3.7%, and included mild to moderate heartburn, dyspepsia, and abdominal pain. Acknowledged limitations include homogeneity of the patient population and a low rate of common metabolic comorbidities such as diabetes. Study results indicate no significant difference in %EWL between active vagal nerve block therapy and treatment with a sham device.

Apovian et al. (2017) reported on the two-year follow-up of the ReCharge study on the subjects who were randomized to vBloc and continued open-label with the therapy. At 24 months subjects (n=103) had a mean excess weight loss (EWL) of 21%; mean total weight loss (TWL) of 8%; 58% had $\geq 5\%$ TWL; and 34% had $\geq 10\%$ TWL. Compared to screening values, significant improvements ($p < 0.05$) were seen in mean LDL, HDL, triglycerides and systolic and diastolic blood pressures. Patients in the sham group who did not cross-over had a mean 4% EWL. Adverse events included heartburn, dyspepsia, and implant site pain. There were four additional revisions between 12 and 24 months of which two were due to pain at the neuroregulator site and one each due to twisted leads and a device that would not recharge. Limitations of the study include the small patient population, short-term follow-up, missing data and lack of a control group.

Sarr et al. (2012) conducted the EMPOWER study, a multicenter double-blind, prospective RCT (n=294) of patients implanted with a vagal blocking system and randomized to the treatment (n=192) or control (n=102) group. Male or female obese subjects, 18-65 years of age, with a BMI of 40-45 kg/m² or 35-39.9 kg/m² with one or more obesity-related, comorbid condition were included. The primary effectiveness objective was to demonstrate a significantly greater %EWL at 12 months in the treated group compared to the control group. At the end of the blinded, 12-month follow-up period, all subjects received open-label VBLOC Therapy and will be followed for an additional four years. The secondary effectiveness objective was to determine if a significantly greater percentage of subjects in the treated group achieved 25% EWL compared to control subjects. Neither endpoint statistically differed between active and sham treatment groups. There were a total of 35 adverse events including infection and pain, with 14 subjects requiring a revision procedure due to an adverse event or to make the device operational. Limitations of the study included compliance issues related to wearing an external device versus a completely implantable system, and the study inclusion of dietary counseling, behavior modification, and exercise training, which may have contributed to the % EWL.

Camilleri et al. (2009) conducted an open-label multicenter study (n=31) to assess the effects of a vagal blocking device on EWL, safety, dietary intake, and vagal function. Electrodes were implanted laparoscopically near the esophagogastric junction to provide intermittent vagal blocking in patients with a BMI range of 35-50 kg/m². The mean EWL at six months follow-up was 14.2% ($p < 0.001$). Calorie intake decreased by $> 30\%$ at six months ($p \leq 0.001$), with earlier satiation ($p < 0.001$) and reduced hunger ($p=0.005$). There were no deaths or device-related serious adverse events. The study is limited by its small sample size and lack of randomization. Additional well-designed studies are needed to further evaluate the role of this therapy in the treatment of obesity.

Evidence evaluating the safety and effectiveness of VNB is limited, not supportive of safety and efficacy at this point, and is therefore insufficient to support use of the procedure for the treatment of obesity.

Professional Societies/Organizations

The American Society for Metabolic and Bariatric Surgery (ASMBS) position statement on VNB for obesity stated that the quantity of the current data and the length of follow-up indicate adequate safety and efficacy in the short term. More prospective studies with longer follow-up are required

to establish the clinically significant efficacy and patient tolerance of this device (Papasavas, et al., 2015).

Vagus Nerve Stimulation (VNS)

VNS provides intermittent electrical stimulation to the tenth cranial nerve, which influences certain patterns of brain activity. The vagus nerve is a major connection between the brain and the rest of the body and as such, carries sensory information from the body to the brain and motor commands from the brain to the body. A potential use of VNS concerns the regulation of brain satiety signals. The brain knows that the stomach is empty or full, largely on the basis of information transmitted by the vagus nerve. Based on the theory the vagus signal could be altered to modify eating behavior, VNS has been proposed as a treatment for obesity. Currently the literature regarding the use of VNS for obesity is limited and therefore conclusions about safety and efficacy cannot be made at this time. Please refer to the Vagus Nerve Stimulation (VNS) Coverage Policy for additional information.

Bariatric Surgery in Adolescents

Two bariatric procedures are established treatment options for those adolescents (age 11–17 years) who have morbid obesity and have been unresponsive to medical and pharmacological management. Consistent low to moderate quality of evidence has shown that Roux-en-Y or sleeve gastrectomy result in significant weight reduction and improvement in co-morbidities postoperatively. Data support the use of these two procedures in adolescents with a BMI ≥ 40 kg/m² or 140% of the 95th percentile (class 3 obesity) (whichever is lower) or a BMI of 35–39.9 kg/m² or 120% of the 95th percentile (class 2 obesity) (whichever is lower) with at least one clinically significant obesity-related comorbidity (e.g., coronary artery disease, diabetes mellitus, idiopathic intracranial hypertension, obstructive sleep apnea, hypertension, gastroesophageal reflux, nonalcoholic steatohepatitis [NASH]). Current data suggest that weight loss outcomes at three years postoperatively are comparable between SG and RYGB in adolescents. Bariatric surgery has been shown to improve risk factors and decrease morbidity and mortality in this age group. However, long-term data are lacking (Inge, 2025; Pratt, 2018; Inge, et al., 2017).

Literature Review

Zhu et al. (2023) published the findings of a multicenter, retrospective, nonrandomized study of 202 obese adolescents with type 2 diabetes treated between June 2017 and May 2019, comparing bariatric surgery (n = 109) with medical therapy (n = 93). Additional propensity score matching yielded 87 individuals per group. The intervention consisted of laparoscopic bariatric procedures (laparoscopic sleeve gastrectomy [n = 63] or laparoscopic Roux-en-Y gastric bypass [n = 46]), while the comparator group received standardized medical management including medications, lifestyle intervention, and behavioral modification. Inclusion criteria were adolescents aged 13–19 years with BMI > 35 kg/m² and type 2 diabetes or glucose intolerance and no contraindication to surgery. Exclusion criteria included type 1 diabetes, substance abuse, secondary obesity, prior abdominal surgery, inflammatory bowel disease, pregnancy, and lack of follow-up data. Outcomes included weight loss (BMI and total weight loss percentage), glycemic control (fasting blood glucose [FBG], HbA1c, HOMA-IR), diabetes remission, and medication use, assessed at baseline and 1, 6, 12, and 24 months, with approximately 2-year follow-up. At 2 years, the surgery group demonstrated greater reductions in BMI compared with controls (entire cohort: 34.9 [4.3] vs. 41.5 [7.1], $p < 0.001$; matched cohort: 31.7 [3.6] vs. 40.9 [6.1], $p < 0.001$), and higher total weight loss (24.8% vs. 5.4% in the entire cohort; 29.6% vs. 5.7% in the matched cohort, $p < 0.001$). Improvements in glycemic indices favored surgery, with significant differences in FBG, HbA1c, and HOMA-IR at 2 years. Diabetes remission occurred in 76.1% of surgical patients versus 6.5% of controls in the entire cohort and 80.5% versus 5.7% after matching ($p < 0.001$ for both comparisons). Medication use also decreased significantly in the surgery group compared with controls. Study limitations include the retrospective, nonrandomized design with potential selection bias despite propensity matching, relatively small sample size, and limited

follow-up duration of two years, which restricts assessment of long-term durability and safety outcomes.

Järholm et al. (2023) conducted a multicenter, open-label randomized controlled trial (AMOS2) of 50 adolescents (age 13–16 years) with severe obesity ($\text{BMI} \geq 35 \text{ kg/m}^2$). Included were individuals with ≥ 1 year of prior obesity treatment, Tanner stage ≥ 3 , and who had passed pediatric psychological and medical evaluation. Key exclusions were monogenic or syndromic obesity, major psychiatric illness, and regular self-induced vomiting. Individuals were assigned 1:1 to metabolic and bariatric surgery (MBS; primarily Roux-en-Y gastric bypass, with sleeve gastrectomy in select cases) or intensive non-surgical management (beginning with an 8-week low-calorie diet and continued comprehensive lifestyle therapy). Outcomes were assessed over a 2-year follow-up. The primary outcome of interest was BMI change at 2 years. Secondary outcomes included metabolic status, quality of life (QoL), and eating behavior. At 2 years, mean BMI had decreased by 12.6 kg/m^2 in the MBS group versus a 0.2 kg/m^2 decrease in the intensive non-surgical group, a statistically significant between-group difference of -12.4 kg/m^2 (95% CI -15.5 to -9.3 ; $p < 0.0001$). The MBS group also showed significantly greater improvements in several metabolic risk markers compared to non-surgical treatment (e.g., improved glycemic control and lipid levels, whereas changes in blood pressure and liver enzyme levels did not differ significantly between groups at 2 years. Physical health-related QoL (e.g., physical functioning, general health) improved significantly more after MBS, while mental health and obesity-specific QoL endpoints were not significantly different between groups at 2 years. In the domain of eating behavior, the MBS group had significantly greater reductions in uncontrolled eating, and by 2 years no surgical participants reported binge eating, versus 2 of 19 available non-surgical participants who still reported binge episodes. There were no significant differences between groups in micronutrient (vitamin and mineral) levels at 2 years. The MBS group experienced a reduction in bone mineral density relative to controls (-0.9 ; $(-1.20$ to $-0.60) < 0.0001$), although their mean bone density remained within the age-appropriate normal range at 2 years. GI symptom profiles were similar between groups, except that the MBS group reported significantly less reflux symptoms. No severe adverse events were reported; four mild complications occurred in the MBS arm (e.g., minor wound issues and one gallstone requiring cholecystectomy) and none in the non-surgical arm. Important limitations include the small sample size, the open-label design, a relatively short 2-year follow-up, and the crossover of 5 (20%) non-surgical participants to MBS during the second year (analyzed by intention-to-treat), which may affect interpretation of long-term outcomes.

Inge et al. (2017) reported 5–12 year outcomes ($n=58$) of the Follow-up of Adolescent Bariatric Surgery at 5 Plus Years (FABS-5+) extension study for severe adolescent obesity ($\text{BMI} \geq 40 \text{ kg/m}^2$). The original prospective, noncomparative study included 74 young people aged 13–21 who underwent Roux-en-Y gastric bypass for clinically severe obesity. At baseline, the mean age of the cohort was 17.1 years and mean BMI was 58.5 kg/m^2 . Inclusion criteria in the original study were adolescents aged ≤ 21 years with a mean BMI of 60.2 kg/m^2 who underwent Roux-en-Y gastric bypass surgery. All but two procedures were laparoscopic. Exclusion criteria were an inability to complete self-report forms because of developmental delay, or death before the long-term study visit. The primary outcome measure was the change in BMI over time. Secondary outcomes measures were prevalence of diabetes, hypertension, and dyslipidemia. Secondary safety outcomes were micronutrient levels and clinical events postoperatively. At the mean follow-up of eight years (range 5.4–12.5 years), the mean age was 25.1 years, mean BMI was 41.7 kg/m^2 and the mean change in BMI was -29.2% . From baseline to long-term follow-up, there was significant improvements in the prevalence of elevated blood pressure ($p=0.001$), dyslipidemia ($p<0.0001$), and type 2 diabetes ($p=0.03$). At follow-up, 25 (46%) patients had mild anemia which required no intervention, 22 (45%) had hyperparathyroidism, and eight (16%) had low amounts of vitamin B12. Although, on average, patients had improvements in bodyweight and health status over time, most patients remained obese ($n=36/57$) having a $\text{BMI} \geq 35 \text{ kg/m}^2$.

Limitations of the study included the small patient population, lack of a control group, larger number of females vs. males, and the number of patients lost to follow-up. The authors noted that surgical intervention soon after the diagnosis of severe obesity (e.g., BMI 35–40 kg/m²) might result in more complete reversal of severe obesity and cardiometabolic risks than when surgery is offered to adolescents who have progressed to higher BMI values. The recommendation for consideration of surgery in adolescents at BMI values of 35–40 kg/m² with other clinical indications is consistent with advice contained in peer-reviewed clinical practice guidelines.

Rajjo et al. (2017) conducted a systematic review of randomized controlled trials (RCTs) (n=133) to evaluate the effectiveness of various medical and surgical interventions for reducing excess body weight in children and adolescents. The RCTs were taken from 16 systematic reviews. RCTs that had enrolled overweight or obese children, age > 2 years; evaluated interventions used to treat pediatric obesity (medication, surgery, lifestyle interventions, and community-based interventions); compared the interventions to usual care or each other; and had a follow-up of at least six months were included. Outcomes included the change in absolute and percentage of change in body mass index (BMI) and weight. Changes in metabolic outcomes were also reviewed. Outcome data revealed the following:

- Multisport and aerobic exercise reduced systolic blood pressure and fasting glucose (low to moderate quality of evidence).
- Low-carbohydrate diets had a similar effect to low-fat diets in terms of body mass index (BMI) reduction (moderate quality of evidence).
- Education-based interventions (compared with usual care) significantly lowered diastolic blood pressure (moderate quality evidence), BMI (low quality evidence), and waist circumference (low quality evidence) but did not substantially reduce systolic blood pressure (very low quality evidence).
- Pharmacotherapy (metformin, sibutramine, orlistat) reduced BMI and waist circumference (sibutramine, orlistat) and increased high-density lipoprotein cholesterol (sibutramine) but also raised systolic and diastolic blood pressure (sibutramine).
- Surgical interventions resulted in an average BMI difference of -13.5 kg/m² from baseline to one year. The BMI loss was greater after Roux-en-Y compared with laparoscopic adjustable gastric banding. Overall, surgical interventions have been mainly restricted to advanced cases with multiple comorbidities and refractory to nonsurgical interventions.
- A combined approach of education and physical activity significantly reduced the BMI (low quality evidence). A Combined approach of dietary modification, physical activity, behavioral therapy, and education substantially reduced systolic and diastolic blood pressure, BMI, and triglycerides but not low-density lipoprotein cholesterol (low quality evidence).
- Family-based interventions, including both the parent and the child, compared with parent-only interventions, did not yield substantial differences in BMI (low quality evidence).

The authors concluded that although several childhood obesity interventions are effective, a comprehensive multicomponent intervention appears to yield the best overall outcomes.

A prospective RCT (n=50) by O'Brien et al. (2010) compared the outcomes of adolescents between the ages of 14 and 18 with a BMI > 35 who were assigned either to a supervised lifestyle intervention or to undergo gastric banding. In the gastric banding group 24/25 participants completed the study versus 18/25 subjects in lifestyle group. An excess weight loss of 78.8% (95% CI, 66.6%-91.0%) was reported in the gastric banding group compared to an excess weight loss of 13.2% (95% CI, 2.6%-21.0%) in the lifestyle group. At 24 months, none of the gastric banding group had the metabolic syndrome (p=0.008) compared to 4/18 (22%) in the lifestyle group (p=0.13). There were no perioperative adverse events. However, surgical revision was required in seven patients for proximal pouch dilatation or tubing injury during follow-up.

Treadwell et al. (2008) performed a systematic review and meta-analysis of the evidence on pediatric obesity and bariatric surgery. Included studies evaluated laparoscopic adjustable gastric banding (LAGB) (n=8 studies; 352 patients), Roux-en-Y gastric bypass (RYGB) (n=6 studies; 131 patients), and other bariatric procedures (n=5 studies; 158 patients). The average patient age was 16.8 years (range, 9-21 years). Meta-analyses of BMI reductions at longest follow-up indicated sustained and clinically significant BMI reductions for both LAGB and RYGB. Comorbidity resolution was infrequently reported, but surgery appeared to resolve some conditions such as diabetes and hypertension. For LAGB, band slippage and micronutrient deficiency were the most frequently reported complications, with sporadic cases of band erosion, port/tube dysfunction, hiatal hernia, wound infection, and pouch dilation. For RYGB, more severe complications have been documented, such as pulmonary embolism, shock, intestinal obstruction, postoperative bleeding, staple line leak, and severe malnutrition.

Professional Societies/Organizations

American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): In a 2022 joint statement the indications for metabolic and bariatric surgery, the ASMBS and IFSO recommend consideration of metabolic and bariatric surgery (MBS) in children/adolescents with BMI > 120% of the 95th percentile (class II obesity) and major co-morbidity, or a BMI > 140% of the 95th percentile (class III obesity) (Eisenberg, et al., 2022).

American Society for Metabolic and Bariatric Surgery (ASMBS) Pediatric Committee:

Based on a systematic review of the literature, ASMBS (2018) updated the 2012 Pediatric Metabolic and Bariatric Surgery (MBS) guidelines. The Society states that MBS is a proven, effective treatment for severe obesity in adolescents and should be considered standard of care. The World Health Organization's (WHO) defines adolescents as age 10-19 years. However, ASMBS states that younger children who meet the other criteria could be considered when benefit outweighs risk. The review included meta-analyses, randomized controlled trials, cohort studies, case reports, and expert opinions. The 2018 guidelines included the following:

- Vertical Sleeve Gastrectomy (VSG) has become the most used and most recommended operation in adolescents with severe obesity due to the near equivalent weight loss to the Roux-en-Y Gastric Bypass (RYGB) as well as fewer reoperations, better iron absorption, and near equivalent effect on comorbidities. However, given the more extensive long-term data available for RYGB, the Society recommends the use of either RYGB or VSG in adolescents. Long term outcomes of GERD after VSG are still not well understood.
- There are no data that the number of weight loss attempts correlates with success after metabolic and bariatric surgery. Compliance with a multidisciplinary preoperative program may improve outcomes after MBS but prior attempts at weight loss should be removed as a barrier to definitive treatment for obesity.
- Use of the most up to date definitions of childhood obesity: a) BMI cut offs of 35 kg/m² or 120% of the 95th percentile with a co-morbidity or b) BMI > 40kg/m² or 140% of the 95th percentile without comorbidity (whichever is less). Requiring adolescents with a BMI over 40 to have comorbidity puts children at a significant disadvantage to attaining a healthy weight. Earlier surgical intervention (at a BMI less than 45 kg/m²) can allow adolescents to reach a normal weight and avoid lifelong medication therapy and end organ damage from co-morbidities. ASMBS stated that there is no data to suggest that a youth's puberty status as measured by Tanner staging, or linear growth, as measured by height, are adversely affected by MBS.
- Certain co-morbidities should be considered in adolescents, specifically the psychosocial burden of obesity, the orthopedic diseases specific to children, GERD, and cardiac risk factors. Given the poor outcomes of medical therapies for type 2 diabetes in children, these comorbidities may be considered an indication for MBS in younger adolescents or those with lower obesity percentiles.

- Regarding when to refer the patient, ASMBS states that since MBS results in better weight loss and resolution of comorbidities in adolescents at lower BMIs with fewer comorbidities, referrals should occur early, as soon as a child is recognized to suffer from severe obesity disease (BMI > 120% of the 95th percentile or BMI of 35). Prior weight loss attempts, tanner stage and bone age should not be considered when referring patients for bariatric surgery.

Contraindications for adolescent MBS include:

- a medically correctable cause of obesity
- an ongoing substance abuse problem (within the preceding year)
- a medical, psychiatric, psychosocial, or cognitive condition that prevents adherence to postoperative dietary and medication regimens
- current or planned pregnancy within 12 to 18 months of the procedure (Pratt, et al., 2018)

Endocrine Society Task Force: The 2017 Endocrine Society Task Force guidelines on pediatric obesity recommended that bariatric surgery be considered only under the following conditions:

- The patient has attained Tanner 4 or 5 pubertal development and final or near-final adult height.
- The patient has a BMI > 40 kg/m² and significant, severe comorbidities.
- Extreme obesity and co-morbidities persist despite compliance with a formal program of lifestyle modification, with or without pharmacotherapy.
- Psychological evaluation confirms the stability and competence of the family unit (psychological distress due to impaired quality of life from obesity may be present, but the patient does not have an underlying untreated psychiatric illness).
- Patient has access to an experienced surgeon in a pediatric bariatric surgery center of excellence providing the necessary infrastructure for patient care, including a team capable of long-term follow-up of the metabolic and psychosocial needs of the patient and family.
- The patient demonstrates the ability to adhere to the principles of healthy dietary and activity habits.

In the 2017 update of the guidelines, the Society placed more emphasis on contraindications in the use of bariatric surgery in growing children and immature teenagers. The Society noted that procedures should only be carried out in those mature pubertal individuals with severe comorbidities of obesity in the presence of a motivated and compliant patient and family and only in the hands of an experienced surgeon with a dedicated and experienced support team.

The Task Force recommended against bariatric surgery for preadolescent children, for pregnant or breastfeeding adolescents, and for those planning to become pregnant within two years of surgery and in any patient who has not mastered the principles of healthy dietary and activity habits; and/or has unresolved substance abuse, eating disorder, or untreated psychiatric disorder.

American Academy of Pediatrics (AAP): The American Academy of Pediatrics first Clinical Practice Guideline for the Evaluation and Treatment of Children and Adolescents With Obesity (Hampl, et al., 2023) made the following Key Action Statements:

- "Pediatricians and other pediatric health care providers (PHCPs) should measure height and weight, calculate body mass index (BMI), and assess BMI percentile using age- and sex-specific Centers for Disease Control (CDC) growth charts or growth charts for children with severe obesity at least annually for all children 2 to 18 years of age to screen for overweight (BMI ≥ 85th percentile to <95th percentile), obesity (BMI ≥ 95th percentile), and severe obesity (BMI ≥ 120% of the 95th percentile for age and sex).
- Pediatricians and other PHCPs should evaluate children 2 to 18 years of age with overweight (BMI ≥ 85th percentile to <95th percentile) and obesity (BMI ≥ 95th percentile) for obesity related comorbidities by using a comprehensive patient history,

mental and behavioral health screening, social determinants of health (SDoH) evaluation, physical examination, and diagnostic studies.

- In children 10 years and older, pediatricians and other PHCPs should evaluate for lipid abnormalities, abnormal glucose metabolism, and abnormal liver function in children and adolescents with obesity (BMI $\geq$ 95th percentile) and for lipid abnormalities in children and adolescents with overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile).
- Pediatricians and other PHCPs should treat children and adolescents for overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile) or obesity (BMI $\geq$ 95th percentile) and comorbidities concurrently.
- Pediatricians and other PHCPs should evaluate for dyslipidemia by obtaining a fasting lipid panel in children 10 years and older with overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile) and obesity (BMI $\geq$ 95th percentile) and may evaluate for dyslipidemia in children 2 through 9 years of age with obesity.
- Pediatricians and other PHCPs should evaluate for prediabetes and/or diabetes mellitus with fasting plasma glucose, 2-hour plasma glucose after 75-gram oral glucose tolerance test (OGTT), or glycosylated hemoglobin (HbA1c).
- Pediatricians and other PHCPs should evaluate for nonalcoholic fatty liver disease (NAFLD) by obtaining an alanine transaminase (ALT) test.
- Pediatricians and other PHCPs should evaluate for hypertension by measuring blood pressure at every visit starting at three years of age in children and adolescents with overweight (BMI $\geq$ 85 to $<$ 95th percentile) and obesity (BMI $\geq$ 95th percentile).
- Pediatricians and other PHCPs should treat overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile) and obesity (BMI $\geq$ 95th percentile) in children and adolescents, following the principles of the medical home and the chronic care model, using a family-centered and nonstigmatizing approach that acknowledges obesity's biologic, social, and structural drivers.
- Pediatricians and other PHCPs should use motivational interviewing (MI) to engage patients and families in treating overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile) and obesity (BMI $\geq$ 95th percentile).
- Pediatricians and other PHCPs should provide or refer children six years and older (Grade B) and may provide or refer children 2 through 5 years of age (Grade C) with overweight (BMI $\geq$ 85th percentile to $<$ 95th percentile) and obesity (BMI $\geq$ 95th percentile) to intensive health behavior and lifestyle treatment. Health behavior and lifestyle treatment is more effective with greater contact hours; the most effective treatment includes 26 or more hours of face-to-face, family-based, multicomponent treatment over a 3 to 12-month period.
- Pediatricians and other PHCPs should offer adolescents 12 years and older with obesity (BMI $\geq$ 95th percentile) weight loss pharmacotherapy, according to medication indications, risks, and benefits, as an adjunct to health behavior and lifestyle treatment.
- Pediatricians and other PHCPs should offer referral for adolescents 13 years and older with severe obesity (BMI $\geq$ 120% of the 95th percentile for age and sex) for evaluation for metabolic and bariatric surgery to local or regional comprehensive multidisciplinary pediatric metabolic and bariatric surgery centers."

A 2019 policy statement issued by the American Academy of Pediatrics (Armstrong et al., 2019) makes the following Practice-Level Recommendations:

- "Recognize that severe obesity (BMI $\geq$ 35 or $\geq$ 120% of the 95th percentile for age and sex, whichever is lower) places the adolescent at higher risk for liver disease, type 2 diabetes mellitus, dyslipidemias, sleep apnea, orthopedic complications, and mental health conditions even when compared with milder degrees of obesity.
- Seek high-quality multidisciplinary centers that are experienced in assessing risks and benefits of various treatments for youth with severe obesity, including bariatric surgery, and provide referrals to where such programs are available.

- Understand the efficacy, risks, benefits, and long-term health implications of the common metabolic and bariatric surgery procedures so that pediatricians can effectively help in family medical decision-making concerning surgical options to manage severe obesity.
- Identify pediatric patients with severe obesity who meet criteria for surgery (Table 1), and provide timely referrals to comprehensive, multidisciplinary, pediatric-focused metabolic and bariatric surgery programs.
- Coordinate pre- and postoperative care with the patient, family, and multidisciplinary, anesthesia, and surgical teams.
- Monitor patients postoperatively for micronutrient deficiencies and consider providing iron, folate, and vitamin B12 supplementation as needed.
- Monitor patients postoperatively for risk-taking behavior and mental health problems.”

The AAP went on to make System-Level recommendations for pediatricians to advocate for increased access for pediatric patients of all racial, ethnic, and socioeconomic backgrounds to multidisciplinary programs that provide high-quality pediatric metabolic and bariatric surgery.

North American Society for Pediatric Gastroenterology, Hepatology and Nutrition

(NASPGHAN): In their 2017 Clinical Practice Guideline Summary: Diagnosis and Treatment of Nonalcoholic Fatty Liver Disease in Children, NASPGHAN stated that bariatric or weight loss surgery can lead to clinically meaningful weight loss in severely obese adolescents (minimum body mass index (BMI) of ≥ 35 kg/m². Within one to two years, this reduction in BMI results in improvement or resolution of many obesity-related comorbid conditions, including dyslipidemia, high blood pressure, insulin resistance, diabetes, and sleep apnea. These conditions are often associated with nonalcoholic fatty liver disease (NAFLD). Because studies in adults have suggested up to 89% of nonalcoholic steatohepatitis (NASH) resolution, it has been proposed as a criterion for adolescent weight loss surgery. NASPGHAN recommends that bariatric surgery be considered for selected adolescents with BMI ≥ 35 kg/m², who have noncirrhotic NAFLD and other serious comorbidities (Vos, et al., 2017).

Systematic Reviews on Bariatric Surgery

Agency for Healthcare Research and Quality (AHRQ) Evidence Report: AHRQ (2018) conducted a technology assessment on short- and long-term outcomes of bariatric surgery in the Medicare population. Studies that included patients age ≥ 65 years were included. Of the 126 eligible studies, 83 described safety and efficacy of bariatric surgery, and 43 described predictors of body weight loss or absolute body weight following surgery. No randomized controlled trials including Medicare-eligible patients were found. There were few direct (head-to-head) comparisons between different surgical procedures with sufficient evidence in nonrandomized studies and none for endoscopic procedures. Studies were primarily observational in design and very few utilized an appropriate design and/or analytical approach that could yield unbiased estimates of causal treatment effects including weight loss and non-weight-loss outcomes. AHRQ stated that “bariatric surgery overall, and in particular the procedures of Roux-en-Y gastric bypass, sleeve gastrectomy, and adjustable gastric banding, resulted in improvements in weight loss outcomes beyond one year after surgery and Roux-en-Y gastric bypass performed better when compared to sleeve gastrectomy or adjustable gastric banding for metabolic, cardiovascular outcomes, renal function outcomes and for postoperative complications; Roux-en-Y gastric bypass also performed better for weight loss outcomes”. According to AHRQ, there are large gaps in regard to comparisons of individual bariatric surgical procedures to each other, and very limited evidence in regard to patient-centered outcomes following including quality of life. (Panagiotou, et al. 2018)

Cochrane Reviews: A systematic review and meta-analysis by Colquitt et al. (2014) evaluated surgical procedures for weight loss in adults. The review included 22 RCTs (n=1798 participants), with sample sizes ranging from 15-250. Most studies followed participants for 12, 24 or 36

months; the longest follow-up was 10 years. A total of seven RCTs compared surgery to non-surgical interventions and found benefits of surgery on measures of weight change at one to two years follow-up. Improvements for some aspects of health-related quality of life (n=2 RCTs) and diabetes (n=5 RCTs) were also found. The overall quality of the evidence was moderate. Five studies reported data on mortality, no deaths occurred. Serious adverse events, reported in four studies, ranged from 0% to 37% in the surgery groups and 0% to 25% in the no surgery groups. Between 2% and 13% of participants required reoperations in the five studies that reported these data. Outcomes were found to be similar between Roux-en-Y gastric bypass (RYGB) and sleeve gastrectomy, with both procedures having better outcomes than adjustable gastric banding. For people with very high BMI, biliopancreatic diversion with duodenal switch (BPD/DS) resulted in greater weight loss than RYGB. Based on one small RCT, duodenojejunal bypass with sleeve gastrectomy and laparoscopic RYGB had similar outcomes. Based on one trial, sleeve gastrectomy led to better weight-loss outcomes than adjustable gastric banding after three years follow-up. Weight-related outcomes were similar between laparoscopic gastric imbrication and laparoscopic sleeve gastrectomy in one trial. Across all studies adverse event rates and reoperation rates were generally poorly reported. It was noted that due to the small number of studies included in the meta-analyses, only limited conclusions can be drawn from them. Also, the long-term effects of surgery remain unclear because the follow-up period in most trials was only one or two years.

O'Brien and Colleagues: O'Brien et al. (Aug 2006) conducted a systematic review of studies evaluating medium-term weight loss after bariatric surgical procedures. Procedures examined in the 43 studies included laparoscopic adjustable gastric banding (LAGB) (n=18), biliopancreatic diversion (BPD) with and without duodenal switch (DS) (n=7), and Roux-en-Y gastric bypass (RYGB) (n=18). Of the LAGB reports, 12 provided data on the LAP-BAND, five on the Obtech® band (Ethicon Endo-Surgery, Inc., Cincinnati, OH), and one study included both devices. Pooled data for all procedures showed a mean excess weight loss in the range of 54–67% with no evidence of loss of effect at 10 years. It was concluded that all current bariatric operations lead to major weight loss in the medium term. BPD and banded RYGBP appear to be more effective than both RYGBP and LAGB, which are equal in the medium term (O'Brien, et al., Aug 2006).

Bariatric Surgery Impact on Health Outcomes

The potential benefits of bariatric surgery on health outcomes include the following:

- The increase in reported morbidity associated with obesity is thought to be mediated primarily by insulin resistance, diabetes, hypertension and lipid disturbances (Sjöström, et al., 2004).
- Diet therapy alone in the absence of surgery is relatively ineffective in treating obesity over the long term (Buchwald, et al., 2004).
- Severely obese patients who undergo bariatric surgery achieve greater short-, intermediate- and long-term (i.e., 10 years) weight loss, more physical activity and lower energy intake than severely obese patients treated with conventional medical interventions, such as very low-calorie diets and pharmacotherapy (Sjöström, et al., 2004; Buchwald, et al., 2004).
- Intermediate- and long-term (i.e., 10 years) incidence rates of recovery from risk factors such as diabetes, hypertriglyceridemia, low levels of high-density lipoprotein cholesterol, hypertension, hyperlipidemia and hyperuricemia are more favorable in surgically-treated patients than in nonsurgical, severely obese patients (Sjöström, et al., 2004; Buchwald, et al., 2004).
- Bariatric surgery reverses, eliminates or significantly improves risk factors of diabetes, hyperlipidemia, hypertension and obstructive sleep apnea (Buchwald, et al., 2004).
- Severely obese diabetic individuals treated with bariatric surgery have shown an 80% reduction in mortality (Sjöström, et al., 2004).
- Weight-loss surgery has been reported to reduce the relative risk of death by 89% with an absolute mortality reduction of 5.49% (Christou, et al., 2004).

- Gastric bypass has been reported to result in more favorable overall health outcomes (i.e., weight loss, risk factor recovery/reduction) relative to other surgical interventions, such as banding procedures (Buchwald, et al., 2004).

Buchwald et al. (2009) performed a meta-analysis of 19 studies with 43 treatment arms and 11,175 patients to determine the impact of bariatric surgery on type 2 diabetes mellitus in association with the procedure performed and the weight reduction achieved. The included studies reported both weight loss and diabetes resolution separately for the 4070 diabetic patients. At baseline, the mean age was 40.2 years with a mean BMI of 47.9 kg/m², and 10.5% had previous bariatric procedures. Meta-analysis of weight loss was 38.5 kg or 55.9% excess weight loss (EWL). Overall, 78.1% of diabetic patients had complete resolution, and diabetes was improved or resolved in 86.6% of patients. Weight loss and diabetes resolution were greatest for patients undergoing biliopancreatic diversion with duodenal switch (BPD/DS), followed by gastric bypass, and least for banding procedures. In the studies reporting only diabetic patients, 82% of patients had resolution of the clinical and laboratory manifestations of diabetes in the first two years after surgery, and 62% remained free of diabetes more than two years after surgery (80% and 75% for the total group) (Buchwald, et al., 2009).

Sjöström et al. (2007) conducted a prospective, matched, surgical interventional trial, referred to as the Swedish Obese Subjects study, which involved 4047 obese subjects. Of these subjects, 2010 underwent bariatric surgery (surgery group) and 2037 received conventional treatment (matched control group). A total of 376 subjects underwent nonadjustable or adjustable banding, 1369 underwent vertical banded gastroplasty, and 265 received gastric bypass. For adjustable banding, the Swedish adjustable Gastric Band was used. Outcome measures included weight change and overall mortality during an average of 10.9 years of follow-up. Vital status was known for all but three subjects at the time of the analysis. In the surgery group, participation rates of subjects at follow-up examination at two, 10, and 15 years were 94%, 84%, and 66%, respectively. Corresponding rates for subjects in the control group were 83%, 75% and 87%. The average weight change in control subjects was less than +/-2% during the period of up to 15 years during which weights were recorded. At 10 years, the weight losses from baseline were stabilized at 25% after gastric bypass, 16% after vertical-banded gastroplasty, and 14% after banding. There were 129 deaths in the control group and 101 deaths in the surgery group. The most common causes of death were myocardial infarction which occurred in 25 subjects in the control group and 13 subjects in the surgery group. Cancer was the most common cause of death from noncardiovascular causes (control group [n=47]; surgery group [n=29]). The main limitation of the study is the lack of randomization; however it is questionable whether randomization is feasible in bariatric surgery trials designed to study mortality. Although study results indicated that bariatric surgery is associated with a reduction in overall mortality, it is undetermined whether the favorable survival effect is explained by weight loss or by other beneficial effects of the surgical procedure (Sjöström, et al., 2007).

The National Institutes of Diabetes and Digestion and Kidney Disease (NIDDK) sponsored the Longitudinal Assessment of Bariatric Surgery (LABS) program. This program involves six clinical centers that have expertise in relevant fields including bariatric surgery, obesity research, endocrinology, epidemiology, and outcomes research. The purpose of the LABS program is to plan and conduct studies that will analyze the risks and benefits of bariatric surgery and its impact on the health and well-being on patients with severe obesity as well as to identify the types of patients who are most likely to benefit from bariatric surgery (NIDDK, 2020). The seven year reported results included:

- Weight loss of an average of 28.4% of body weight was experienced after gastric bypass surgery and 14.9% after laparoscopic gastric band surgery

- Most participants maintained the weight loss. Gastric bypass surgery patients that regained weight at three to seven years post-surgery, regained an average of 3.9% of their body weight, while gastric band surgery patients regained an average of 1.4%.
- High cholesterol was less common after both gastric bypass and gastric band surgery
- Diabetes and high blood pressure were less common after gastric bypass surgery with a small number of patients experiencing a reoccurrence of diabetes over time
- Alcohol use disorders increased after gastric bypass surgery but not after gastric band surgery
- Pain and physical function improved after bariatric surgery

Reoperation/Revisional Bariatric Surgery

Revisional bariatric surgery (RBS) includes a variety of abdominal operations performed on patients who have complications, weight loss failure and/or weight regain, or poor resolution of comorbidities after bariatric surgery for severe obesity. Approximately 10%–17% of patients who undergo bariatric surgery experience complications and approximately 7% undergo RBS. Previous bariatric operative approaches may fail for functional or technical reasons, causing inadequate weight loss or severe complications. The literature indicates that re-operative procedures may be required for severe gastroesophageal reflux disease (GERD), staple line breakdown, esophageal mobility issues, metabolic complications of jejunoileal bypass, obstruction, alkaline or acid reflux esophagitis, band erosion, stricture, anastomotic ulcer, or gastric pouch dilatation following gastric restrictive procedures. Following Roux-en-Y the most frequent complications concern the gastrojejunal anastomosis with anastomotic stenoses and marginal ulcers being the most common complications (Ellsmere, 2025; Lim, et al., 2018; Fringeli, et al., 2005).

Weight loss and comorbidity resolution following a bariatric operation is typically rapid in the first year. After this initial period of success, there is a gradual increase in weight and a new balance is reached at a somewhat higher threshold over the next two to three years, but at a level that still contributes to good resolution of comorbidity and improved quality of life. Some patients do not achieve satisfactory weight loss after the primary operation. In others, weight regain occurs with return of comorbid conditions after initial success, requiring re-evaluation for additional surgical intervention. Such failure may be the result of a leak in the band, a large stomach pouch, or a gastrogastic fistula that can be corrected with a reoperation. Although noncompliance with diet and exercise regimens plays a role, weight gain and recurrence of comorbid conditions may occur despite patient compliance due to individual biology. In these cases, a more aggressive bariatric procedure may be indicated to provide effective therapy (Sudan, et al., 2015).

There are three main categories of revisional bariatric surgery (Brethauer et al., 2014):

- Conversion: a change from one type of procedure to a different type.
- Corrective: a procedure that attempts to remedy complications or incomplete treatment effects of a previous bariatric operation.
- Reversal: a procedure that restores the original anatomy.

The type of revisional bariatric surgery procedure performed is determined by factors such as type of primary procedure, patient anatomy, medical history and indications for RBS. Weight loss and comorbidity outcomes of laparoscopic adjustable gastric banding (LAGB) patients converted to Roux-en-Y Gastric bypass (RYGB), sleeve gastrectomy (SG), and biliopancreatic diversion with duodenal switch (BPD/DS) have been reported to have results similar to the outcomes for primary bariatric procedures (Handojo, et al., 2023). Conversion to RYGB or BPD/DS has been performed for patients who need additional therapy for weight loss or regain weight after SG (Sharples, et al., 2017; Brethauer et al., 2014). Less commonly performed is the revision of a gastric bypass via placement of an adjustable gastric band. This revision, referred to as “[band over bypass](#)” or “salvage banding”, is a less invasive option to control pouch size compared to the other limited

options such as a conversion to a longer limb bypass procedure with the associated adverse effect of severe malnutrition. Further weight loss after salvage banding has been reported in the literature as varying from 55.9%–94.2% excess body mass index loss (EBMIL) after 12–42 months of follow-up (Vijgen, et al., 2012). Similarly, banded sleeve gastrectomy or “[band over sleeve](#)” has been proposed as an option to counteract sleeve dilatation and ameliorate weight loss over time (Karcz, et al., 2014). There is insufficient evidence in the published peer-reviewed medical literature to support the safety and effectiveness of band over bypass or band over sleeve procedure.

Reoperation by surgical reversal (i.e., “takedown”) or surgical revision of bariatric surgery is generally considered to be medically necessary at any time following the original surgery when the patient experiences complications from the original surgery, such as stricture, obstruction, pouch dilatation, erosion, or band slippage.

Bariatric Surgery for the Treatment of Other Conditions

Type 2 Diabetes Mellitus: Bariatric surgery is currently being evaluated as a treatment and potential cure for Type 2 Diabetes Mellitus (T2DM). Studies reporting the results of bariatric surgery on T2DM have primarily included morbidly obese patients (i.e., a BMI ≥ 40 or a BMI 35–39.9 with a clinically significant obesity-related comorbidity) and have demonstrated that obese diabetic patients who undergo bariatric surgery experience complete T2DM remission. As an example, Roux-en-Y gastric bypass (RYGB) reduces the storage capacity of the stomach, induces malabsorption, and causes hormonal changes which may lead to improvement in diabetic symptoms. Sleeve gastrectomy (SG) is surgical reduction of the stomach only, which is proposed to improve T2DM by inducing weight loss, some hormonal changes, and modification of gastrointestinal motility, bile acids, and gut microbiota. Few studies have investigated the safety and efficacy of bariatric surgery, also referred to as metabolic surgery, in patients with a BMI < 35 (class I obesity). Although bariatric surgery has also been proposed as a potential treatment for type 1 diabetes mellitus (T1DM), the published peer-reviewed medical literature contains limited evidence regarding T1DM.

Literature Review-Type 2 Diabetes Mellitus:

Courcoulas et al. (2024) reported results of a pooled analysis of four randomized trials (ARMMS-T2D) to evaluate the long-term impact of bariatric surgery versus lifestyle management on type 2 diabetes. A total of 305 participants were available for long-term follow-up, of whom 262 (86%) enrolled, including 96 assigned to medical/lifestyle intervention and 166 to bariatric surgery (Roux-en-Y gastric bypass, sleeve gastrectomy, or adjustable gastric banding). The median follow-up was 11 years. Outcomes were reported at 7 years (primary endpoint) and up to 12 years. Key inclusion criteria across trials included age 18–65 years, body mass index (BMI) 27–45, and a diagnosis of type 2 diabetes. The primary outcome was change in hemoglobin A1c (HbA1c) from baseline to 7 years, with secondary outcomes including diabetes remission (HbA1c $< 6.5\%$ without medications), glycemic control thresholds, weight change, cardiometabolic risk factors, medication use, and adverse events. At 7 years, HbA1c decreased by 0.2% (95% CI, -0.5% to 0.2%) in the medical/lifestyle group and by 1.6% (95% CI, -1.8% to -1.3%) in the bariatric surgery group, with a between-group difference of -1.4% (95% CI, -1.8% to -1.0% ; $p < 0.001$); this difference remained significant at 12 years (-1.1% ; 95% CI, -1.7% to -0.5% ; $p=0.002$). Diabetes remission at 7 years occurred in 6.2% of the medical/lifestyle group versus 18.2% of the bariatric surgery group ($p=0.02$), and at 12 years in 0.0% versus 12.7%, respectively ($p < 0.001$). Additional outcomes favored bariatric surgery, including a higher proportion achieving HbA1c $< 7.0\%$ (54.1% vs 26.7%; $p < 0.001$), greater weight loss (-19.9% vs -8.3% ; $p < 0.001$), and reduced diabetes medication and insulin use ($p < 0.001$), while differences in fasting glucose, blood pressure, and LDL cholesterol were not statistically significant at 7 years ($p \geq 0.05$ for these outcomes). Adverse events showed no differences in mortality or

major cardiovascular events; however, anemia, fractures, and gastrointestinal events were more frequent in the surgical group. Study limitations include the open-label design, heterogeneity across pooled trials (including differing protocols and interventions), crossover from medical/lifestyle to surgery (25%), potential selection and attrition bias, nonuniform assessment of some outcomes, assumptions regarding missing data, limited power for subgroup and safety endpoints, and evolving surgical and medical therapies over time, all of which may affect internal validity and generalizability.

Schauer et al. (2014) published an RCT (n=150) of obese patients with uncontrolled T2DM randomized to receive either intensive medical therapy alone (n=40) or intensive medical therapy plus Roux-en-Y gastric bypass (n=48) or sleeve gastrectomy (n=49). The Surgical Treatment and Medications Potentially Eradicate Diabetes Efficiently (STAMPEDE) trial included patients between the ages of 20–60 years, with a glycated hemoglobin level > 7.0%, and a BMI of 27–43. The primary outcome was a glycated hemoglobin level of 6.0% or less, with or without the use of diabetes medications. A total of 91% of the patients completed 36 months of follow-up. At three years, the criterion for the primary end point was met by 5% of the patients in the medical-therapy group, compared to 38% of those in the gastric-bypass group ($p < 0.001$) and 24% of those in the sleeve-gastrectomy group ($p=0.01$). Study results indicated that for obese patients with uncontrolled type 2 diabetes, bariatric surgery was associated with improved glycemic control and weight reduction compared to intensive medical therapy alone. It was noted that limitations to the study included an inadequate sample size and duration to detect differences in the incidence of diabetes complications, such as myocardial infarction, stroke, or death. The study protocol specifies further follow-up at years for all patients, which should allow additional assessment of even longer-term efficacy (Schauer, et al., 2014). Schauer et al. (2017) published five-year outcomes for the STAMPEDE trial. At five years, the criterion for the primary end point was met by 2/38 patients (5%) who received medical therapy alone versus 14/49 patients (29%) who underwent gastric bypass ($p=0.01$), and 11/47 patients (23%) who underwent sleeve gastrectomy ($p=0.03$). Patients who underwent surgical procedures had a greater mean percentage reduction from baseline in glycated hemoglobin level than did patients who received medical therapy alone ($p=0.003$). A single major late surgical complication (i.e., reoperation) was reported.

Maglione et al. (2013) performed an Agency for Healthcare Research and Quality (AHRQ) review of the evidence (n=24 studies) on efficacy, safety, and comparative effectiveness of various types of bariatric surgery for treating adult patients with a body mass index (BMI) of 30.0 to 34.9 kg/m² and diabetes or impaired glucose tolerance (IGT). The review compared effectiveness of surgery versus nonsurgical interventions in this population. Included studies were primarily observational (n=19 studies). Two trials comparing different procedures (n=2 studies), and three trials comparing surgical versus nonsurgical interventions were also included. Studies for the analysis had to report on laparoscopic adjustable gastric banding (LAGB), Roux-en-Y gastric bypass (RYGB), biliopancreatic diversion with duodenal switch (BPD/DS), sleeve gastrectomy, or nonsurgical treatment, and had to include patients with a BMI of at least 30 kg/m² but less than 35 kg/m² with diabetes or IGT. Excluded were nonsurgical studies already included in previous systematic reviews or with less than one year follow-up; those with no outcomes of efficacy, effectiveness, or safety/adverse events; and studies with a sample size of less than three. Outcomes measured were weight and blood glucose levels. Based primarily on glucose control outcomes, moderate strength evidence of efficacy of bariatric surgery in treating diabetes in patients with a BMI of at least 30 but less than 35 kg/m² in the short term was found. At one-year follow-up, surgery patients showed much greater weight loss than usually seen in studies of diet, exercise, or other behavioral interventions. The overall evidence was rated as moderate due to paucity of data. Observational data, which start as low strength evidence, were upgraded due to consistency of results regarding BMI and blood sugar. The strength of evidence of efficacy for RYGB, LAGB, and SG in treating diabetes and IGT in patients with a BMI of between 30 and 35 in

the short term (i.e., up to 2 years) was rated as moderate. For BPD, both the number of studies and their sample sizes are much lower; thus, the strength of evidence of efficacy for this procedure was rated low. Evidence on comparative effectiveness of surgical procedures is insufficient. The strength of evidence for short-term harms was low for all four surgical procedures and insufficient for long-term adverse events. It was concluded that the literature on bariatric surgery for diabetes or IGT patients with BMI of at least 30 kg/m² and less than 35 kg/m² has many limitations. There is minimal data on long-term efficacy and safety, as few studies of this target population have long-term follow-up. No evidence was found on major clinical endpoints such as all-cause mortality, cardiovascular mortality or morbidity, or peripheral arterial disease. The studies of bariatric surgery in this population have measured only intermediate or surrogate endpoints regarding glucose control. While control of glucose is certainly important, the available evidence from the diabetes literature indicates it may be premature to assume that controlling glucose to normal or near normal levels completely mitigates the risk of microvascular and macrovascular events. Thus, claims of a "cure" for diabetes based on glucose control within one or two years require longer term data before they can be substantiated.

Ikramuddin et al. (2013) conducted a multicenter unblinded randomized trial (n=120) to compare Roux-en-Y gastric bypass with lifestyle and intensive medical management (n=60) with intensive management alone (n=60). Subjects with a hemoglobin A1c (HbA1c) level of $\geq 8.0\%$, BMI 30.0–39.9, C peptide level of > 1.0 ng/mL, and type 2 diabetes for at least six months were included. The primary endpoint was a composite goal of HbA1c $< 7.0\%$, low-density lipoprotein cholesterol < 100 mg/dL, and systolic blood pressure < 130 mm Hg. Secondary outcome measures included weight loss, medication use, and adverse events. After 12 months of follow-up, 28 participants (49%) in the gastric bypass group and 11 (19%) in the lifestyle-medical management group achieved the primary end points ($p < 0.01$). Participants in the gastric bypass group required 3.0 fewer medications and lost 26.1% vs 7.9% of their initial body weight compared with the lifestyle-medical management group. There were 22 serious adverse events in the gastric bypass group, including a single cardiovascular event, and 15 in the lifestyle-medical management group. The gastric bypass group experienced more nutritional deficiency than the lifestyle-medical management group. Study limitations include the small patient population and short-term follow-up.

Lee et al. (2011) randomized 60 patients with T2DM, HbA1c $> 7.5\%$, c-peptide ≥ 1.0 , and a BMI > 25 and < 35 kg/m² to either gastric bypass (n=30) or sleeve gastrectomy (n=30) performed laparoscopically. The primary outcome was remission of diabetes defined as HbA1c $< 6.5\%$ and fasting glucose < 126 mg/dL on no diabetes medications at the one-year follow-up. Follow-up was 100% in both groups at one year. The average age of participants was 45 years, with an average BMI of 30 kg/m² (range 25–34), and an average HbA1c of 10.0%. The diabetes remission rate was higher in the RYGB group (93% versus 47%, $p=0.02$). The average reduction in HbA1c at one year was also higher in the RYGB group (4.2% versus 3.0%, $p < 0.001$). At the one-year follow-up, the average HbA1c was lower in the RYGB group (5.7% versus 7.2%, $p < 0.001$), as was the average fasting glucose level (99 versus 140, $p < 0.001$), the LDL-cholesterol (97 versus 137, $p<0.001$), and BMI (22.8 versus 24.4, $p=0.009$). This study is limited by the small number of participants and short-term follow-up.

Dixon et al. (2008) conducted an unblinded randomized controlled trial to determine if surgically induced weight loss resulted in better glycemic control and less need for diabetes medications than conventional approaches to weight loss and diabetes control. This study included 60 obese patients with a BMI range of 30–40, recently diagnosed (i.e., < 2 years) type 2 diabetes, and with no evidence of renal impairment or diabetic retinopathy. The surgical group (n=30) underwent laparoscopic adjustable gastric banding (LAGB) along with conventional diabetes care and the conventional-therapy group received diabetes therapy with a focus on weight loss by lifestyle change. The primary outcome measure was remission of type 2 diabetes demonstrated by a

fasting glucose level <126 mg/dL [7.0 mmol/L] and glycated hemoglobin [HbA1c] value < 6.2% while taking no glycemic therapy. Secondary measures included weight and components of the metabolic syndrome. Of the 60 patients enrolled, 55 (92%) completed the two-year follow-up. Remission of type 2 diabetes was achieved by 22 (73%) in the surgical group (n=30) and four (13%) in the conventional-therapy group (p < 0.001). Relative risk of remission for the surgical group was 5.5 (95% confidence interval, 2.2-14.0). The surgical group achieved a mean 20% body weight loss at two years compared to a 1.4% body weight loss among the conventional-therapy group (p < 0.001). The reduction in metabolic syndrome was significant in the surgical group (p < 0.001), but not in the conventional-therapy group (p=0.23). It was noted that although study results suggested that patients who received surgical intervention were more likely to achieve remission of type 2 diabetes through greater weight loss, these results need to be confirmed in a larger study with a more diverse population and an assessment of long-term efficacy.

Case series with patient populations ranging from 18–42 and follow-up periods of 12–24 months have also demonstrated promising results, with reversal rates of type 2 diabetes mellitus ranging from 62%–88%. However, these studies are limited by their design, small patient populations and short-term follow-ups (Gianos, et al., 2012; Abbatini, et al., 2012; Huang, et al., 2011; Boza, et al., 2011; Serrot, et al., 2011).

Gastroesophageal reflux disease (GERD): Gastroesophageal reflux disease is a condition that occurs when the contents of the stomach come up into the esophagus. Symptoms can be mild or severe and range from heartburn, regurgitation, dysphagia, chest pain, water brash, globus sensation, odynophagia, extraesophageal symptoms (e.g. chronic cough, hoarseness, wheezing) and nausea (Kahrilas, 2024). Medical management includes lifestyle and dietary modifications, and pharmacologic therapy (Kahrilas, 2025). Refractory GERD is diagnosed when patients continue to have reflux symptoms or endoscopic evidence of esophagitis despite pharmacologic treatment. GERD is associated with obesity. For morbidly obese patients with GERD, the reported surgical treatment of choice is Roux-en-Y gastric bypass (RYGB) (Schwaitzberg, 2023). Bariatric surgery has been proposed to treat refractory gastroesophageal reflux disease in the absence of morbid obesity. There is insufficient evidence in the peer reviewed literature demonstrating the safety and efficacy of bariatric surgery for the treatment of GERD when there is an absence of morbid obesity.

Gastroparesis: Gastroparesis is when the stomach is delayed in emptying solid contents when there is no mechanical obstruction. Symptoms include nausea, vomiting, early satiety, belching, bloating and/or upper abdominal pain. Initial therapy includes dietary modification, hydration, glycemic control and pharmacologic therapy. Patients with refractory symptoms may be treated with a jejunostomy and venting gastrostomy tube (Camilleri, 2025). Bariatric surgery has also been proposed to treat gastroparesis in the absence of morbid obesity. Studies in the peer reviewed literature investigating the safety and efficacy of bariatric surgery for this condition are lacking.

Professional Societies/Organizations

American Diabetes Association (ADA): The 2025 American Diabetes Association Standards of Medical Care in Diabetes discusses metabolic surgery for the treatment of diabetes. In the discussion of obesity management, the ADA recommendations are assigned ratings of **A**, **B**, or **C**, depending on the quality of the evidence in support of the recommendation with expert opinion **E** as a separate category for recommendations in which there is no evidence from clinical trials, clinical trials may be impractical, or there is conflicting evidence. The ADA recommendations include the following:

- “Consider metabolic surgery as a weight and glycemic management approach in people with diabetes with BMI ≥ 30.0 kg/m² (or ≥ 27.5 kg/m² in Asian American individuals) who are otherwise good surgical candidates. **A**
- Metabolic surgery should be performed in high-volume centers with interprofessional teams knowledgeable about and experienced in managing obesity, diabetes, and gastrointestinal surgery. **E**
- People being considered for metabolic surgery should be evaluated for comorbid psychological conditions and social and situational circumstances that have the potential to interfere with surgery outcomes. **B**
- People who undergo metabolic surgery should receive long-term medical and behavioral support and routine micronutrient, nutritional, and metabolic status monitoring. **B**
- If post-metabolic surgery hypoglycemia is suspected, clinical evaluation should exclude other potential disorders contributing to hypoglycemia, and management should include education, medical nutrition therapy with a registered dietitian nutritionist experienced in post-metabolic surgery hypoglycemia, and medication treatment, as needed. **A** In individuals with post-metabolic surgery hypoglycemia, use continuous glucose monitoring to improve safety. **C**
- In people who undergo metabolic surgery, routinely screen for psychosocial and behavioral health changes and refer to a qualified behavioral health professional as needed. **C**
- Monitor individuals who have undergone metabolic surgery for insufficient weight loss or weight recurrence at least every 6–12 months. **E** In those who have insufficient weight loss or experience weight recurrence, assess for potential predisposing factors and, if appropriate, consider additional weight loss interventions (e.g., weight management pharmacotherapy). **C**

ADA noted that although metabolic surgery has been shown to improve the metabolic profiles of morbidly obese patients with type 1 diabetes, establishing the role of metabolic surgery in such patients will require larger and longer studies.

The ADA states that “health disparities adversely affect people who have systematically experienced greater obstacles to health based on their race or ethnicity, socioeconomic status, gender, disability, or other factors. Overwhelming research shows that these disparities may significantly affect health outcomes, including increasing the risk for obesity, diabetes and diabetes-related complications (ADA, 2025).”

In the discussion of children and adolescents, ADA stated that type 2 diabetes disproportionately impacts youth of ethnic and racial minorities and can occur in complex psychosocial and cultural environments, which may make it difficult to sustain healthy lifestyle changes and self-management behaviors. ADA stated that small retrospective studies and a nonrandomized study suggest that bariatric surgery may have similar benefits in obese adolescents with type 2 diabetes compared to outcomes in adults. However, no randomized trials have compared the safety and effectiveness of surgery to conventional treatment options. The 2025 ADA Standards of Medical Care in Diabetes for children and adolescents recommendations state:

- “metabolic surgery may be considered for the treatment of adolescents with type 2 diabetes who have class 2 obesity or higher (BMI > 35 kg/m² or $> 120\%$ of 95th percentile for age and sex, whichever is lower) and who have elevated A1C and/or serious comorbidities despite lifestyle and pharmacologic intervention.
- metabolic surgery should be performed only by an experienced surgeon working as part of a well-organized and engaged multidisciplinary team, including a surgeon, endocrinologist, registered dietitian nutritionist, behavioral health specialist, and nurse.”

American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): In a 2022 joint

statement on the indications for metabolic and bariatric surgery (Eisenberg, et al., 2022), the ASMBS and IFSO make the following recommendations:

- metabolic and bariatric surgery (MBS) is recommended for individuals with a body mass index (BMI) ≥ 35 kg/m², regardless of presence, absence, or severity of co-morbidities.
- BMI thresholds be adjusted in the Asian population such that a BMI ≥ 25 kg/m² suggests clinical obesity, and individuals with BMI ≥ 27.5 kg/m² should be offered MBS
- MBS is recommended in patients with type 2 diabetes (T2D) and BMI ≥ 30 kg/m²

American Association of Clinical Endocrinologists (AACE), The Obesity Society (TOS), and the American Society for Metabolic and Bariatric Surgery (ASMBS): The 2013 update (updated 2019) of the AACE/TOS/ASMBS practice guidelines for the perioperative nutritional, metabolic, and nonsurgical support of the bariatric surgery patient included the following recommendations:

- Patients with a BMI ≥ 40 kg/m² without coexisting medical problems and for whom bariatric surgery would not be associated with excessive risk should be eligible for one of the procedures.
- Patients with a BMI ≥ 35 kg/m² and one or more severe obesity-related co-morbidities, including type 2 diabetes (T2DM), high risk for T2D (insulin resistance, prediabetes, and/or metabolic syndrome), hypertension, hyperlipidemia, obstructive sleep apnea (OSA), obesity-hypoventilation syndrome (OHS), Pickwickian syndrome (a combination of OSA and OHS), nonalcoholic fatty liver disease (NAFLD) or nonalcoholic steatohepatitis (NASH), idiopathic intracranial hypertension, gastroesophageal reflux disease (GERD), venous stasis disease, severe urinary incontinence, debilitating arthritis, impaired mobility due to obesity, or considerably impaired quality of life, may also be offered a bariatric procedure.
- Patients with BMI of 30–34.9 kg/m² with diabetes or metabolic syndrome may also be offered a bariatric procedure. The current evidence is limited by the number of subjects studied and lack of long-term data demonstrating net benefit.
- There is insufficient evidence for recommending bariatric surgery specifically for glycemic control alone, lipid lowering alone or cardiovascular disease risk reduction alone, independent of BMI criteria. In their discussion, the Society stated that there were no compelling studies that supported recommending bariatric surgery for management of T2DM in the absence of obesity (BMI < 30 kg/m²).
- Preoperative weight loss or medical nutritional therapy may be used in selected cases to improve co-morbidities, reduce liver volume and/or help improve the technical aspects of the surgery.

Regarding the various procedures, the Societies stated that in general, laparoscopic bariatric procedures are preferred in order to lower early postoperative morbidity and mortality. Laparoscopic adjustable gastric banding (LAGB), laparoscopic sleeve gastrectomy (LSG), laparoscopic Roux-en-Y gastric bypass (RYGB), and laparoscopic biliopancreatic diversion (BPD), BPD/duodenal switch (BPD-DS), or related procedures are primary bariatric and metabolic procedures that may be performed in patients requiring weight loss and/or metabolic control (Mechanick, et al., 2013, updated 2020).

American Society for Metabolic and Bariatric Surgery (ASMBS): ASMBS published a 2018 updated position statement on class I obesity (e.g., BMI 30-35 kg/m²). The review included systematic reviews, randomized controlled trials, observational studies and retrospective reviews that investigated bariatric surgery for patients with a BMI < 35 kg/m². ASMBS' updated recommendations included the following:

- Current nonsurgical treatments for class I obesity are often ineffective at achieving major, long-term weight reduction and resolution of co-morbidities.

- For patients with BMI 30-35 kg/m² and obesity-related co-morbidities who do not achieve substantial, durable weight loss and co-morbidity improvement with reasonable nonsurgical methods, bariatric surgery should be offered as an option for suitable individuals. In this population, surgical intervention should be considered after failure of nonsurgical treatments.
- Particularly given the presence of high-quality data in patients with type 2 diabetes, bariatric and metabolic surgery should be strongly considered for patients with BMI 30 to 35 kg/m² and type 2 diabetes.
- The safety and efficacy of adjustable gastric banding (AGB), sleeve gastrectomy (SG), and Roux-en-Y gastric bypass (RYGB) in low-BMI patients appear to be similar to results in patients with severe obesity and well-tolerated, effective treatments.
- The best evidence for bariatric and metabolic surgery for patients with class I obesity and co-morbid conditions exists for patients in the 18 to 65 age group.

American Association of Clinical Endocrinologists (AACE)/The Obesity Society (TOS)/American Association of Clinical Endocrinologists (AACE):

The Societies updated their joint bariatric surgery guidelines in 2013. This guidance stated that patients with BMI of 30-34.9 kg/m² with diabetes or metabolic syndrome may also be offered a bariatric procedure although current evidence is limited by the number of subjects studied and lack of long-term data demonstrating net benefit. There is insufficient evidence for recommending a bariatric surgical procedure specifically for glycemic control alone, lipid lowering alone, or cardiovascular disease risk reduction alone, independent of BMI criteria (Mechanick, et al., 2013). Clinical practice guidelines issued by the AACE and American College of Endocrinology (ACE) in 2016 reaffirm these findings (Garvey, et al., 2016). In 2019, the Societies updated their guidance stating that there is an increasing body of evidence supporting a sustained improvement in glycemic control concomitant with reductions in diabetes medications in patients with BMI 30 to 34.9 kg/m²; however, the current evidence is limited by the number of subjects studied and lack of long-term data demonstrating net benefit. The authors noted that future studies will need to elucidate the differential impact of multiple current surgical treatments for efficacy and safety. (Mechanick, et al., 2020). Additional studies are needed to validate these findings.

International Diabetes Organizations: Recommendations from a 2016 joint position statement by the International Diabetes Organizations include the following (Rubino, et al., 2016):

- Metabolic surgery should be a recommended option to treat T2D in appropriate surgical candidates with class III obesity (BMI > 40 kg/m²), regardless of the level of glycemic control or complexity of glucose-lowering regimens, as well as in patients with class II obesity (BMI 35.0–39.9 kg/m²) with inadequately controlled hyperglycemia despite lifestyle and optimal medical therapy.
- Metabolic surgery should also be considered to be an option to treat T2D in patients with class I obesity (BMI 30.0–34.9 kg/m²) and inadequately controlled hyperglycemia despite optimal medical treatment by either oral or injectable medications (including insulin).
- All BMI thresholds should be reconsidered depending on the ancestry of the patient. For example, for patients of Asian descent, the BMI values above should be reduced by 2.5 kg/m².
- Metabolic surgery should be performed in high-volume centers with multidisciplinary teams that understand and are experienced in the management of diabetes and GI surgery.

It was further noted that “although additional studies are needed to further demonstrate long-term benefits, there is sufficient clinical and mechanistic evidence to support inclusion of metabolic surgery among antidiabetes interventions for people with T2D and obesity” (Rubino, et al., 2016).

Liver Biopsy or Upper Endoscopy

Routine Liver Biopsy at the Time of Bariatric Surgery

Steatotic (fatty) liver disease is a broad term defined as hepatic steatosis diagnosed on radiologic imaging or by liver biopsy. This is further subdivided by:

- Metabolic dysfunction-associated steatotic liver disease (MASLD; previously termed nonalcohol-associated fatty liver disease [NAFLD]): refers to the presence of liver steatosis (> 5 percent hepatic steatosis) without any other primary causes (e.g., drug-induced liver injury) and at least one metabolic risk factor (e.g., obesity, diabetes mellitus, dyslipidemia, or hypertension). These patients have minimal (i.e., < 20 g daily for females and < 30 g daily for males) to no alcohol consumption.
- MASLD with metabolic dysfunction-associated steatohepatitis (MASH): have identified histologic evidence of inflammation and hepatocellular injury, such as ballooning of hepatocytes, with or without fibrosis. This was previously known as nonalcoholic steatohepatitis (NASH).
- MASH cirrhosis: current or previous histologic evidence of MASH or history of MASLD with cirrhosis
- Metabolic dysfunction- and alcohol-associated liver disease (MetALD): with liver steatosis, at least one metabolic risk factor, and a history of moderate (but not heavy) alcohol use. Moderate amounts of alcohol are defined as 20 to 50 grams (g) daily (140 to 350 g per week) for females and 30 to 60 g daily (210 to 420 g per week) for males.

MASLD may progress to cirrhosis and is likely an important cause of cryptogenic cirrhosis. Most patients with MASLD/NAFLD are asymptomatic, although some may complain of fatigue, malaise, and vague right upper abdominal discomfort. MASLD/NAFLD often comes to attention because laboratory testing revealed elevated liver aminotransferases or hepatic steatosis that was detected incidentally on abdominal imaging. A definitive diagnosis of MASLD/NAFLD requires the following (Sheth and Chopra, 2025; Rinella, et al., 2023):

- demonstration of hepatic steatosis by imaging or biopsy
- presence of one out of five cardiometabolic criteria for adults
 - BMI ≥ 25 kg/m² (23 Asian decent) **OR** waist circumference (WC) > 94 cm male (M) 80 cm female (F) **OR** ethnicity adjusted equivalent
 - Fasting serum glucose ≥ 5.6 mmol/L (100 mg/dl) **OR** 2-hour post-load glucose levels ≥ 7.8 mmol/L (≥ 140 mg/dl) **OR** HbA1c $\geq 5.7\%$ (39 mmol/L) **OR** type 2 diabetes **OR** treatment for type 2 diabetes
 - Blood pressure $\geq 130/85$ mmHg **OR** specific antihypertensive drug treatment
 - Plasma triglycerides ≥ 1.70 mmol/L (150 mg/dl) **OR** lipid lowering treatment
 - Plasma HDL-cholesterol ≤ 1.0 mmol/L (40 mg/dl) (M) and ≤ 1.3 mmol/L (50 mg/dl) (F) **OR** lipid lowering treatment
- presence of one out of five cardiometabolic criteria for pediatric patients
 - BMI $\geq 85^{\text{th}}$ percentile for age/sex (BMI z score $\geq +1$) **OR** WC > 95th percentile **OR** ethnicity adjusted equivalent
 - Fasting serum glucose ≥ 5.6 mmol/L (≥ 100 mg/dl) **OR** serum glucose ≥ 11.1 mmol/L (≥ 200 mg/dl) **OR** 2-hour post-load glucose levels ≥ 7.8 mmol (140 mg/dl) **OR** HbA1c $\geq 5.7\%$ (39 mmol/L) **OR** already diagnosed/treated type 2 diabetes **OR** treatment for type 2 diabetes
 - Blood pressure age < 13 yr, BP $\geq 95^{\text{th}}$ percentile **OR** $\geq 130/80$ mmHg (whichever is lower); age ≥ 13 yr, 130/85 mmHg **OR** specific antihypertensive drug treatment
 - Plasma triglycerides age < 10 yr, ≥ 1.15 mmol/L (≥ 100 mg/dl); age ≥ 10 yr, ≥ 1.70 mmol/L (≥ 150 mg/dl) **OR** lipid lowering treatment
 - Plasma HDL-cholesterol ≤ 1.0 mmol/L (≤ 40 mg/dl) **OR** lipid lowering treatment

Literature Review

Dolce et al. (2009) presented a series of 108 patients undergoing bariatric surgery who had routine intraoperative liver biopsy. The aim of this study was to determine the relationship between the intraoperative liver appearance and the histopathologic findings during laparoscopic bariatric surgery. An intraoperative liver visual score was recorded according to the size, tan-speckling, and contour. The liver histologic findings were categorized into 3 groups: (1) normal; (2) bland steatosis; and (3) nonalcoholic steatohepatitis (NASH). The liver visual score was compared with the liver histologic findings. The prevalence of NASH was found to be 23% (n=25). Of the 25 patients with NASH, 12 (48%) had normal-appearing livers. Of the 50 normal-appearing livers, 12 (24%) had NASH and 14 (28%) had bland steatosis. The authors noted that the correlation between the general appearance of the liver and the presence of NASH is poor, limiting the sensitivity of selective liver biopsy.

Shalhub et al. (2004) analyzed prospective data on 242 patients who underwent open and laparoscopic Roux-en-Y gastric bypass (RYGB) to determine the role of routine liver biopsy in managing bariatric patients. The same pathologist graded all liver biopsies as mild, moderate or severe steatohepatitis. NASH was defined as steatohepatitis without alcoholic or viral hepatitis. Consecutive liver biopsies were compared to those liver biopsies selected because of grossly fatty livers. Selective liver biopsies were performed in 86 of the first 174 patients and routine liver biopsies were done in the remaining 68 consecutive patients. The two groups were reported to have to have similar findings of steatosis, but more patients were categorized as having moderate and severe NASH based on routine liver biopsy compared to selective biopsy ($p < 0.05$). Both groups had a similar prevalence of cirrhosis. There was no correlation found between BMI, abnormal liver tests, and the severity of NASH. Study results indicate that liver biopsy is the gold standard for diagnosing NASH. However, additional data from well-designed RCTs are needed to support the need for routine liver biopsy during bariatric surgical procedures.

Some surgeons support the use of concurrent routine liver biopsy in all patients undergoing bariatric surgery. Like prophylactic cholecystectomy, routine liver biopsy in the absence of clinical findings at the time of bariatric surgery continues to be debated. Just what role routine liver biopsy plays in patients undergoing bariatric surgery is not known. Impact on health outcomes has not been established through well-designed clinical trials. At this time, there is not sufficient evidence to support routine liver biopsy in patients undergoing bariatric surgery.

Professional Societies/Organizations

American Association for the Study of Liver Diseases (AASLD) and the European Association for the Study of the Liver (EASL), in collaboration with the Latin American Association for the Study of the Liver (ALEH): In 2023, the above professional societies recommended a new nomenclature inclusive of updated definitions for the conditions formerly encompassed by the term nonalcoholic fatty liver disease (NAFLD) (Rinella, et al., 2023). The societies conducted a global Delphi consensus process involving various national hepatology societies, endocrinology societies, and patient advocacy organizations, and representing 56 countries to produce the new nomenclature and a change in the definition. Kanwal et al. (2024) summarized key points for clinical application:

- metabolic dysfunction–associated steatotic liver disease (MASLD) replaces nonalcoholic fatty liver disease (NAFLD)
- metabolic dysfunction–associated steatohepatitis (MASH) replaces nonalcoholic steatohepatitis (NASH)
- Studies indicate a near complete overlap (99%) between the MASLD-defined population and the historical NAFLD-defined populations
- All recommendations in the AASLD Practice Guidance on the clinical assessment and management of NAFLD can be applied to patients with MASLD and MASH

American Association for the Study of Liver Diseases (AASLD): The 2023 updated practice guidance on the diagnosis and management of nonalcoholic fatty liver disease stated that metabolic abnormalities such as insulin resistance, dyslipidemia, central obesity, and hypertension precede the development of nonalcoholic fatty liver disease (NAFLD). The entire spectrum of obesity, ranging from overweight to obese and severely obese, is associated with NAFLD and disease progression. According to the guidelines, in patients with severe obesity undergoing bariatric surgery, > 95% will have NAFLD. There is also a very high prevalence of NAFLD in individuals with type 2 diabetes mellitus. The recommendations included the following:

- Liver biopsy should be considered in patients with NAFLD who are at increased risk of having steatohepatitis and/or advanced fibrosis.
- The presence of metabolic syndrome (MetS), NAFLD fibrosis score (NFS) or fibrosis-4 index (FIB-4), or liver stiffness measured by vibration controlled transient elastography (VCTE) or MR elastography (MRE) may be used for identifying patients who are at risk for steatohepatitis and advanced fibrosis.
- Liver biopsy should be considered in patients with suspected NAFLD in whom competing etiologies for hepatic steatosis and the presence and/or severity of coexisting chronic liver diseases cannot be excluded without a liver biopsy.
- Bariatric surgery can be considered in otherwise eligible obese individuals with NAFLD or NASH.
- It is premature to consider bariatric surgery as an established option to specifically treat nonalcoholic steatohepatitis (NASH).
- The type, safety, and efficacy of bariatric surgery in otherwise eligible obese individuals with well-compensated NASH cirrhosis due to NAFLD are not established. In otherwise eligible patients with compensated NASH or cryptogenic cirrhosis, bariatric surgery may be considered on a case-by-case basis by an experienced bariatric surgery program.

Hiatal Hernia Repair at the Time of Bariatric Surgery

Hiatal or hiatus hernia refers to the protrusion of an organ, typically the stomach, through the esophageal opening in the diaphragm into the chest. Hiatal hernia is often associated with obesity and gastroesophageal reflux disease (GERD) and its complications. Hiatal hernias are broadly divided into two main types, sliding and paraesophageal. However, the most comprehensive classification of hiatal hernia includes the following:

- Type I are sliding hiatal hernias, where the gastroesophageal junction migrates above the diaphragm. There is a widening of the muscular hiatal tunnel and circumferential laxity of the phrenoesophageal membrane, allowing a portion of the gastric cardia to herniate upward. The stomach remains in its usual longitudinal alignment and the fundus remains below the gastroesophageal junction.
- Type II are pure paraesophageal hernias (PEH). The gastroesophageal junction remains in its normal anatomic position but a portion of the fundus herniates through the diaphragmatic hiatus adjacent to the esophagus. The gastric fundus then serves as the leading point of herniation.
- Type III are a combination of Types I and II, with both the gastroesophageal junction and the fundus herniating through the hiatus. With progressive enlargement of the hernia through the hiatus, the phrenoesophageal membrane stretches, displacing the gastroesophageal junction above the diaphragm, thereby adding a sliding element to the type II hernia. The fundus lies above the gastroesophageal junction.
- Type IV hiatal hernias are associated with a large defect in the phrenoesophageal membrane, allowing other organs, such as colon, spleen, pancreas and small intestine to enter the hernia sac.

Typically, type 1 hiatal hernias are asymptomatic. However, with a large hernia the patient may have symptoms of gastroesophageal reflux disease (GERD) (e.g., heartburn, regurgitation, dysphagia). Many patients with a type II hernia are either asymptomatic or have only vague, intermittent symptoms. When present, symptoms are generally related to ischemia or partial or complete obstruction. The most common symptoms of type II hernia are epigastric or substernal pain, postprandial fullness, substernal fullness, nausea, and retching. A type II hernia can progressively enlarge so that the entire stomach eventually herniates, with the pylorus juxtaposed to the gastric cardia, forming an upside-down, intrathoracic stomach. Paraesophageal hernias are associated with abnormal laxity of structures normally preventing displacement of the stomach (gastrosplenic and gastrocolic ligaments). As the hernia enlarges, the greater curvature of the stomach rolls up into the thorax. Because the stomach is fixed at the gastroesophageal junction, the herniated stomach tends to rotate around its longitudinal axis resulting in an organoaxial volvulus. Gastric volvulus may lead to acute gastric obstruction, incarceration, and perforation (Kahrilas, 2025; Society of American Gastrointestinal and Endoscopic Surgeons [SAGES], 2024; Kahrilas et al., 2008)

Diagnosis is based on symptoms of GERD, surgical history (e.g., esophagomyotomy, partial gastrectomy), and diagnostic studies (e.g., upper endoscopy, barium swallow, endoscopy, esophageal manometry). Some physicians evaluate patients prior to bariatric surgery with an esophagogastroduodenoscopy or upper gastrointestinal study to detect conditions such as hiatal hernias and esophageal mucosal abnormalities related to gastroesophageal reflux (Mechanick, et al., 2008).

Symptoms of GERD are medically managed with medications that neutralize or reduce stomach acid. Surgery is generally reserved for emergency situations and for those who are not responsive to medications. Surgical repair of hiatal hernia by laparoscopy, laparotomy or thoracotomy is often combined with surgery for GERD. Nissen fundoplication is one method of repair used to treat GERD when it is caused by a hiatal hernia. Surgical repair of a paraesophageal hernia is typically not performed because the annual risk of developing acute symptoms requiring emergent surgery is less than 2% and the risk decreases exponentially after 65 years. The mortality rate from elective paraesophageal hernia repair is approximately 1.4%. Some propose that younger and healthier patients with a life expectancy of > 10 years should consider surgery to prevent both the risk of acute gastric volvulus and potentially progressive symptoms. Elective surgical repair of paraesophageal hernia is indicated in patients with subacute symptoms such as gastroesophageal reflux disease (GERD), refractory to medical therapy, dysphagia, early satiety, postprandial chest or abdominal pain, anemia, or vomiting. Emergent repair is required in patients with a gastric volvulus, uncontrolled bleeding, obstruction, strangulation, perforation, and/or respiratory compromise secondary to the hernia. The underlying surgical principles for successful repair include reduction of hernia contents, removal of the hernia sac, closure of the hiatal defect, and an antireflux procedure (Kahrilas, 2025; SAGES, 2024; Schieman, et al., 2009; Kahrilas et al., 2008). Hiatal hernia repair performed at the time of the primary bariatric procedure is considered integral to the procedure.

Literature Review

The few studies investigating the effectiveness and long-term outcomes of hiatal hernia repair performed at the time of bariatric surgery are primarily in the form of retrospective reviews, case reports and case series with small patient populations and short-term follow-ups. In some cases simultaneous hiatal hernia repair and bariatric surgery were proposed to prevent postoperative GERD. Studies included patients who were diagnosed preoperatively and those who were diagnosed intra-operatively (Mahawar, et al., 2015). There is insufficient evidence to support hiatal hernia repair in conjunction with bariatric surgery in an asymptomatic patient. Patient selection criteria for simultaneous procedures have not been established.

Professional Societies/Organizations

Society of American Gastrointestinal and Endoscopic Surgeons (SAGES): Based on a systematic review of the literature SAGES (2013) developed guidelines for the management of hiatal hernia. The guidelines included the following strong recommendations for surgical intervention for hiatal hernias:

- Repair of a type I hernia in the absence of reflux disease is not necessary. The indication for repair of a sliding (Type I) hiatal hernia is gastroesophageal reflux disease. The hernia is not the indication for the procedure but must be repaired. A fundoplication to address the reflux disease is mandatory. Outside of this situation, Type I sliding hiatal hernias have been thought to be almost inconsequential and not warranting surgical repair.
- All symptomatic paraesophageal hiatal hernias should be repaired particularly those with acute obstructive symptoms or which have undergone volvulus.
- Acute gastric volvulus requires reduction of the stomach with limited resection if needed.

Two weak recommendations by the Society stated that routine elective repair of completely asymptomatic paraesophageal hernias may not always be indicated. Consideration for surgery should include the patient's age and co-morbidities. Secondly, during operations for Roux-en-Y gastric bypass, sleeve gastrectomy and the placement of adjustable gastric bands, all detected hiatal hernias should be repaired because of the association with gastroesophageal reflux symptoms. This advice must be tempered by other reports which show that placement of an adjustable gastric band may relieve reflux symptoms, even without reduction of a hiatal hernia. Retrospective reviews and small case series suggested possible benefits of hiatal hernia repair combined with other types of bariatric surgery (e.g., adjustable gastric band placement; gastric bypass and sleeve gastrectomy).

Upper Endoscopy at the Time of Bariatric Surgery

The role of routine upper endoscopy in obese patients prior to bariatric surgery is controversial. The rationale for performing an upper endoscopy before bariatric surgery is to detect and/or treat lesions that might potentially affect the type of surgery performed, cause complications in the immediate postoperative period, or result in symptoms after surgery (American Society for Gastrointestinal Endoscopy [ASGE], 2015).

The American Association of Clinical Endocrinologists, The Obesity Society, and the American Society for Metabolic and Bariatric Surgery (AACE/TOS/ASMBS) guidelines for bariatric surgery stated that all gastrointestinal symptoms should be evaluated and treated before bariatric surgery. According to these guidelines, although it is commonplace for surgeons to perform a routine upper gastrointestinal study or endoscopy to screen for peptic ulcer disease before many other types of surgical procedures, this practice has been questioned for bariatric surgery. After bariatric surgery, upper intestinal endoscopy is the preferred diagnostic procedure for the evaluation of persistent and severe gastrointestinal symptoms (e.g., nausea, vomiting, abdominal pain). In many circumstances, upper endoscopy can also incorporate a therapeutic intervention with transendoscopic dilation of a recognized stricture (Mechanick, et al., 2008; updated 2020).

The 2015 updated guideline on the role of endoscopy in the bariatric surgery patient by the American Society for Gastrointestinal Endoscopy (ASGE) in conjunction with representatives from the Society of Gastrointestinal and Endoscopic Surgeons (SAGES) and the American Society for Metabolic and Bariatric Surgery (ASMBS) included the following statements:

- "We recommend water-soluble contrast radiography rather than endoscopy as the initial investigation in the postoperative bariatric patient suspected of having a leak or fistula.
- We recommend endoscopy as a first-line diagnostic study in the evaluation of the postoperative bariatric patient with abdominal pain, nausea, or vomiting. In the immediate postoperative period consultation with the surgeon is recommended.

- We recommend that endoscopic dilation of symptomatic stomal stenoses be planned in accordance with the type of anastomosis created during the original bariatric operation. Generally, dilation should be limited to 15 mm and should be avoided after LAGB and VBG procedures.”

The guideline does not discuss any indications for upper endoscopy performed during bariatric surgery

Professional society guidance suggests that upper endoscopy is warranted when performed in symptomatic patients prior to bariatric surgery. Well-designed prospective studies are needed to further evaluate the utility of preoperative routine upper endoscopy in bariatric surgery patients. Upper endoscopy performed at the time of bariatric surgery is not supported in the peer-reviewed medical literature and is not considered medically necessary.

Professional Societies/Organizations

Society of American Gastrointestinal Endoscopic Surgeons (SAGES): According to the 2008 Sages guideline for clinical application of laparoscopic bariatric surgery, preoperative weight loss may be useful to reduce liver volume and improve access for laparoscopic bariatric procedures, but mandated preoperative weight loss does not affect postoperative weight loss or comorbidity improvements. Laparoscopic Roux-en-Y gastric bypass (LRYGB), gastric banding by vertical banded gastroplasty or adjustable gastric banding, and biliopancreatic diversion with and without duodenal switch are established and validated bariatric procedures that provide effective long-term weight loss and resolution of co-morbid conditions. Laparoscopic sleeve gastrectomy (LSG) is validated as providing effective weight loss and resolution of comorbidities to 3-5 years. Laparoscopic revisional procedures may be performed safely, but with more complications than primary bariatric procedures, therefore the relative risks and benefits of laparoscopy should be considered on a case-by-case basis.

Health Equity Considerations

Health equity is the highest level of health for all people; health inequity is the avoidable difference in health status or distribution of health resources due to the social conditions in which people are born, grow, live, work, and age.

Social determinants of health are the conditions in the environment that affect a wide range of health, functioning, and quality of life outcomes and risks. Examples include safe housing, transportation and neighborhoods; racism, discrimination and violence; education, job opportunities and income; access to nutritious foods and physical activity opportunities; access to clean air and water; and language and literacy skills.

Individuals living with obesity, particularly those considering bariatric surgery, experience a greater prevalence of mental health challenges compared to the general population. The mental health risks associated with this encompass psychological distress, disordered eating patterns, and a decline in health-related quality of life. While bariatric surgery is recognized for its effectiveness in facilitating weight loss, it is important to acknowledge that it also presents considerable mental health challenges that should not be overlooked. It is crucial to approach this matter with sensitivity and understanding, as the implications for mental well-being can be profound and far-reaching (Francois, 2025; Morledge and Pories, 2020; Reddy, et al., 2020; Kubik, et al., 2013).

Epidemiologic data has shown that lower BMI values are correlated with risk of type 2 diabetes, cardiometabolic risk factors and increased risk of mortality in South Asian, Southeast Asian, and East Asian populations when compared to other ethnic groups (Ntuk, et al., 2014; Razak et al., 2007; Zhou, 2002). In 2000, the World Health Organization proposed the following weight

classification in adult Asians: BMI < 18.5 kg/m² indicates underweight, 18.5 to 22.9 kg/m² healthy weight, 23 to 24.9 kg/m² overweight, 25 to 29.9 kg/m² obese class I, and ≥ 30 kg/m² obese class II. The U.S. Census Bureau collects race data according to U.S. Office of Management and Budget guidelines and this data is based on self-identification. According to the U.S. Census Bureau, an Asian is a person with origins from the Far East (China, Japan, Korea, and Mongolia), Southeast Asia (Cambodia, Malaysia, the Philippine Islands, Thailand, Vietnam, Indonesia, Singapore, Laos, etc.), or the Indian subcontinent (India, Pakistan, Bangladesh, Bhutan, Sri Lanka, and Nepal) (2010). The National Center for Health Statistics (NCHS), a division of the Centers for Disease Control (CDC), also collects data on race and states that race is based on a respondent's description of their own racial background, regardless of Hispanic or Latino origin (2019-2023).

The U.S. Department of Health and Human Services set 10-year, measurable public health objectives for the nation, most recently with Healthy People 2030. Healthy People 2030 retained the Healthy People 2020 goal to reduce the proportion of adults with obesity. According to Healthy People 2030, 41.8% of American adults are obese. Healthy People 2020 states, "Obesity is a problem throughout the population. However, among adults, the prevalence is highest for middle-aged people and for non-Hispanic black and Mexican American women (Flegal, et al., 2010). Among children and adolescents, the prevalence of obesity is highest among older and Mexican American children and non-Hispanic black girls (Ogden, et al., 2010). The association of income with obesity varies by age, gender, and race/ethnicity (Ogden, et al., 2018)." According to the National Health and Nutrition Examination Survey 2013-2016 data, more women (40.8%) than men (36.5%) were obese, with non-Hispanic black women having the highest prevalence (55.9%) (Hales, et al., 2018).

Health disparities have been identified in outcomes of bariatric surgery among ethnic groups. Sheka et al. (2019) reported on an analysis of 108,198 patients from the 2015 Metabolic and Bariatric Surgery Accreditation and Quality Improvement Program national database to identify differences in mortality, length of stay, readmission, and reintervention by race in patients undergoing laparoscopic Roux-en-Y gastric bypass or laparoscopic sleeve gastrectomy (SG). Black patients had a higher body mass index (BMI) preoperatively (laparoscopic Roux-en-Y gastric bypass: 48.0 kg/m² vs. 45.7 kg/m²; SG 46.8 kg/m² vs. 44.9 kg/m²). In both the laparoscopic Roux-en-Y gastric bypass and SG groups, black patients had significantly longer length of stay and higher rates of readmission. Black patients had significantly higher 30-day mortality (0.2% versus 0.1%, $p < .001$) and higher rates of reoperation or reinterventions in the SG group. Amirian et al. (2020) compared the 30-day postoperative outcomes of 106,932 patients from the 2016 Metabolic and Bariatric Surgery Accreditation and Quality Improvement Program (MBSAQIP) database who underwent primary laparoscopic Roux-en-Y gastric bypass (LRYGB) or laparoscopic sleeve gastrectomy (LSG). The majority of the patients were white (79.5%), followed by 19.3% African American (AA), 0.5% Asian, 0.4% American Indian or Alaska Native, 0.3% Native Hawaiian or other Pacific Islander. After controlling for other covariates in multivariate logistic regression and selecting whites as reference, AA was the only race associated with a higher risk of postoperative complications and readmissions. Additionally, AA and American Indian or Alaska Natives were associated with a higher reintervention rate. For postoperative complications, AA had higher rates of pulmonary embolism and longer length of stay; Asian patients had higher wound disruption, urinary tract infections, and myocardial infarction.

Mocanu et al. (2020) conducted a retrospective review of the Metabolic and Bariatric Surgery Accreditation and Quality Improvement Program data registry (MBSAQIP) patients who underwent primary laparoscopic sleeve gastrectomy (LSG) or Roux-en-Y gastric bypass (RYGB) from 2015 to 2017 to identify rates of postoperative complications based on specific patient populations. A total of 430,936 patients were identified with 79.3% female, 73.1% white, 17.6% African-American and 9.3% other. When compared to females, males were more likely to develop complications (3.7% versus 3.45%; $p=0.002$), had increased reoperation rates (1.33% vs. 1.18%; $p < 0.001$)

and a 2-fold greater mortality (0.18% vs. 0.07; $p < 0.001$). At 30 days, female patients had increased intervention rates (1.34% vs. 1.18%; $p < 0.001$) and readmission rates (3.89% vs. 3.53%; $p < 0.001$). Black patients had higher rates of serious complications (4.14% vs. 3.41%; $p < 0.001$), mortality (0.13% vs. 0.09%; $p < 0.001$), intervention (1.74% vs. 1.24%; $p < 0.001$), and readmission (5.03% vs. 3.56%; $p < 0.001$) at 30 days when compared with white patients. Independent predictors of major complications were female sex ($p < 0.001$) and black race ($p < 0.001$). Black race was one of the greatest independent predictors of mortality ($p < 0.001$). As identified in these studies, there are significant differences in outcomes following bariatric surgery. The factors that underlie these disparities are unclear and require further study to optimize bariatric surgery outcomes.

Global Burden of Disease (GBD) 2015 Obesity Collaborators (2017) performed a systematic evaluation of prospective observational studies on the health effects of overweight and obesity in 195 countries over 25 years. The researchers found sufficient evidence supporting a causal relationship between high BMI and cancers of the esophagus, colon and rectum, liver, gallbladder and biliary tract, pancreas, breast, uterus, ovary, kidney, thyroid, and leukemia.

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Revision Details

Type of Revision	Summary of Changes	Date
Annual review	No clinical policy statement changes.	7/15/2026
Focused review	No clinical policy statement changes.	2/15/2026
Annual review	- Expanded coverage by lowering the body mass index (BMI) criteria for adults to: - BMI of ≥ 35 kg/m² (regardless of the presence, absence, or severity of obesity-associated medical problems) (BMI ≥ 27.5 kg/m² in Asian population) - BMI of 30–34.9 kg/m² with type 2 diabetes mellitus and/or other obesity-associated medical problems (BMI ≥ 25 kg/m² in Asian population) - Updated terminology: - metabolic dysfunction–associated steatotic liver disease (MASLD) replaces nonalcoholic fatty liver disease (NAFLD) - metabolic dysfunction–associated steatohepatitis (MASH) replaces nonalcoholic steatohepatitis (NASH) - Added coverage for endoscopic sleeve gastrectomy for the adult population	8/15/2025
Focused review	No clinical policy statement changes.	2/15/2025
Annual review	No clinical policy statement changes.	1/15/2025

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